測圓海鏡

Sea Mirror of Circle Measurements

卷一—卷三　|　Volumes I –III

Bilingual Edition —雙語版

Original Chinese • English Translation

元　李冶　撰

(Li Ye / Li Zhi, 1192–1279)

以天元術解勾股容圓一百七十問

Using the Method of the Celestial Element (Tian Yuan Shu)
to Solve 170 Problems on the Inscribed Circle

A foundational text of medieval Chinese algebra,
completed in the year 1248 of the Common Era.

Contents

1 Introduction / 引言

測圓海鏡簡介

《測圓海鏡》乃李冶（李治，1192–1279）之傑作。李冶為宋元四大數學家之一，與楊輝、秦九韶、朱世傑齊名。此書成於 1248 年，系統運用**天元術**（多項式代數方法）求解圓與直角三角形相關之一百七十問。
全書結構如下：

- **卷一**：理論基礎——圓城圖式、總率名號、今問正數、識別雜記（含六百九十二條幾何公式）
- **卷二至卷十二**：一百七十問，以天元術求解，方程次數自一次至六次。

本文檔包含**卷一至卷三**之全文，即全部基礎理論及首批問題。

Introduction

The *Ceyuan Haijing* (測圓海鏡, literally "Sea Mirror of Circle Measurements") is the masterpiece of Li Ye (李冶, also known as Li Zhi, 1192–1279), one of the four great mathematicians of the Song–Yuan period of China, alongside Yang Hui, Qin Jiushao, and Zhu Shijie.
Completed in 1248, this work systematically applies the **tian yuan shu** (天元術, "method of the celestial element")—an algebraic method for setting up and solving polynomial equations—to a total of 170 geometric problems concerning a circle inscribed in or related to a right triangle (勾股容圓).
The book is organized as follows:

- **Volume 1** establishes the theoretical foundation: the *Circle City Diagram* (圓城圖式), the *General Rate Names* (總率名號), the *Current Question Numbers* (今問正數), and the *Identification Miscellaneous Records* (識別雜記) containing 692 geometric formulas.
- **Volumes 2–12** present 170 problems, solved by the tian yuan shu method. The problems involve equations of degrees ranging from 1 to 6.

The present document contains the complete text of **Volumes 1–3**, comprising all foundational material and the first suite of problems.

2 Original Preface / 原序

原序

數本難窮，吾欲以力強窮之，彼其數不惟不能得其凡，而吾之力且憊矣。然則數果不可以窮耶？既已名之數矣，則又何為而不可窮也。

故謂數為難窮斯可，謂數為不可窮斯不可。何則？彼其冥冥之中，固有昭昭者存，夫昭昭者其自然之數也。非自然之數，其自然之理也。數一出於自然，吾欲以力強窮之，使隸首復生亦末如之何也已。

苟能推自然之理，以明自然之數，則雖遠而乾端坤倪，幽而神情鬼狀，未有不合者矣。

Original Preface

Numbers are fundamentally difficult to exhaust; if I try to force them by effort, not only will I fail to apprehend their essence, but my strength will be depleted as well. Yet can numbers truly not be exhausted? Since they are already called *numbers*, why then could they not be exhausted?

It is acceptable to say that numbers are difficult to exhaust; but it is unacceptable to say they cannot be exhausted. Why? In the midst of darkness, there is indeed brightness. That brightness is the natural number. It is not merely the natural number—it is the natural principle. Since numbers arise from nature, if I try to force them by effort, even if Li Shou (the legendary inventor of arithmetic) were reborn, he could do no better.

If one can extend natural principles to illuminate natural numbers, then even things as distant as the corners of heaven and earth, or as mysterious as spirits and ghosts, will all be in agreement.

予自幼喜算數，恆病夫考圓之術例出於牽強，殊乖於自然，如古率、徽率、密率之不同，截弧、截矢、截背之互見，內外諸角，析剖支條，莫不各自名家，與世作法。及反覆研究，卒無以當吾心焉。

老大以來，得洞淵九容之說，日夕玩繹，而向之病我者，使爆然落去而無遺余。山中多暇，客有從余求其說者，於是乎又為衍之，遂累一百七十問。

既成編，客復目之《測圓海鏡》，蓋取夫天臨海鏡之義也。

Since youth I have delighted in arithmetic. I was always troubled that the methods for investigating circles seemed forced and artificial, contrary to nature—the ancient rate, Hui's rate, and the close rate all differing; the arc, sagitta, and back each being considered separately; the internal and external angles analyzed into minute subdivisions—each school establishing its own name and method for the world. After repeated study, none could satisfy my mind.

In old age, I obtained the *Dongyuan Nine Containments* (洞淵九容) theory, and pondered it day and night. What had previously troubled me suddenly fell away completely, leaving no trace. With leisure time in the mountains, when guests asked me to explain this theory, I expanded upon it, eventually accumulating **one hundred and seventy problems**.

When the compilation was complete, a guest named it *Ceyuan Haijing*—taking the meaning of "heaven overlooking the sea mirror."

昔半山老人集《唐百家詩選》，自謂廢日力於此，良可惜。明道先生以上蔡謝君記誦為玩物喪志。夫文史尚矣，猶之為不足貴，況九九賤技能乎！

嗜好酸鹹，平生每痛自戒勑，竟莫能已，類有物憑之者，吾亦不知其然而然也。

故嘗私為之解曰：由技兼於事者言之，夷之禮，夔之樂，亦不免為一技。由技進乎道者言之，石之斤，扁之輪，非聖人之所與乎。

覽吾之編，察吾苦心，其憫我者當百數，其笑我者當千數。乃若吾之所自得，則自得焉耳，寧復為人憫笑計哉。

李冶序

In the past, the Old Man of Banshan compiled the *Anthology of One Hundred Tang Poets*, saying that to waste his days on such a thing was truly a pity. Master Mingdao criticized Xie of Shangcai for memorizing texts as "playing with things and losing one's ambition." Literature and history are esteemed subjects, yet even they are considered not worth valuing—how much less the lowly skill of multiplication!

My fondness for such sour and salty pursuits—throughout my life I have sternly admonished myself against them, yet in the end I could not stop, as if some force were driving me. I myself do not know why it is so.

So I have privately explained it thus: speaking from the perspective of skills combined with affairs, even the rites of Yi and the music of Kui were not exempt from being a single skill. Speaking from the perspective of advancing from skill to the Way, was not the stone-axe of [the sage] and the wheel of Bian [the wheelwright] praised by the sages?

Those who read my compilation and perceive my earnest efforts—those who pity me will number in the hundreds, those who laugh at me will number in the thousands. As for what I myself have gained, I have simply gained it; how could I calculate for the sake of being pitied or laughed at?

— Preface by Li Ye

3 Volume 1 / 卷一

卷一為全書之理論基礎，分為四大部分：

1. **圓城圖式**——主幾何圖形
2. **總率名號**——十五個直角三角形及其元素之定義
3. **今問正數**——各線段之數值
4. **識別雜記**——六百九十二條幾何公式

Volume 1 is the theoretical foundation of the entire work. It comprises four major sections:

1. **Circle City Diagram** (圓城圖式) — the master geometric configuration
2. **General Rate Names** (總率名號) — definitions of all 15 right triangles and their elements
3. **Current Question Numbers** (今問正數) — numerical values for all line segments
4. **Identification Miscellaneous Records** (識別雜記) — 692 geometric formulas

3.1 Circle City Diagram / 圓城圖式

圓城圖式為一百七十問所基之主幾何圖形。其構為一大直角三角形（勾股形），內切一圓，並設特定之點與線。

其情境如下：

> 假令有圓城一所，不知周徑，四面開門，門外縱橫各有十字大道。其西北十字道頭定為乾地，其東北十字道頭定為艮地，其東南十字道頭定為巽地，其西南十字道頭定為坤地。所有測望雜法一一設問如後。

The *Circle City Diagram* is the master geometric configuration upon which all 170 problems are based. It consists of a large right triangle (called 勾股形, “leg–leg shape”) with its inscribed circle, together with specific points and lines.

The scenario Li Ye constructs is:

> Suppose there is a circular city whose circumference and diameter are unknown. It has gates on all four sides, and outside each gate there are crossroads running north–south and east–west. The northwest crossroad is designated **Qian** (乾), the northeast **Gen** (艮), the southeast **Xun** (巽), and the southwest **Kun** (坤). All surveying methods are set forth as questions below.

二十二名點 / The 22 Named Points

最大直角三角形之頂點為**天**、**地**、**乾**。內切圓之心稱**心**。各點以單字命名：

天	地	乾	心	日	南	北	川
東	西	艮	坤	山	月	巽	青
泉	朱	金	泛	旦	夕		

The largest right triangle has vertices **Tian** (天, Heaven), **Di** (地, Earth), and **Qian** (乾). The center of the inscribed circle is called **Xin** (心, Heart). Key points include intersections of horizontal and vertical lines through the center and other constructed lines. Each point is designated by a single Chinese character:

Tian	Di	Qian	Xin	Ri	Nan	Bei	Chuan
Dong	Xi	Gen	Kun	Shan	Yue	Xun	Qing
Quan	Zhu	Jin	Fan	Dan	Xi		

3.2 General Rate Names / 總率名號

全書研究**十五個直角三角形**，皆以天地線之一段為弦（**弦**，即斜邊）。總率名號定義各三角形之三邊。
每個三角形：

- **弦**（c）——斜邊
- **股**（b）——較長之直邊
- **勾**（a）——較短之直邊

The work studies **15 right triangles**, all having a segment of the *Tiandi* (天地) line as their hypotenuse (弦, “string”). The *General Rate Names* define these triangles and their three sides.
In each triangle:

- **弦** (c) — the hypotenuse (斜邊)
- **股** (b) — the longer vertical leg
- **勾** (a) — the shorter horizontal leg

十五三角形表 / Table of 15 Triangles

No.	名稱 / Name	頂點 / Vertices	弦	股	勾
1	通 (Tong/General)	天–地–乾	通弦	通股	通勾
2	邊 (Bian/Side)	天–西–川	邊弦	邊股	邊勾
3	底 (Di/Base)	日–地–北	底弦	底股	底勾
4	黃廣 (Huangguang)	天–山–金	黃廣弦	黃廣股	黃廣勾
5	黃長 (Huangchang)	月–地–泉	黃長弦	黃長股	黃長勾
6	上高 (Shanggao)	天–日–旦	上高弦	上高股	上高勾
7	下高 (Xiagao)	日–山–朱	下高弦	下高股	下高勾
8	上平 (Shangping)	月–川–青	上平弦	上平股	上平勾
9	下平 (Xiaping)	川–地–夕	下平弦	下平股	下平勾
10	大差 (Dacha)	天–月–坤	大差弦	大差股	大差勾
11	小差 (Xiaocha)	山–地–艮	小差弦	小差股	小差勾
12	皇極 (Huangji)	日–川–心	皇極弦	皇極股	皇極勾
13	太虛 (Taixu)	月–山–泛	太虛弦	太虛股	太虛勾
14	明 (Ming)	日–月–南	明弦	明股	明勾
15	叀 (Zhuan)	山–川–東	叀弦	叀股	叀勾

注：上高 = 下高，上平 = 下平，故十五三角形中實僅十三個互異。

Note: 上高 = 下高 and 上平 = 下平, so among 15 triangles, only 13 are distinct.

3.3 Current Question Numbers / 今問正數

此節給出圖中每條線段之數值。內切圓之**半徑定為一百二十步**，作為標準單位。

This section gives the numerical value of every line segment in the diagram. The **radius of the inscribed circle is taken as 120 steps** (步), which serves as the standard unit.

1. 通（Tong / 通用）

弦六百八十	勾三百二十	股六百
勾股和九百二十	較二百八十	
勾和一千	較三百六十	
股和一千二百八十	較八十	
較和九百六十	較四百	
和和一千六百	較二百四十	

1. Tong (General)

Hypotenuse 680	Short leg 320, Long leg 600
Sum of legs 920	Difference 280
Leg-hypotenuse sum 1000	Difference 360
Leg-hypotenuse sum 1280	Difference 80
Difference-sum 960	Difference 400
Sum-sum 1600	Difference 240

That is: hypotenuse $c = 680$, short leg $a = 320$, long leg $b = 600$.

2. 邊（Bian / Side）

弦五百四十四	勾二百五十六	股四百八十
勾股和七百三十六	較二百二十四	
勾和八百	較二百八十八	
股和一千零二十四	較六十四	
較和七百六十八	較三百二十	
和和一千二百八十	較一百九十二	

2. Bian (Side)

Hypotenuse 544	Short leg 256, Long leg 480
Sum of legs 736	Difference 224
Leg-hypotenuse sum 800	Difference 288
Leg-hypotenuse sum 1024	Difference 64
Difference-sum 768	Difference 320
Sum-sum 1280	Difference 192

3. 底（Di / Base）

弦四百二十五	勾二百	股三百七十五
勾股和五百七十五	較一百七十五	
勾和六百二十五	較二百二十五	
股和八百	較五十	
較和六百	較二百五十	
和和一千	較一百五十	

3. Di (Base)

Hypotenuse 425	Short leg 200, Long leg 375
Sum of legs 575	Difference 175
Leg-hypotenuse sum 625	Difference 225
Leg-hypotenuse sum 800	Difference 50
Difference-sum 600	Difference 250
Sum-sum 1000	Difference 150

4. **黃廣（Huang Guang）**

弦五百一十　勾二百四十（即城徑也）　股四百五十
勾股和六百九十　較二百一十

其餘十一三角形依同樣模式，每個十三個數量（弦、勾、股、勾股和、勾股較、勾弦和、勾弦較、股弦和、股弦較、弦較和、弦較較、弦和和、弦和較），共 $15 \times 13 = 195$ 個數值條目。

4. **Huang Guang**

Hypotenuse 510　Short leg 240, Long leg 450 (this is the city diameter)
Sum of legs 690　Difference 210

The remaining 11 triangles follow the same pattern, each with 13 numerical quantities (弦, 勾, 股, 勾股和, 勾股較, 勾弦和, 勾弦較, 股弦和, 股弦較, 弦較和, 弦較較, 弦和和, 弦和較), giving a total of $15 \times 13 = 195$ numerical entries.

3.4 Identification Miscellaneous Records / 識別雜記

此為全書之理論核心，含**六百九十二條幾何公式**，關聯圖中各線段。此等公式純為幾何定理，與具體數值無關。後續各卷之問題皆以此等公式配合天元術求解。
本分為八部分：

1. **諸雜名目**——一般定義及三十餘條定理
2. **五和五較**
3. **諸弦**
4. **大小差**
5. **諸差**
6. **諸率互見**
7. **四位拾遺**
8. **拾遺**

This is the theoretical heart of the work. It contains **692 geometric formulas** relating the various line segments in the diagram. These formulas are purely geometric theorems, independent of any particular numerical values. The problems in all subsequent volumes are solved by using these formulas together with the tian yuan shu.
The section is divided into eight parts:

1. **Various Names** (諸雜名目) – general definitions and 30+ theorems
2. **Five Sums and Five Differences** (五和五較)
3. **Various Hypotenuses** (諸弦)
4. **Large and Small Differences** (大小差)
5. **Various Differences** (諸差)
6. **Mutual Relations of Rates** (諸率互見)
7. **Four-Place Supplement** (四位拾遺)
8. **Supplement** (拾遺)

諸雜名目示例：

Sample Definitions from Various Names:

名稱 / Name	**定義** / Definition
內率	明勾 × 叀股 / Ming 勾 × Zhuan 股
外率	明股 × 叀勾 / Ming 股 × Zhuan 勾
虛率	虛勾 × 虛股 / Xu 勾 × Xu 股
角差	高股 − 平勾 / Gao 股 − Ping 勾
次差	(明股 − 明勾) + (叀股 − 叀勾)
混同和	黃長股 + 黃廣勾 / Huangchang 股 + Huangguang 勾
傍差	(明股 − 明勾) − (叀股 − 叀勾)

定理示例：

Sample Theorems:

- 凡大差小差相乘為半段徑冪。
- 大差勾小差股相乘同上。
- 黃廣股黃長勾相乘為經冪。
- 勾股和即弦黃和。

- The product of the large difference and small difference equals half the square of the diameter.
- The product of the large-difference short-leg and the small-difference long-leg equals the same.
- The product of the Huangguang long-leg and Huangchang short-leg equals the square of the diameter.
- The sum of the two legs equals the sum of the hypotenuse and the diameter of the inscribed circle: $a + b = c + d$, where d is the diameter of the inscribed circle.

4 Volume 2 / 卷二

正率一十四問

此十四問構成「洞淵九容」序列，源自道教大師李思聰（號洞淵）之著作。其呈現直角三角形中九種基本容圓類型，另加五題。

Fourteen Questions on Standard Rates

These fourteen problems form the "Dongyuan Nine Containments" (洞淵九容) sequence, derived from the earlier work of the Daoist master Li Sicong (李思聰, known as "Master Dongyuan"). They present the nine fundamental types of circles contained in or related to a right triangle, plus five additional problems.

4.0.1 第一問 / Problem 1

問 甲乙二人俱在乾地，乙東行三百二十步而立，甲南行六百步望見乙，問徑幾里?

答 城徑二百四十步。

術 此為勾股容圓也。以勾股相乘，倍之為實，並勾股冪以求弦，復加入勾股共，以為法。

草 置甲南行六百步在地，以乙東行三百二十步乘之，得一十九萬二千步，倍之得三十入萬四千步為實。以乙東行步自之，得一十萬零二千四百步為勾冪；以甲南行步自之，得三十六萬步為股冪。二冪相並，得四十六萬二千四百步為方實，以平方開之，得六百八十步，則弦也。以弦加勾股共，共得一千六百步以為法。如法而一，得二百四十步，則城徑也。合問。

Problem. Two people, A and B, both stand at Qian. B walks 320 steps east and stops. A walks 600 steps south and sees B. What is the diameter?

Answer. The city diameter is 240 steps.

Method. This is the "circle contained in the right triangle" case. Multiply the two legs and double it to obtain the dividend. Combine the squares of the legs to find the hypotenuse, then add the sum of the legs to form the divisor.

Working. Place A's 600 steps south. Multiply by B's 320 steps east, obtaining 192,000 steps; double to get 384,000 as the dividend. Square B's eastward steps to get 102,400 as the short-leg square; square A's southward steps to get 360,000 as the long-leg square. Add the two squares: 462,400. Extract the square root to get 680, which is the hypotenuse. Add the sum of the legs to the hypotenuse: 1,600 as the divisor. Divide: 240 steps, the city diameter. This agrees with the question.

4.0.2 第二問 / Problem 2

問 甲乙二人俱在西門，乙東行二百五十六步，甲南行四百八十步望見乙，問答同前。

答 城徑二百四十步。

術 此為勾上容圓也。以勾股相乘，倍之為實，並勾股冪以求弦，加入股以為法。

草 置甲南行四百八十步在地，以乙東行二百五十六步乘之，得一十二萬二千八百八十步，倍之得二十四萬五千七百六十步為實。以乙東行步自之，得六萬五千五百三十六步為勾冪；以甲南行步自之，得二十三萬零四百步為股冪。勾股冪相並，得二十九萬五千九百三十六步為方實，以平方開之，得五百四十四步為弦也。以弦加入南行步，共得一千零二十四步以為法。而一，得二百四十步則城徑。合問。

Problem. Two people both stand at the West Gate. B walks 256 steps east, A walks 480 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on the short leg" case. Multiply the two legs and double as dividend. Combine the squares of the legs to find the hypotenuse; add the long leg as divisor.

Working. Place A's 480 steps south. Multiply by B's 256 steps east, obtaining 122,880; double to get 245,760 as dividend. Square B's eastward steps: 65,536 as the short-leg square. Square A's southward steps: 230,400 as the long-leg square. Combine: 295,936. Extract square root: 544, the hypotenuse. Add the southward steps: 1,024 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.3 第三問 / Problem 3

問 甲乙二人俱在北門，乙東行二百步而止，甲南行三百七十五步望見乙，問答同前。

答 城徑二百四十步。

術 此為股上容圓也。以勾股相乘，倍之為實，以勾股冪求弦，加入勾以為法。

草 置甲南行三百七十五步，以乙東行二百步乘之，得七萬五千步，倍之得一十五萬步為實。以乙東行自之，得四萬步為勾冪；以甲南行自之，得一十四萬零六百二十五步為股冪。勾股冪相並，得一十八萬零六百二十五步為方實。如平方而一，得四百二十五步則弦也。加入乙東行二百步，共得六百二十五步以為法。以法除之，得二百四十步，則城徑也。合問。

Problem. Two people both stand at the North Gate. B walks 200 steps east and stops. A walks 375 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on the long leg" case. Multiply the two legs and double as dividend. Use the squares of the legs to find the hypotenuse; add the short leg as divisor.

Working. Place A's 375 steps south. Multiply by B's 200 steps east: 75,000; double to get 150,000 as dividend. Square B's eastward steps: 40,000 as the short-leg square. Square A's southward steps: 140,625 as the long-leg square. Combine: 180,625. Extract square root: 425, the hypotenuse. Add B's 200 eastward steps: 625 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.4 第四問 / Problem 4

問 甲乙二人俱在圓城中心而立，乙穿城向東行一百三十六步而止，甲穿城南行二百五十五步望見乙，問答同前。

答 城徑二百四十步。

術 此為勾股上容圓也。以勾股相乘，倍之為實，並勾股冪如法求弦，以為法。

草 以二行步相乘，得三萬四千六百八十步，倍之得六萬九千三百六十步為實。置乙東行自之，得一萬八千四百九十六步為勾冪；又以甲南行自之，得六萬五千零二十五步為股冪。二冪相並，得八萬三千五百二十一步為方實，以平方開之，得二百八十九步即弦也。便以為法。如法除實，得二百四十步，即圓城之徑也。合問。

Problem. Two people both stand at the center of the circular city. B goes through the city, walking 136 steps east and stops. A goes through the city, walking 255 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on both legs" case. Multiply the two legs and double as dividend. Combine the squares of the legs to find the hypotenuse as divisor.

Working. Multiply the two walking distances: 34,680; double to get 69,360 as dividend. Square B's eastward steps: 18,496 as the short-leg square. Square A's southward steps: 65,025 as the long-leg square. Combine: 83,521. Extract square root: 289, the hypotenuse. This serves as the divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.5 第五問 / Problem 5

問 甲乙二人同立於乾地，乙東行一百八十步遇塔而止，甲南行三百六十步，回望其塔正居城徑之半，問答同前。

答 城徑二百四十步。

術 此為弦上容圓也。以勾股相乘，倍之為實，以勾股和為法。

草 以二行步相乘，得六萬四千八百步，倍之得一十二萬九千六百步為實。並二行步，得五百四十步以為法。除實，得二百四十步，即城徑也。合問。

Problem. Two people stand together at Qian. B walks 180 steps east and reaches a pagoda. A walks 360 steps south and looks back: the pagoda lies exactly at half the city diameter. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle on the hypotenuse" case. Multiply the two legs and double as dividend. Use the sum of the legs as divisor.

Working. Multiply the two walking distances: 64,800; double to get 129,600 as dividend. Add the two distances: 540 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.6 第六問 / Problem 6

問 甲乙二人俱在坤地，乙東行一百九十二步而止，甲南行三百六十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為勾外容圓也。以勾股相乘，倍之為實，以弦較和為法。

草 以二行步相乘，得六萬九千一百二十步，倍之得一十三萬八千二百四十步為實。置乙東行自之，得三萬六千八百六十四步為勾冪；又置甲南行自之，得一十二萬九千六百步為股冪。二冪相並，得一十六萬六千四百六十四步為方實，以平方開之，得四百零八，即弦也。又置甲南行步內減乙東行步，餘一百六十八步，即較也。以較加弦，共得五百七十六步以為法。實如法而一，得二百四十步，為城徑也。合問。

〔按〕此題用勾股求得弦，即可加減得弦較較為城徑。今必以勾股相乘倍積為實，求得弦較和為法，而後始得弦較較為城徑者，蓋欲因此並明勾股相乘之倍積為弦較較、弦較和相乘之積，非故為紆回也。

Problem. Two people both stand at Kun. B walks 192 steps east and stops. A walks 360 steps south and sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle outside the short leg" case. Multiply the two legs and double as dividend. Use the sum of hypotenuse and difference as divisor.

Working. Multiply the two walking distances: 69,120; double to get 138,240 as dividend. Square B's eastward steps: 36,864 as the short-leg square. Square A's southward steps: 129,600 as the long-leg square. Combine: 166,464. Extract square root: 408, the hypotenuse. Subtract B's steps from A's: 168, the difference. Add difference to hypotenuse: 576 as divisor. Divide: 240 steps, the city diameter. This agrees.

[Translator's note: This problem uses the right triangle to find the hypotenuse, then by addition and subtraction obtains the "hypotenuse-difference-difference" as the city diameter. The method deliberately uses the product of the legs (doubled) as dividend to demonstrate that this product equals the product of (hypotenuse−difference) and (hypotenuse+difference).]

4.0.7 第七問 / Problem 7

問 甲乙二人同立於艮地，甲南行一百五十步而止，乙東行八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為股外容圓也。以勾股相乘，倍之為實，以弦和較為法。

Problem. Two people stand together at Gen. A walks 150 steps south and stops. B walks 80 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle outside the long leg" case. Multiply the two legs and double as dividend. Use the difference of hypotenuse-sum as divisor.

4.0.8 第八問 / Problem 8

問 甲乙二人同立於坤地，甲南行一百九十二步而止，乙東行七十二步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為弦外容圓也。以勾股相乘，倍之為實，以勾股較為法。

Problem. Two people stand together at Kun. A walks 192 steps south and stops. B walks 72 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "circle outside the hypotenuse" case. Multiply the two legs and double as dividend. Use the difference of the legs as divisor.

4.0.9 第九問 / Problem 9

問 甲乙二人俱在巽地，乙東行一百九十二步而止，甲南行三百六十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為勾外容半圓也。以勾股相乘，倍之為實，以股弦和為法。

Problem. Two people both stand at Xun. B walks 192 steps east and stops. A walks 360 steps south. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "semicircle outside the short leg" case. Multiply the two legs and double as dividend. Use the sum of long leg and hypotenuse as divisor.

4.0.10 第十問 / Problem 10

問 甲乙二人俱在艮地，甲南行一百五十步而止，乙東行八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為股外容半圓也。以勾股相乘，倍之為實，以勾弦和為法。

Problem. Two people both stand at Gen. A walks 150 steps south and stops. B walks 80 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the "semicircle outside the long leg" case. Multiply the two legs and double as dividend. Use the sum of short leg and hypotenuse as divisor.

以上十問構成「洞淵九容」，第五問（弦上容圓）為補充。卷二其餘四問繼續以天元術求解之標準率問題。

The above ten problems constitute the "Dongyuan Nine Containments" (洞淵九容), supplemented by Problem 5 (circle on the hypotenuse). The remaining four problems of Volume 2 continue with additional standard-rate problems using the tian yuan shu method.

5 Volume 3 / 卷三

邊股一十七問

卷三呈十七問，所給數據涉及邊股（b_2），定義為「邊」三角形（表中第二）之長直邊，等於四百八十步。本卷所有問題之未知數皆為圓城直徑（城徑 = 二百四十步）。此等問題展示天元術之漸增精妙，方程次數自二次至三次。

Seventeen Questions on the Side-Long-Leg

Volume 3 presents 17 problems in which the given data involve the *side long-leg* (邊股, b_2), defined as the long leg of the "Side" triangle (邊 triangle, No. 2 in the table), equal to 480 steps. In all problems of this volume, the unknown to be found is the diameter of the circular city (城徑 = 240 steps). These problems demonstrate the increasing sophistication of the tian yuan shu method, with equations ranging from quadratic to cubic degree.

5.0.1 第一問 / Problem 1

問 乙出東門南行，不知步數而止。甲出西門南行四百八十步，望見乙，復就乙行五百一十步與乙相會，問答同前。

答 城徑二百四十步。

術 倍相減步，以乘二之甲南行步，為平方實，得城徑。

草 識別得二行相減，餘三十步，即乙出東門南行步也。倍相減步，得六十步，以乘二之甲南行步九百六十步，得五萬七千六百步，為平方實。如法開之，得二百四十步，即城徑也。合問。

Problem. B goes out the East Gate and walks south, an unknown number of steps, then stops. A goes out the West Gate and walks 480 steps south, sees B, then walks 510 steps to reach B. Answer as before.

Answer. The city diameter is 240 steps.

Method. Double the difference of the two walking distances; multiply by twice A's southward steps to obtain the square-root dividend; this gives the city diameter.

Working. Recognize: subtracting the two distances leaves 30 steps, which is B's southward distance from the East Gate. Double the difference: 60 steps. Multiply by twice A's southward steps (960): 57,600, the square-root dividend. Extract the square root: 240 steps, the city diameter. This agrees.

5.0.2 第二問 / Problem 2

問 甲出西門南行四百八十步而止，乙從艮隅東行八十步，望見甲，問答同前。

答 城徑二百四十步。

術 倍南行步，以東行步乘之為實，東行步為從方，一步常法，得全徑。

草 立天元一為全徑，以減於二之甲南行步，得為兩個大差也。以乙東行步乘之，得為圓徑冪，寄左。然後以天元冪與左相消，得以帶縱平方開之，得二百四十步，即城徑也。合問。

又法： 半之乙東行步，乘南行步為實，半之乙東行步為從，一步常法，得半徑。

草 立天元一為半城徑，減甲南行步，得為大差也。以半之東行步乘之，得即半徑冪，寄左。然後以天元冪為同數，與左相消，得開帶縱平方，得一百二十步。倍之，即城徑也。合問。

Problem. A goes out the West Gate and walks 480 steps south, then stops. B walks 80 steps east from the Gen corner and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Double the southward steps; multiply by the eastward steps as dividend; the eastward steps as the “following square”; unity as the constant method; obtain the full diameter.

Working. Set the celestial element unity as the full diameter. Subtract from twice A’s southward steps to obtain two large differences. Multiply by B’s eastward steps to obtain the circle-diameter square; store on the left. Then cancel the celestial element square with the left; extract the square root with a following edge: 240 steps, the city diameter. This agrees.

Alternative Method: Halve B’s eastward steps; multiply by the southward steps as dividend; halved eastward steps as the following term; unity as constant method; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from A’s southward steps to obtain the large difference. Multiply by halved eastward steps to obtain the radius square; store on the left. Then use the celestial element square as the equal number; cancel with the left; extract the square root with a following edge: 120 steps. Double it: the city diameter. This agrees.

5.0.3 第三問 / Problem 3

問 甲出西門南行四百八十步而止，乙從艮隅亦南行一百五十步，望見甲，問答同前。

答 城徑二百四十步。

術 兩行步相乘為實，南行步為從方，一為隅，得半徑。

草 立天元一為半城徑，以減乙南行步，得為半梯頭；以甲行步為梯底，以乘之，得為半徑冪，寄左。然後以天元冪與左相消，得開帶縱平方，得一百二十步。倍之，即城徑也。合問。

Problem. A goes out the West Gate and walks 480 steps south, then stops. B also walks 150 steps south from the Gen corner and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply the two walking distances as dividend; the southward steps as the “following square”; unity as the corner; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from B’s southward steps to obtain half the trapezoid head. Use A’s steps as the trapezoid base; multiply to obtain the radius square; store on the left. Then cancel the celestial element square with the left; extract the square root with a following edge: 120 steps. Double it: the city diameter. This agrees.

5.0.4 第四問 / Problem 4

問 甲出西門南行四百八十步，乙出東門直行一十六步，望見甲，問答同前。

答 城徑二百四十步。

術 以四之東行步，乘南行冪為實，從空，東行為廉，一步為隅法，得全徑。

草 立天元一為圓徑，加乙東行步，得為中勾；其甲南行即中股也。置東行步為小勾，以中股乘之，得，合以中勾除，今不受除，便以為小股也。內寄中勾分母。乃復以中股乘之，得三百六十八萬六千四百，又四之，得一千四百七十四萬五千六百，為一段圓徑冪，寄中勾分母，寄左。然後以天元徑自之，又以中勾乘之，得為同數，與左相消，得以帶縱立方開之，得二百四十步，為城徑也。合問。

〔按〕不受除者，無可除之理也。凡二數，此數於彼數有可除之理，則受除；無可除之理，則不受除也。蓋除有法有實，實可二，法不可二。此題以中勾為法，而中勾內有一元，又有十六步，其為數已二矣，又何以均分不一之數乎？故曰不受也。寄分者，姑寄其應除之數也。俟求得兩相等數，而此數內尚少一除，不除此而轉乘彼，則兩數仍相等，猶之受除者也。此所謂以乘代除也。

Problem. A goes out the West Gate and walks 480 steps south. B goes out the East Gate and walks 16 steps straight, and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply four times the eastward steps by the square of the southward steps as dividend; the "following" term is empty; the eastward steps as the edge; unity as the corner method; obtain the full diameter.

Working. Set the celestial element unity as the circle diameter. Add B's eastward steps to obtain the middle short-leg. A's southward steps are the middle long-leg. Place the eastward steps as the small short-leg; multiply by the middle long-leg. This should be divided by the middle short-leg, but it is "not accepted for division"—use it directly as the small long-leg, with the middle short-leg as denominator. Multiply again by the middle long-leg: 3,686,400; multiply by 4: 14,745,600 as a segment of the circle-diameter square, with the middle short-leg as denominator; store on the left. Then square the celestial element diameter and multiply by the middle short-leg as the equal number; cancel with the left; extract the cube root with a following edge: 240 steps, the city diameter. This agrees.

[Translator's note: "Not accepted for division" means there is no valid way to divide. When two numbers are such that one can be divided by the other, the division is "accepted"; otherwise it is "not accepted." For division has divisor and dividend; the dividend can be compound, but the divisor cannot. Here the middle short-leg serves as divisor, but it contains both the celestial element and 16 steps—it is already compound. How can one evenly divide a non-uniform number? Hence "not accepted." Storing the denominator means temporarily retaining the number that should be divided. When two equal quantities are found, and one still lacks a division, instead of dividing this, multiply that; the two quantities remain equal, as if the division had been accepted. This is the so-called "substituting multiplication for division."]

5.0.5 第五問 / Problem 5

問 乙出南門東行七十二步而止，甲出西門南行四百八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 以乙東行冪，乘甲南行為實；乙東行冪為從方；甲南行步內減二之東行步為益廉；一步常法，得半徑。

草 立天元一為半城徑，以減南行步，得為小股。又以天元加乙東行步，得為小勾。又以天元加南行步，得為大股。乃置大股在地，以小勾乘之，得下式，合以小股除之，今不受除，便以為大勾，內寄小股分母。又置天元半徑，以分母小股乘之，得以減大勾，得為半個梯底於上。以乙東行七十二步為半個梯頭，以乘上位，得為半徑冪，內寄小股分母，寄左。然後置天元冪，又以分母小股乘之，得為同數，與左相消，以立方開之，得一百二十步。倍之，即城徑也。合問。

又法： 以二數相乘為實，相減為從，一虛法，平開，得半徑。

Problem. B goes out the South Gate and walks 72 steps east, then stops. A goes out the West Gate and walks 480 steps south. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply the square of B's eastward steps by A's southward steps as dividend; the square of B's eastward steps as the "following square"; subtract twice the eastward steps from A's southward steps as the "beneficial edge"; unity as the constant method; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from the southward steps to obtain the small long-leg. Add the celestial element to B's eastward steps to obtain the small short-leg. Add the celestial element to the southward steps to obtain the large long-leg. Place the large long-leg; multiply by the small short-leg. This should be divided by the small long-leg, but it is "not accepted for division"—use it directly as the large short-leg, with the small long-leg as denominator. Place the celestial element radius; multiply by the denominator (small long-leg); subtract from the large short-leg to obtain half the trapezoid base on top. Use B's 72 eastward steps as half the trapezoid head; multiply the upper quantity to obtain the radius square, with the small long-leg as denominator; store on the left. Then place the celestial element square; multiply by the denominator (small long-leg) as the equal number; cancel with the left; extract the cube root: 120 steps. Double it: the city diameter. This agrees.

Alternative Method: Multiply the two numbers as dividend; their difference as the following term; unity as the empty method; extract the square root; obtain the radius.

5.0.6 第六問 / Problem 6

問 乙出南門東行，不知步數而立。甲出西門南行四百八十步，望見乙與城參相直，又就乙行四百零八步與乙相會，問答同前。

答 城徑二百四十步。

[其餘十一問依同樣模式，所給數據涉及邊股（邊股 = 四百八十步）及其他各種測量，每問皆以天元術建立多項式方程，求解得城徑二百四十步。]

Problem. B goes out the South Gate and walks east, an unknown number of steps, then stands. A goes out the West Gate and walks 480 steps south, sees B aligned with the city, then walks 408 steps to reach B. Answer as before.

Answer. The city diameter is 240 steps.

[The remaining 11 problems of Volume 3 continue in the same pattern, with given data involving the side long-leg (邊股 = 480 steps) and various other measurements, each leading to polynomial equations solved by the tian yuan shu method to yield the city diameter of 240 steps.]

6 The Tian Yuan Shu Method / 天元術

天元術

天元術（「天元術」，「method of the celestial element」）為《測圓海鏡》全書所運用之代數技法。此乃中國數學史上最早之系統方法，用以建立並求解一元多項式方程。

The Tian Yuan Shu Method

The *tian yuan shu* (天元術, "method of the celestial element") is the algebraic technique employed throughout the *Ceyuan Haijing*. It is the earliest systematic method in Chinese mathematics for setting up and solving polynomial equations with one unknown.

6.1 Methodology / 方法論

其程序如下：

1. **立天元一為** X（「設天元一為 X」）——宣告未知量。以現代記號，等同於「令 $x = X$」。
2. 以天元 x 表達所有已知量及關係。
3. 構造兩個相等之表達式（常以不同方法計算同一幾何量），一為「左」式暫存，另一為「同數」。
4. **相消**（「mutual cancellation」）——令兩式相等並化簡，得多項式方程之標準形式。
5. 以開方法求解：開平方、開立方、或更高次之開方。

The procedure follows these steps:

1. **Set the celestial element unity to be** X ("立天元一為 X") — This declares the unknown quantity. In modern notation, it is equivalent to "Let $x = X$."
2. Express all given quantities and relationships in terms of the celestial element x.
3. Construct two equal expressions (often representing the same geometric quantity computed by different methods), one designated as the "left" (左) expression and stored temporarily, the other as the "equal number" (同數).
4. **Mutual cancellation** (相消) — Set the two expressions equal and simplify to obtain the polynomial equation in standard form.
5. Solve the equation by root-extraction methods (開方): square-root extraction (開平方), cube-root extraction (開立方), or higher-degree root extraction.

6.2 Polynomial Notation / 多項式記號

李冶運用籌算板上的位置記法：

- 係數縱向排列，常數項（實）在上。
- 其下：x 之係數（方，「square」），繼以 x^2（廉，「edge」），繼以 x^3（隅，「corner」）。
- 更高次：第一廉、第二廉等。
- 負係數以末位斜劃表示。

Li Ye uses a positional notation on a counting-rod board:

- Coefficients are arranged vertically, with the constant term (實) at the top.
- Below it: the coefficient of x (方, "square"), then x^2 (廉, "edge"), then x^3 (隅, "corner").
- For higher degrees: first 廉, second 廉, etc.
- Negative coefficients are indicated by a diagonal stroke through the last digit.

6.3 Example: Quadratic Equation / 示例：二次方程

第一問導出相當於下式之方程：

$$384000 = 1600 \cdot d$$

其中 d 為直徑。但更複雜之問題產生真正的多項式方程。例如卷三第四問產生**三次方程**：

$$(4 \cdot 16)(480)^2 = (480 + 16) \cdot d^2 - d^3$$

以現代形式，李冶以「帶縱立方」解之。

The first problem leads to the equation equivalent to:

$$384000 = 1600 \cdot d$$

where d is the diameter. But more complex problems yield true polynomial equations. For instance, Problem 4 of Volume 3 produces a **cubic equation**:

$$(4 \cdot 16)(480)^2 = (480 + 16) \cdot d^2 - d^3$$

in modern form, which Li Ye solves by "opening the cube with an edge" (帶縱立方).

6.4 Historical Significance / 歷史意義

《測圓海鏡》為現存最早系統運用天元術之著作。所解方程包括：

次數 / **Degree**	**方程數量** / **Number**
一次 / Linear (degree 1)	31
二次 / Quadratic (degree 2)	106
三次 / Cubic (degree 3)	24
四次 / Quartic (degree 4)	20
六次 / Sextic (degree 6)	1
總計 / **Total**	**182**

The *Ceyuan Haijing* is the earliest surviving work that systematically employs the tian yuan shu. The equations solved include:

Degree	**Number of Equations**
Linear (degree 1)	31
Quadratic (degree 2)	106
Cubic (degree 3)	24
Quartic (degree 4)	20
Sextic (degree 6)	1
Total	**182**

注：本排印版呈現《測圓海鏡》卷一至卷三之全文。卷一包含全部定義、數值數據及六百九十二條幾何公式。卷二、卷三呈現一百七十問中之首批三十一問，以天元術求解。其餘九卷（卷四–卷十二）繼續一百三十九問，複雜度漸增，至卷七達六次方程。

Note: This typeset edition presents the complete text of Volumes 1–3 of the Ceyuan Haijing. Volume 1 contains all definitions, numerical data, and 692 geometric formulas. Volumes 2 and 3 present the first 31 of 170 problems, solved by the tian yuan shu method. The remaining nine volumes (卷四–卷十二) continue with 139 further problems of increasing complexity, culminating in a sixth-degree equation in Volume 7.

— End of Volumes 1–3 / 卷一至卷三終 *—*

欽定四庫全書・Imperial Siku Quanshu

子部・Mathematics and Astronomy

測圓海鏡　Ceyuan Haijing

卷四至卷六　Volumes 4–6

底勾、大股、大勾三卷全編

Double Difference, Large Leg, and Large Base: Complete

元　李冶　撰

Yuan Dynasty　Li Ye (Renqing, Jingzhai)

字仁卿，號敬齋，真定欒城人

Jinshi of the Late Jin; Hanlin Academician under Yuan

Bilingual English–Chinese Edition

Original Chinese with English Translation

Parallel Text: Chinese (Left) ・English (Right)

依《欽定四庫全書》本・Based on the Imperial Siku Quanshu Edition

卷四：第五十一問至第六十七問・Vol. 4: Problems 51–67

卷五：第六十八問至第八十五問・Vol. 5: Problems 68–85

卷六：第八十六問至第一百三問・Vol. 6: Problems 86–103

Contents

引言・Introduction

《測圓海鏡》十二卷，元李冶撰。冶字仁卿，號敬齋，眞定欒城人。金末進士，入元官翰林學士。此書乃其所著天元術之傑作，以勾股測圓之法，設問答以明天元一之術。全書凡一百七十問，皆設城池圓形，以人物行步之數推求圓徑。本編收錄卷四、卷五、卷六三卷，凡五十三問：

- 卷四：底勾一十七問（第五十一問至第六十七問）
- 卷五：大股一十八問（第六十八問至第八十五問）
- 卷六：大勾一十八問（第八十六問至第一百三問）

書中設問之例：假令有圓城一座，甲乙二人各從城門出行，或東或南，行若干步，遙相望見，或斜行相會。只知若干步數，餘皆不知，以求圓城之徑。每問皆有問、答、草、法四部分。

Ceyuan Haijing (Sea Mirror of Circle Measurements), in twelve volumes, was written by Li Ye of the Yuan Dynasty. Li Ye, styled Renqing, sobriquet Jingzhai, was from Luancheng, Zhending. A Jinshi of the late Jin period, he served as Hanlin Academician under the Yuan. This work is his masterpiece on the *tian yuan* (celestial element) method, using right-triangle circle-measurement techniques to elucidate the art of *tian yuan yi*. The complete work contains 170 problems, each positing a circular city wall, from whose walkers' paces the diameter is sought.

This edition contains Volumes 4, 5, and 6—53 problems in total:

- **Volume 4**: 17 problems on the base leg (51–67)
- **Volume 5**: 18 problems on the large leg (68–85)
- **Volume 6**: 18 problems on the large base (86–103)

Each problem presents a scenario: A and B depart from city gates, walking east or south, viewing each other diagonally or meeting obliquely. Given certain paces, all else unknown, find the city diameter. Each problem has four parts: **Question**, **Answer**, **Working**, and **Method**.

天元術記法說明：書中以籌算式記多項式，自下而上排列，常數項在下，一次項在上，二次又在上，依此類推。如 $\frac{480}{1}$ 表示 $480-x$，$\frac{0\quad 0\quad 48000}{1\quad 0\quad 0}$ 表示 $48000-x^2$ 之類。所謂「立天元一為某數」，即設未知數 x 為所求之量。

On Tian Yuan Notation: Polynomials are recorded in counting-rod style, arranged bottom-to-top: constant term lowest, linear term above, quadratic above that, etc. Thus $\frac{480}{1}$ denotes $480-x$, and $\frac{0\quad 0\quad 48000}{1\quad 0\quad 0}$ denotes $48000-x^2$. "Let the celestial element unity be [some quantity]" means: let unknown x equal the sought quantity.

1 卷四：底勾一十七問・Vol. 4: Double Difference (17 Problems)

底勾一十七問・Double Difference

（第五十一問至第六十七問・Problems 51–67）

【題意總說】卷四論「底勾」之術。所謂底勾，乃勾股形之最下勾股，與圓徑相切之基本勾股形也。凡一十七問，皆以底勾為基，求圓城之徑。底勾者，通勾股形之勾邊，亦即大勾，為諸勾股形之根本。諸問或知甲東行之數，或知乙南行之數，或知斜行之數，三者之中知其二，即可立天元一，建方程而開方得圓徑。

[General Summary] Volume 4 treats the “base leg” (底勾) method. The base leg is the lowest side of the right triangle formed tangent to the circular diameter—the fundamental right triangle. All 17 problems use the base leg as foundation to find the city diameter. It is the base of the universal right triangle, i.e., the large base, the root of all sub-triangles. Given any two of: A’s eastward walk, B’s southward walk, or the diagonal walk, one may establish the celestial element and solve.

第一問： 或問：乙出南門東行不知步數而立，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問圓城之徑幾何?

答曰：城徑二百四十步。

草曰：識別得二行相減餘，即乙出南門東行數也。以甲東行二百步減於就乙斜行二百七十二步，餘七十二步，為乙東行數。以乘甲東行步，得 $72 \times 200 =$ 一萬四千四百步，又四之，得五萬七千六百步為實。以平方開之，得 $\sqrt{57600} =$ 二百四十步，即城徑也。
驗算：城徑二百四十步，半徑一百二十步。乙東行七十二步，甲東行二百步，甲比乙多行 $200 - 72 = 128$ 步。以勾股術驗之，斜行為弦，$\sqrt{128^2 + 120^2} = \sqrt{16384 + 14400} = \sqrt{30784}$，復以半徑加之，合得斜行步數，其術驗矣。

法曰：二行差數乘甲東行又四之，為平方實，開方得全徑。
術曰：以就乙斜行步內減甲東行步，餘為乙東行數；以此數乘甲東行步為實，四之；開平方，即得城徑。

Problem 51: Suppose B exits the south gate and walks east an unknown number of paces, while A exits the north gate and walks east 200 paces to see him. A then walks diagonally 272 paces to meet B. What is the diameter of the circular city wall?

Answer: The city diameter is 240 paces.

Working: By inspection, the difference of the two walks gives B’s eastward walk. Subtracting A’s 200 paces from the diagonal 272 paces leaves 72 paces as B’s eastward walk. Multiplying by A’s 200 paces gives $72 \times 200 = 14{,}400$ paces. Quadrupling gives $57{,}600$ as the dividend. Extracting the square root: $\sqrt{57600} = 240$ paces, the city diameter.
Verification: Diameter 240, radius 120. B walks 72 east, A walks 200 east; difference $200 - 72 = 128$. By the gougu (Pythagorean) theorem: diagonal is the hypotenuse; $\sqrt{128^2 + 120^2} = \sqrt{16384 + 14400} = \sqrt{30784}$. Adding the radius gives the diagonal walk, confirming the result.

Method: The difference of the two walks multiplied by A’s eastward walk, then quadrupled, is the dividend for square-root extraction. Method: From the diagonal walk subtract A’s eastward walk, remainder is B’s eastward walk. Multiply this by A’s eastward walk as dividend; quadruple; extract square root to obtain the city diameter.

第二問：或問：乙出南門東行一百二十步而立，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑，以減於二之甲東行步，得$\frac{480}{1}$為兩个小差。以乙東行一百二十步乘之，得$\frac{57600}{1}$為城徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，得$\frac{57600}{1\quad 0}$（負上正下），以平方開之，得二百四十步，即城徑也。
釋義：二之甲東行者，以甲東行為勾，其倍即通勾與圓徑之差。以乙東行乘之者，乙東行即小勾，乘大差得大勾之冪，亦即城徑冪之半。故以天元冪相消，開平方即得。

法曰：半之乙東行步乘甲東行為實，半乙東行為從，一步常法，得半徑。術曰：置乙東行步折半，以乘甲東行步為實；以半乙東行步為從法；一步為隅法；開平方得半徑，倍之即城徑。

Problem 52: Suppose B exits the south gate and walks east 120 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtracting it from twice A's eastward walk gives $\frac{480}{1}$ as the two small differences. Multiplying by B's 120 paces gives $\frac{57600}{1}$ as the city diameter-squared, deposited on the left. Then canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{57600}{1\quad 0}$ (negative above, positive below). Extracting the square root yields 240 paces, the city diameter.
Explanation: Twice A's eastward walk: A's walk being the base, its double is the difference between the universal base and the diameter. Multiplying by B's walk (the small base) gives the square of the large base, i.e., half the city-diameter squared. Cancelling with the celestial element and extracting the root gives the result.

Method: Half of B's eastward walk multiplied by A's eastward walk is the dividend; half of B's eastward walk is the coefficient; unity is the constant divisor. This yields the radius. Method: Halve B's eastward walk, multiply by A's eastward walk as dividend; use half B's walk as the coefficient, unity as the corner divisor; extract square root to obtain the radius; double it for the diameter.

第三問：或問：乙從坤隅南行三百六十步，甲出北門東行二百步見之。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，減於乙南行三百六十步之半，得$\frac{180}{1}$為小差。半乙南行步，得一百八十步，以乘小差，得$\frac{32400}{180}$為半徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，得$\frac{32400}{180\quad 1\quad 0}$（上負下正），開平方得一百二十步，倍之得全徑二百四十步。
釋義：此問以乙南行為股，甲東行為勾。半乙南行即大股之半，去半徑餘為小差。以小差乘半股得半徑冪，此天元術之妙用也。

法曰：兩行步相乘為實，甲東行為從，乙南行為隅，開平方得半徑。術曰：以甲東行步乘乙南行步為實，以甲東行步為從，乙南行步為隅法；開平方得一百二十步，即半徑；倍之即城徑。

Problem 53: Suppose B walks south 360 paces from the Kun corner, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting it from half of B's southward walk (360 paces) gives $\frac{180}{1}$ as the small difference. Half of B's southward walk is 180 paces; multiplying by the small difference gives $\frac{32400}{180}$ as the radius-squared, deposited on the left. Canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{32400}{180\quad 1\quad 0}$ (negative above, positive below). Extracting square root gives 120 paces; doubling gives the full diameter of 240.
Explanation: Here B's southward walk is the leg, A's eastward walk is the base. Half B's walk is half the large leg; less the radius gives the small difference. Multiplying small difference by half the leg gives the radius-squared—the subtle efficacy of the tian yuan method.

Method: The product of the two walks is the dividend; A's eastward walk is the coefficient; B's southward walk is the corner divisor. Extract square root to obtain the radius. Method: Multiply A's eastward by B's southward as dividend; A's walk as coefficient, B's walk as corner; extract square root to obtain 120 paces, the radius; double for the diameter.

第四問： 或問：乙從坤隅南行三百六十步而止，甲出北門東行二百步見之，就乙斜行四百零八步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行冪得四萬步，以乙南行冪得一十二萬九千六百步，並之得一十七萬三千六百步為弦冪之實。又以斜行四百零八步自之，得一十六萬六千四百六十四步。以此減弦冪實，餘七千一百三十六步。以四之得二萬八千五百四十四步為城徑冪之較實。
別求之：立天元一為城徑，以減於二之甲東行，餘即乙出南門東行數也。以乙南行乘甲東行冪，又四之為實。從空，乙行為廉，一步常法，得城徑。開立方得二百四十步。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。術曰：置甲東行步自乘，又以乙南行步乘之，四之為實；乙南行步為廉法；一步為隅法；開立方即得城徑。

Problem 54: Suppose B walks south 360 paces from the Kun corner and stops, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 408 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Squaring A's eastward walk gives 40,000; squaring B's southward walk gives 129,600. Their sum, 173,600, is the dividend for the hypotenuse-squared. The diagonal 408 squared gives 166,464. Subtracting from the dividend leaves 7,136. Quadrupling gives 28,544 as the comparison-dividend for the diameter-squared.
Alternatively: Let the celestial element be the diameter. Subtracting from twice A's eastward walk gives B's eastward walk from the south gate. Multiply B's southward walk by A's walk-squared; quadruple as dividend. Empty coefficient; B's walk as corner; unity as constant. Extracting cube root gives 240 paces.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant divisor. Extract cube root for the diameter. Method: Square A's eastward walk, multiply by B's southward walk, quadruple as dividend; B's walk as corner divisor; unity as corner; extract cube root.

第五問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑，加乙南行一百三十五步，得$\frac{135}{1}$為股率。其甲東行二百步即勾率也。其乙南行為小股，以勾率乘之，合以大弦率除。今不受除，便以此為小勾，為母，寄股率。
乃先以小弦乘大和，得下式$\frac{0\quad 0\quad 12000}{1\quad 0\quad 0}$寄左。又以倍天元減斜步，得$\frac{100}{2}$為小和，以乘大弦，得$\frac{12000}{1}$為同數，與左相消。開平方得二百四十步。
按：此草以率法為主，以勾股之率通諸數，建立天元方程，為天元術之正統用法。立天元一為城徑，以諸率配之，求其等式，相消開方，即得所求。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 55: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Adding B's southward 135 paces gives $\frac{135}{1}$ as the leg-rate. A's eastward 200 paces is the base-rate. B's southward walk is the small leg; multiplying by the base-rate should be divided by the large hypotenuse-rate. Not dividing, this becomes the small base as numerator, with the leg-rate deposited.
First multiply the small hypotenuse by the large sum, obtaining the expression $\frac{0\quad 0\quad 12000}{1\quad 0\quad 0}$, deposited left. Then subtract twice the celestial element from the diagonal, obtaining $\frac{100}{2}$ as the small sum; multiply by the large hypotenuse, obtaining $\frac{12000}{1}$ as the equal number, which cancels with the left. Extracting square root gives 240 paces.
Note: This working uses rate-methods to connect all quantities, establishing tian yuan equations in the orthodox manner.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第六問：或問：乙從坤隅南行三百六十步，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為半徑。以甲東行二百步減斜行二百七十二步，餘七十二步，為乙出南門東行數。以乙南行三百六十步折半，得一百八十步，為大股之半。以此半股減天元半徑，得$\frac{180}{1}$為小差。以小差乘半股，得$\frac{32400}{180}$為半徑冪，寄左。以天元冪相消，開平方得一百二十步，倍之得全徑二百四十步。

法曰：兩行步相乘倍之為實，乙南行為從，一步常法，得半徑。術曰：以甲東行步乘乙南行步，倍之為實；乙南行步為從法；一步為隅法；開平方得半徑，倍之即城徑。

Problem 56: Suppose B walks south 360 paces from the Kun corner, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting A's 200 paces from the diagonal 272 leaves 72 paces as B's eastward walk from the south gate. Halving B's 360-pace southward walk gives 180 paces, half the large leg. Subtracting the radius from this half-leg gives $\frac{180}{1}$ as the small difference. Multiplying small difference by half the leg gives $\frac{32400}{180}$ as the radius-squared, deposited left. Canceling with the celestial element-squared and extracting square root gives 120 paces; doubling gives the full diameter of 240.

Method: The product of the two walks, doubled, is the dividend; B's southward walk is the coefficient; unity is the constant divisor; this yields the radius. Method: Multiply A's eastward by B's southward, double as dividend; B's walk as coefficient, unity as corner; extract square root for the radius; double for the diameter.

第七問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之，復就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步為小股，甲東行二百步為小勾。以斜行二百七十二步為小弦。驗之：$135^2+72^2=18225+5184=23409$，而 $272^2=73984$，非直股勾股也。當以天元術正之。
立天元一為城徑，以減於二之甲東行，得 $\frac{400}{1}$ 為兩个小差之和。以乙南行乘之，又四之，得大實。以天元冪與左相消，開平方得二百四十步。

法曰：以乙南行乘甲東行冪為實，從空，乙行為廉，一步常法，得城徑。

Problem 57: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. B's southward 135 paces is the small leg; A's eastward 200 is the small base; the diagonal 272 is the small hypotenuse. Verification: $135^2+72^2=18225+5184=23409$, while $272^2=73984$—not a direct right triangle. We must correct by the tian yuan method.
Let the celestial element be the diameter. Subtracting from twice A's eastward walk gives $\frac{400}{1}$ as the sum of the two small differences. Multiplying by B's southward walk and quadrupling gives the large dividend. Canceling with the celestial element-squared and extracting square root gives 240 paces.

Method: B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第八問：或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。以減於倍甲東行四百步，餘一百三十步，為大小差之較。以此較乘甲東行冪，得 $130\times 40000=$ 五百二十萬步為實。從空，倍乙行為廉，一步常法，開立方得城徑。
又法：立天元一為半徑，以半甲東行一百步減斜行半數一百三十六步，餘三十六步為半較。以乙南行半數六十七步半乘之，得半徑冪之較，與天元冪相消，開平方即得。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 58: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270 paces. Subtracting from twice A's eastward walk (400 paces) leaves 130 paces as the difference between the large and small differences. Multiplying this difference by A's walk-squared: $130\times 40000=$ 5,200,000 paces as dividend. Empty coefficient; twice B's walk as corner; unity as constant; extracting cube root gives the diameter.
Alternative method: Let the celestial element be the radius. Subtracting half A's walk (100) from half the diagonal (136) leaves 36 as half the difference. Multiplying by half B's walk (67.5) gives the radius-squared difference; cancel with the celestial element-squared; extract square root.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter.

第九問：或問：乙出南門直行一百三十五步，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：此問與前問類而異者，甲見乙之後復斜行相會。立天元一為城徑，以兩行相減，餘乘甲東行又四之為實。兩行差為廉，一步常法，開立方得城徑。
細草：以乙南行一百三十五步減甲東行二百步，餘六十五步，為兩行之差。以乘甲東行二百步，得一千三百步，又四之得五千二百步為實。兩行差六十五步為廉法，一步為隅法，開立方得二百四十步。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 59: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: This problem is similar to the previous but differs: after A sees B, he then walks diagonally to meet him. Let the celestial element be the city diameter. Subtract the two walks; multiply the remainder by A's eastward walk and quadruple as dividend. The difference of the two walks is the corner divisor; unity is the constant; extract cube root for the diameter.
Detailed working: Subtract B's 135 from A's 200, leaving 65 as the difference. Multiply by A's 200: 1,300; quadruple: 5,200 as dividend. The difference 65 is the corner divisor; unity is the constant corner. Extracting cube root gives 240 paces.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend. Empty coefficient; the walk-difference as corner; unity as constant; extract cube root for the diameter.

第十問：或問：乙出南門直行一百三十五步，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步，又四之得二千一百六十萬步為實。從空，乙行一百三十五步為廉法，一步為隅法，開立方得二百四十步。
按：此草以乙南行為小股，甲東行為小勾。以勾冪乘股，蓋以大勾之冪比於小勾之冪，若大股之冪比於小股之冪。以率言之，大勾即城徑與小勾之關係，故以天元建立而開立方。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 60: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Multiplying B's southward 135 paces by A's walk-squared (40,000) gives 5,400,000 paces; quadrupled gives 21,600,000 as dividend. Empty coefficient; B's 135 paces as corner divisor; unity as constant corner; extracting cube root gives 240 paces.
Note: This working takes B's southward as the small leg, A's eastward as the small base. Multiplying the base-squared by the leg: the ratio of the large base-squared to the small base-squared equals that of the large leg-squared to the small leg-squared. The large base is the relation between diameter and small base, hence establishing tian yuan and extracting cube root.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第十一問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑與斜行各幾何?

答曰：城徑二百四十步。斜行二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。以甲東行冪四萬步乘之，得一千零八萬步，又倍之得二千零一十六萬步為實。倍乙南行二百七十步為廉法，一步為隅法，開立方得城徑二百四十步。
既得城徑，求斜行：以城徑二百四十步減於二之甲東行四百步，餘一百六十步，為乙東行與半徑之差。以此差冪與半徑冪相加開方，得斜行二百七十二步。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。術曰：倍乙南行步，以乘甲東行冪為實；倍乙南行步為廉法；一步為隅法；開立方得城徑。

Problem 61: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What are the city diameter and the diagonal walk?

Answer: City diameter: 240 paces. Diagonal walk: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270 paces. Multiplying by A's walk-squared (40,000) gives 10,080,000 paces; doubled gives 20,160,000 as dividend. Twice B's southward (270) is the corner divisor; unity is the constant corner; extracting cube root gives diameter 240 paces.
To find the diagonal: Subtract diameter 240 from twice A's eastward walk (400), leaving 160 as the difference between B's eastward walk and the radius. Adding the square of this difference to the square of the radius and extracting root gives the diagonal of 272 paces.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter. Method: Double B's southward walk, multiply by A's walk-squared as dividend; twice B's walk as corner; unity as corner; extract cube root.

第十二問：或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以兩行相減，甲東行二百步減乙南行一百三十五步，餘六十五步，為兩行之較。以較乘甲東行，得一千三百步，又四之，得五千二百步為實。從空，兩行較六十五步為廉法，一步為隅法，開立方得城徑二百四十步。
驗草：城徑二百四十步，半徑一百二十步。以半徑減甲東行，餘八十步為甲至城心之東行。以乙南行一百三十五步減半徑，餘十五步，為乙至城心之南行。以此二數各自乘并之，$80^2+15^2=6400+225=6625$，開方得 81.39 不盡，須以天元正術求之。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 62: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces, then diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtract the two walks: A's 200 minus B's 135 leaves 65 as the difference. Multiply the difference by A's 200: 1,300; quadruple: 5,200 as dividend. Empty coefficient; the difference 65 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
Verification: Diameter 240, radius 120. Radius subtracted from A's walk leaves 80 as A's distance from the center eastward. B's 135 minus radius leaves 15 as B's distance southward from the center. Squaring and summing: $80^2 + 15^2 = 6400 + 225 = 6625$; root ≈ 81.39, inexact— must use the orthodox tian yuan method.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend. Empty coefficient; the walk-difference as corner; unity as constant; extract cube root for the diameter.

第十三問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑及乙東行步數各幾何？

答曰：城徑二百四十步，乙東行七十二步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步為實。從空，乙行為廉，一步常法，開平方得城徑。
求乙東行：以城徑二百四十步減於二之甲東行四百步，餘一百六十步，以減甲東行二百步，餘七十二步，即乙東行步數。
釋天元：此問以乙南行乘甲東行冪為實者，乙南行為股率，甲東行為勾率，以勾冪乘股率者，以大勾之冪與大股相乘之義也。

法曰：以乙南行乘甲東行冪為實，從空，乙行為廉，一步常法，得城徑。

Problem 63: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What are the city diameter and B's eastward paces?

Answer: City diameter: 240 paces. B's eastward walk: 72 paces.

Working: Let the celestial element unity be the city diameter. Multiplying B's southward 135 paces by A's walk-squared (40,000) gives 5,400,000 as dividend. Empty coefficient; B's walk as corner; unity as constant; extract square root for the diameter.
To find B's eastward walk: Subtract diameter 240 from twice A's eastward (400), leaving 160. Subtract this from A's 200 paces, leaving 72 paces as B's eastward walk.
Tian Yuan explained: Using B's southward (the leg-rate) multiplied by A's walk-squared (the base-squared): the base-squared multiplied by the leg-rate represents the meaning of the large base-squared multiplied by the large leg.

Method: B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract square root for the diameter.

第十四問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之，復斜行與乙相會。問城徑及斜行步數。

答曰：城徑二百四十步，斜行二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步，乘甲東行冪（四萬步），得一千零八十萬步為實。從空，倍乙南行為廉，一步為隅法，開平方得城徑二百四十步。
求斜行：以城徑減於二之甲東行，餘一百六十步。以此與甲東行相加，得三百六十步，折半得一百八十步，為弦之半。以自乘得三萬二千四百步，加入半徑冪一萬四千四百步，得四萬六千八百步，即非弦冪也。當以天元別求之：以半徑冪加甲乙東行之較冪，合半之，開方得斜行。$120^2+128^2=14400+16384=30784$，而 $272^2=73984$，知須以天元方程正求之。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 64: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him, then walks diagonally to meet B. What are the city diameter and diagonal walk?

Answer: City diameter: 240 paces. Diagonal walk: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270; multiplying by A's walk-squared (40,000) gives 10,800,000 as dividend. Empty coefficient; twice B's walk as corner; unity as constant; extracting square root gives diameter 240 paces.
To find the diagonal: Subtract diameter from twice A's eastward, leaving 160. Add to A's eastward: 360; halve: 180 as half the hypotenuse. Squaring gives 32,400; adding radius-squared 14,400 gives 46,800—not the hypotenuse-squared. Must use tian yuan separately: add radius-squared to the difference of A-B eastward walks squared, halve, extract root for diagonal. $120^2+128^2=14400+16384=30784$, while $272^2=73984$; hence must use tian yuan equations correctly.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract square root for the diameter.

第十五問： 或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以兩行相減，甲東行二百步減乙南行一百三十五步，餘六十五步為較。以較乘甲東行冪（四萬步），得二百六十萬步，又四之得一千零四十萬步為實。從空，較六十五步為廉法，一步為隅法，開立方得城徑二百四十步。
按：此問以較為主，蓋斜行相會，須以兩行差建立方程。較者，大小差之較也，為天元術中重要之率。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 65: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces, then diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtract the two walks: A's 200 minus B's 135 leaves 65 as the difference. Multiply the difference by A's walk-squared (40,000): 2,600,000; quadrupled: 10,400,000 as dividend. Empty coefficient; the difference 65 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
Note: This problem uses the difference as key. For diagonal meetings, one must establish equations from the walk-difference. The difference is the difference of large and small differences, an important rate in tian yuan.

Method: The walk-difference multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; walk-difference as corner; unity as constant; extract cube root for the diameter.

第十六問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步，又四之得二千一百六十萬步為實。從空，乙行一百三十五步為廉法，一步為隅法，開立方得城徑二百四十步。
細釋：甲東行為勾，乙南行為股。以勾冪乘股為實者，大勾之冪當與大股相乘，而乙南行即大股之表示，故以此為實。從空者，無從法也。乙行為廉，即二次項之係數也。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 66: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Multiply B's southward 135 paces by A's walk-squared (40,000): 5,400,000; quadrupled: 21,600,000 as dividend. Empty coefficient; B's 135 paces as corner divisor; unity as constant corner; extracting cube root gives diameter 240 paces.
Detailed explanation: A's eastward walk is the base, B's southward is the leg. Using base-squared multiplied by leg as dividend: the large base-squared ought to multiply the large leg, and B's southward represents the large leg, hence used as dividend. Empty coefficient means no linear coefficient. B's walk as corner is the quadratic coefficient.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第十七問： 或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑幾何？又問甲乙相望之斜距幾何？

答曰：城徑二百四十步。斜距二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。乘甲東行冪（四萬步），得一千零八十萬步，又倍之得二千一百六十萬步為實。從空，倍乙南行為廉法，一步為隅法，開立方得城徑二百四十步。
求斜距：既得城徑，則半徑一百二十步。甲東行二百步減半徑一百二十步，餘八十步為甲至城邊之距。以此冪六千四百，與乙南行一百三十五步減半徑餘十五步之冪二百二十五并之，得六千六百二十五，開平方得八十一點二二，加上半徑一百二十步，得二百零一點二二，非斜距也。當以天元正術：以甲東行減乙南行之差冪，加入甲乙各自去城邊之冪，并而開方，合加半徑，即得斜距二百七十二步。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 67: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter? Also, what is the diagonal distance between A and B?

Answer: City diameter: 240 paces. Diagonal distance: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270; multiplying by A's walk-squared (40,000) gives 10,800,000; doubled: 21,600,000 as dividend. Empty coefficient; twice B's walk as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
To find the diagonal distance: Diameter 240 gives radius 120. A's 200 minus radius 120 leaves 80 as A's distance to the wall. Its square 6,400 plus B's 135 minus radius 15 (square 225) gives 6,625; square root ≈ 81.22; plus radius 120 gives ≈ 201.22—not the diagonal. Must use orthodox tian yuan: the difference of A's and B's walks squared, plus each distance to the wall squared, summed and rooted, plus radius, gives diagonal 272.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter.

測圓海鏡卷四終 — 凡一十七問

End of Ceyuan Haijing Volume 4 — 17 Problems

2 卷五：大股一十八問・Vol. 5: Large Leg (18 Problems)

大股一十八問・Large Leg

（第六十八問至第八十五問・Problems 68–85）

【題意總說】卷五論「大股」之術。所謂大股，乃最大勾股形之股邊，即通股也。以大股為主，配以諸勾股形之關係，求圓城之徑。凡一十八問。大股者，勾股形中最長之股，即甲從乾隅直南至坤隅之步數，為六百步之率也。諸問或知乙南行之數，或知甲南行之數，或知斜望之數，皆以大股為基本建立天元方程。此卷方程多為三次、四次，須開立方、開三乘方以求之，為天元術之精深處。

[General Summary] Volume 5 treats the “large leg” (大股) method. The large leg is the leg-side of the largest right triangle—the universal leg. Taking the large leg as principal, combined with relations among sub-triangles, we seek the city diameter. All 18 problems. The large leg is A’s full southward walk from the Qian corner to the Kun corner—the 600-pace standard. Given B’s southward walk, or A’s southward walk, or diagonal sightings, we establish tian yuan equations. This volume features third- and fourth-degree equations requiring cube-root and fourth-root extraction—the subtle core of tian yuan.

第十八問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之減於甲南行六百步，得 $\frac{600}{2}$ 為大差也。以自之，得 $\frac{360000}{2400\quad 4}$ 為大差冪也。置甲南行冪 $\frac{360000}{1}$ 內減大差冪，而半之，得 $\frac{0}{1200\quad 2}$ 為大弦也，內帶大差為分母。

又置甲南行冪內減大差冪，而半之，得 $\frac{0}{600\quad 1}$ 為大勾也，帶大差分母。又以大差乘股六百步，得 $\frac{360000}{1}$ 併入和，得下式 $\frac{360000\quad 0\quad 0}{1\quad 1\quad 1200\quad 1}$ 為小和，以乘大弦，得 $\frac{0\quad 0}{0\quad 0\quad 1\quad 0\quad 1200\quad 1}$ $\frac{0\quad 0}{0\quad 0\quad 1\quad 0\quad 0\quad 1}$ 寄左。

又以倍天元減斜步，得 $\frac{100}{2}$ 為小和，以乘大弦，得同數，與左相消。開立方得一百二十步，即半徑也；倍之得全徑二百四十步。

釋義：此草以大差為主，大差者甲南行減圓徑也。甲從乾隅南行六百步，為大股之全數。減去圓徑，餘為大差。以大差冪與甲南行冪相減，半之得大弦，此即勾股求弦之法。又以大差乘股併入和，為小和以乘大弦，建立天元方程。其術精深，為全書之難問。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實。倍二行差減甲南行步即甲南行也。四之甲南行步內減於甲南行餘二百四十步，即城徑也。術曰：以甲南行步自之為冪，以乙南行步乘之為實；以甲南行步為從，乙南行步為廉；一步為隅法；開立方得城徑。

Problem 68: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned between them. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting twice it from A’s 600-pace southward walk gives $\frac{600}{2}$ as the large difference. Squaring: $\frac{360000}{2400\quad 4}$ as the large-difference-squared. Place A’s walk-squared $\frac{360000}{1}$, subtract large-diff-squared, halve: $\frac{0}{1200\quad 2}$ as the large hypotenuse, carrying large difference as denominator.

Also subtract and halve to obtain $\frac{0}{600\quad 1}$ as the large base, with large-diff denominator. Multiply large difference by the 600-pace leg: $\frac{360000}{1}$; combine into the sum, obtaining the expression $\frac{360000\quad 0\quad 0}{1\quad 1\quad 1200\quad 1}$ as the small sum. Multiply by the large hypotenuse, deposit left.

Also subtract twice the celestial element from the diagonal, obtaining $\frac{100}{2}$ as small sum; multiply by large hypotenuse for the equal number; cancel with the left. Extract cube root: 120 paces, the radius; double for the full diameter 240.

Explanation: This working uses the large difference as key—A’s walk minus the diameter. The 600-pace walk is the full large leg. Subtract the diameter for the large difference. The method is among the most profound in the entire work.

Method: Twice the walk-difference minus A’s southward walk, then multiplied by A’s walk, is the dividend. Method: Square A’s southward walk, multiply by B’s southward walk as dividend; A’s walk as coefficient, B’s walk as corner; unity as corner; extract cube root for the diameter.

第十九問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以減於甲南行六百步，得$\frac{600}{1}$為大差。置甲南行冪內減大差冪，而半之，得大弦也。又以大差乘股六百步，併入和，得下式為小和，以乘大弦，寄左。又以倍天元減斜步，得小和，以乘大弦，得同數，與左相消。開立方得半徑一百二十步，倍之得城徑。
細草：甲南行六百步自之，得三十六萬步為甲南行冪。大差$\frac{600}{1}$自之，得$\frac{360000}{1200\quad 1}$。以甲南行冪減之，得$\frac{0}{1200\quad 1}$，半之得$\frac{0}{600\quad \frac{1}{2}}$為大弦，帶大差分母。以大差乘股，得$\frac{360000}{1}$，併入得小和。以小和乘大弦，寄左。以倍天元減斜步得小和，乘大弦為同數，與左相消，開立方得一百二十步。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 69: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting from A's 600-pace southward walk gives $\frac{600}{1}$ as the large difference. Subtract large-diff-squared from A's walk-squared and halve to obtain the large hypotenuse. Multiply large difference by the 600-pace leg, combine into the sum, obtaining the small sum; multiply by large hypotenuse, deposit left. Subtract twice the celestial element from the diagonal for the small sum, multiply by large hypotenuse for the equal number; cancel with the left; extract cube root for radius 120; double for the diameter.
Detailed working: A's 600 squared: 360,000. Large diff squared: $\frac{360000}{1200\quad 1}$. Subtracting from A's walk-squared gives $\frac{0}{1200\quad 1}$; halved: $\frac{0}{600\quad \frac{1}{2}}$ as large hypotenuse with large-diff denominator. Multiply large diff by leg, $\frac{360000}{1}$, combine for small sum. Cancel and extract.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract cube root for the diameter.

第二十問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑，以二之減於甲南行六百步，餘$\frac{600}{2}$為大差。以自之得大差冪$\frac{360000}{2400\quad 4}$。置甲南行冪$\frac{360000}{1}$內減大差冪而半之，得大弦，內帶大差為分母。又置甲南行冪內減大差冪而半之，得大勾，帶大差分母。又以大差乘股六百步，併入和，得下式為小和，以乘大弦，寄左。又以倍天元減斜步，得小和，以乘大弦，得同數，與左相消。開立方得一百二十步，即半徑；倍之得城徑二百四十步。
按：此問與首問相類，惟立天元一為城徑而非半徑，故方程中諸數皆倍之，開立方後須倍之乃得城徑，學者宜辨之。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實。倍二行差減甲南行步即甲南行也。四之甲南行步內減於甲南行餘，即城徑也。

Problem 70: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtracting twice it from A's 600-pace walk leaves $\frac{600}{2}$ as the large difference. Squaring gives large-diff-squared $\frac{360000}{2400\quad 4}$. From A's walk-squared $\frac{360000}{1}$ subtract large-diff-squared and halve to obtain the large hypotenuse, carrying large-diff denominator. Also halve to obtain the large base with large-diff denominator. Multiply large diff by the 600-pace leg, combine for small sum, multiply by large hypotenuse, deposit left. Subtract twice the celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: 120 paces, the radius; double for diameter 240.
Note: Similar to the first problem, but the celestial element is the diameter, not the radius. Hence all quantities are doubled; after extracting cube root, double again for the diameter. Students should distinguish.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. The remainder is the city diameter.

第二十一問： 或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相減，得四百六十五步為大小股之差。以此差乘甲南行冪（三十六萬步），得一億六千七百四十萬步為實。從空，兩行差為廉，一步常法，開立方得城徑二百四十步。
釋義：此草以兩股差為主。大股甲南行六百步，小股乙南行一百三十五步。兩股之差四百六十五步，即大小差之較也。以此建立天元方程，較之以前數問更為簡捷，乃法之變通也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 71: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600-pace southward walk is the large leg; B's 135 is the small leg. Subtracting gives 465 paces as the leg-difference. Multiplying this difference by A's walk-squared (360,000) gives 167,400,000 as dividend. Empty coefficient; the walk-difference as corner; unity as constant; extracting cube root gives diameter 240 paces.
Explanation: This working uses the leg-difference as key. Large leg 600, small leg 135; difference 465 is the comparison of large and small differences. Establishing the tian yuan equation this way is simpler than previous methods—a flexible variation of the technique.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract cube root for the diameter.

第二十二問： 或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍二行差（甲南行六百步減乙南行一百三十五步，得四百六十五步，倍之得九百三十步），內減甲南行步，得三百三十步。復以乘甲南行步，得 $330 \times 600 = 198000$ 步為實。從空，倍二行差減甲南行步為廉，一步常法，開立方得城徑二百四十步。
細草：甲南行六百步，乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減六百步，餘三百三十步。以乘六百步，得一十九萬八千步為實。以三百三十步為廉法，一步為隅法，開立方得二百四十步，即城徑。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 72: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice the walk-difference (A's 600 minus B's 135 gives 465; doubled: 930), subtract A's southward walk: 330. Multiply by A's walk: $330 \times 600 = 198{,}000$ paces as dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extracting cube root gives diameter 240 paces.
Detailed working: A's 600, B's 135, difference 465. Doubled: 930; subtract 600: 330. Multiply by 600: 198,000 as dividend. 330 as corner divisor; unity as constant; extract cube root: 240 paces, the diameter.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十三問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘，得 $600 \times 135 = 81000$ 步為實。從空，兩行步相加得七百三十五步為廉，一步常法，開平方得城徑二百四十步。
按：此問開平方即得，不須開立方，較之他問為簡易。蓋以兩股相乘為實，兩股和為從，適合天元平方之式也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 73: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Multiply: $600 \times 135 = 81{,}000$ as dividend. Empty coefficient; sum of walks 735 as corner; unity as constant; extracting square root gives diameter 240 paces.
Note: This problem requires only square-root extraction, not cube root— simpler than others. The product of the two legs as dividend and their sum as coefficient perfectly fits the tian yuan quadratic pattern.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十四問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。以三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
驗草：城徑二百四十步，半徑一百二十步。甲南行六百步減圓徑二百四十步，餘三百六十步為大差。以大差與半徑各為股勾，$360^2 + 120^2 = 129600 + 14400 = 144000$，開方得 379.47 不盡。當以三乘方正之，其術繁矣。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 74: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135 gives 465 as the difference. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Verification: Diameter 240, radius 120. A's 600 minus diameter 240 leaves 360 as the large difference. Using large difference and radius as leg and base: $360^2 + 120^2 = 129{,}600 + 14{,}400 = 144{,}000$; root ≈ 379.47, inexact. Must use fourth-root correction—complex.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十五問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘，得八萬一千步為實。以兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。
按：此問以兩股相乘為實，蓋大股與小股相乘之積，當等於以城徑為高之矩形面積。以兩股和為從者，大小股之和與城徑之關係式也。開平方即得，術最簡明。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 75: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Multiply: 81,000 as dividend. The leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Note: This problem uses the product of the two legs as dividend. The product equals the rectangular area with the diameter as one dimension. Using the leg-sum as coefficient expresses the relation between diameter and leg sum. Square-root extraction gives the result directly—the clearest method.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十六問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步減乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減甲南行步，餘三百三十步。以乘甲南行冪（三十六萬步），得一億一千八百八十萬步為實。以三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
細釋：此問以差為主，倍差內減大股餘三百三十步，此數即大差之半與小差之關係數也。以此數為廉法，乘大股冪為實，開立方而得城徑，天元之妙用也。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 76: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's walk-squared (360,000): 118,800,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Detailed explanation: Using the difference as key: twice the difference minus the large leg leaves 330—the relation-number between half the large difference and the small difference. Using this as corner, multiplied by large-leg-squared as dividend, and extracting cube root—the marvelous efficacy of tian yuan.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十七問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。
又草：立天元一為半徑。以甲南行減圓徑，餘為大差。以大差自之為大差冪，以甲南行冪減之半之為大弦。又以大差乘股併入和為小和，以小和乘大弦，寄左。以倍天元減斜步為小和，乘大弦為同數，與左相消，開立方得半徑一百二十步。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 77: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Alternative working: Let the celestial element be the radius. Subtract diameter from A's southward walk for the large difference. Square for large-diff-squared; subtract from A's walk-squared and halve for the large hypotenuse. Multiply large diff by leg, combine for small sum; multiply by large hypotenuse, deposit left. Subtract twice celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: radius 120 paces.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十八問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍二行差，甲南行六百步減乙南行一百三十五步，差四百六十五步，倍之九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
天元釋義：此草所云天元一者，即代數之未知數 x 也。以大股六百為甲南行，小股一百三十五為乙南行。兩行之差為大差之表示。倍差內減大股，乃大差與圓徑之關係。以此關鍵數建立方程，開立方即得。此天元術之所以勝於幾何也。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 78: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice the walk-difference: A's 600 minus B's 135 gives 465; doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Tian Yuan explained: The "celestial element unity" is the algebraic unknown x. Large leg 600 is A's walk; small leg 135 is B's walk. The difference represents the large difference. Twice the difference minus the large leg is the relation between large difference and diameter. Using this key number to establish the equation and extracting cube root—this is why tian yuan surpasses pure geometry.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十九問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。
釋率：大股為通股，即甲從乾隅直南至坤隅之全步。小股為乙從南門直南至某處之步。兩股相乘之實，即以大股為長、小股為寬之長方形面積。以此面積為天元方程之實，而以兩股和為從法，開平方即得城徑，其理精妙。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 79: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large (universal) leg—A's full southward walk from Qian to Kun. B's 135 is the small leg—B's walk from the south gate. Their product, 81,000, is the area of a rectangle with large leg as length and small leg as width. Using this area as the tian yuan dividend and the leg-sum as coefficient, extracting square root gives the diameter—subtle and exquisite reasoning.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為兩行差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉法，一步為隅法，開立方得城徑二百四十步。
術解：倍差內減大股餘三百三十步，此數即大差之半與小差之關係數也。以此數為廉法，乘大股為實，開立方而得城徑。天元術中以三次方程為主，此問即典型的天元三次方程之應用。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 80: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: walk-difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.
Method explained: Twice the difference minus the large leg leaves 330— the relation-number between half the large difference and the small difference. Using this as corner, multiplying by the large leg as dividend, extracting cube root for the diameter. This is the classic application of tian yuan cubic equations.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十一問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從法，一步為隅法，開平方得城徑二百四十步。
細草：兩股相乘 $135\times600=81000$ 為實。兩股和 $135+600=735$ 為從。設城徑為 x，則有方程：$x^2+735x=81000$。以天元術開平方：置實八萬一千，從七百三十五，隅一步。以隅法一約實，初商二百四十。以初商乘從法，得 $240\times735=176400$。以減實不盡，知初商即為城徑之數，蓋實較從乘商之數為小也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 81: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Detailed working: Product $135\times600=81000$ as dividend. Sum $135+600=735$ as coefficient. Let diameter be x; the equation is $x^2+735x=81000$. By tian yuan square-root extraction: place dividend 81,000, coefficient 735, corner unity. Divide: initial quotient 240. Multiply quotient by coefficient: $240\times735=176{,}400$. Subtracting exceeds the dividend, so the initial quotient 240 is the diameter—the dividend is smaller than the coefficient times the quotient.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十二問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
三次方程釋：此問所得為天元三次方程。設城徑為 x，則方程形如 $x^3+330x^2=198000$。以天元術開立方：置實一十九萬八千，廉三百三十，隅一步。以隅約實，初商二百。以初商自乘乘隅，得四萬；又以初商乘廉，得六萬六千；并之得十萬六千，以減實餘九萬二千。再以初商二百乘隅，得二百，加入廉三百三十，得五百三十為下廉。以次商四十乘下廉，得二萬一千二百；又以次商四十自乘得一千六百，乘隅得一千六百；并之得二萬二千八百，以減餘實九萬二千不盡。復以次商加入下廉，再以三商乘之，開盡得二百四十步為城徑。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 82: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Cubic equation explained: This yields a tian yuan cubic. Let diameter be x; the equation has form $x^3+330x^2=198{,}000$. By tian yuan cube-root extraction: place dividend 198,000, corner 330, corner divisor unity. Divide: initial quotient 200. Square 200, multiply by unity: 40,000; multiply 200 by 330: 66,000; sum 106,000; subtract from dividend: remainder 92,000. Multiply 200 by unity: 200; add to 330: 530 as lower corner. Next quotient 40 times 530: 21,200; plus 40 squared (1,600) times unity; total 22,800; subtract from remainder. Continue until exhausted: 240 paces.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十三問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。按：此草簡明直捷，為大股問中之平易者。以兩股之積為實，兩股之和為從，一步為隅，恰合天元平方方程之式。開平方即得，不須繁複之運算，乃初學天元者之津梁也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 83: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Note: This working is direct and concise—the simplest among large-leg problems. Product as dividend, sum as coefficient, unity as corner: exactly the tian yuan quadratic pattern. Square-root extraction suffices; no complex computation needed. The gateway for beginners in tian yuan.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十四問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步減乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。
天元源流：天元術者，中國古代代數學之最高成就也。始見於《九章算術》之開方術，經唐宋諸賢之發展，至李冶先生而集大成。立天元一為所求之數，以各種條件建立方程，相消開方而得答數。此問以三次方程求城徑，正天元術之精妙所在。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 84: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.
Origins of Tian Yuan: The tian yuan method is the supreme achievement of ancient Chinese algebra. Originating in the root-extraction methods of *Jiuzhang Suanshu*, developed through the Tang and Song dynasties, and brought to full flower by Master Li Ye. Letting the celestial element be the sought quantity, establishing equations from given conditions, canceling and extracting roots for the answer. This problem uses a cubic equation to find the diameter—the very essence of tian yuan subtlety.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十五問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。又法：立天元一為半徑。甲南行六百步減二之天元（圓徑），得$\frac{600}{2}$為大差。自之得大差冪，以甲南行冪減之半之為大弦。以大差乘甲南行併入和為小和，乘大弦寄左。以倍天元減斜步為小和，乘大弦為同數，與左相消。開立方得半徑一百二十步，倍之得城徑二百四十步。此以半徑為天元之法也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 85: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.
Alternative: Let the celestial element be the radius. A's 600 minus twice the celestial element (the diameter) gives $\frac{600}{2}$ as the large difference. Square for large-diff-squared; subtract from A's walk-squared and halve for the large hypotenuse. Multiply large diff by A's walk, combine for small sum; multiply by large hypotenuse, deposit left. Subtract twice the celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: radius 120; double for diameter 240.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

測圓海鏡卷五終 — 凡一十八問

End of Ceyuan Haijing Volume 5 — 18 Problems

3 卷六：大勾一十八問・Vol. 6: Large Base (18 Problems)

大勾一十八問・Large Base

(第八十六問至第一百三問・Problems 86–103)

【題意總說】卷六論「大勾」之術。所謂大勾，乃最大勾股形之勾邊，即通勾也。以大勾為主，配以諸勾股形之關係，求圓城之徑。凡一十八問。大勾者，甲從乾隅直東至艮隅之步數，為三百二十步之率也。與大股相對而言，大股為南北向，大勾為東西向。此卷之問多設乙從東門直行若干步，甲從乾隅東行若干步望乙之狀，以建立天元方程。方程之次數較高，有三次、四次之方程，須開立方、開三乘方求之，為全書最精深之卷。

[General Summary] Volume 6 treats the “large base” (大勾) method. The large base is the base-side of the largest right triangle—the universal base. Taking the large base as principal, combined with sub-triangle relations, we seek the city diameter. All 18 problems. The large base is A’s eastward walk from the Qian corner to the Gen corner—the 320-pace standard. Contrasted with the large leg (north-south), the large base is east-west. This volume mostly posits B walking east from the east gate and A walking east from the Qian corner to view B, establishing tian yuan equations. Equations here reach higher degrees—third and fourth—requiring cube-root and fourth-root extraction. The most profound volume in the entire work.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之加乙東行一十六步，得$\frac{32}{2}$為小差。半甲東行三百二十步，得一百六十步，為大勾之半。以乘小差，得$\frac{5120}{320}$為半徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，得$\frac{5120}{320\quad 1\quad 0}$（上負下正），開平方得一百二十步，即半徑；倍之得全徑二百四十步。

釋義：此草以小差為主。小差者，半徑之倍加以乙東行之數也。甲東行三百二十步為大勾，其半即大勾之半。以小差乘大勾之半，得半徑冪，此天元術之巧妙處。與天元冪相消，開平方即得半徑。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之東行內減二之乙東行為從，四益隅，得半徑。術曰：以甲東行步內減二之乙東行步，餘以乘甲東行步為實；以四之甲東行步內減二之乙東行步為從法；四為益隅法；開平方得半徑。

Problem 86: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned between them. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Twice it plus B’s 16 paces gives $\frac{32}{2}$ as the small difference. Half of A’s 320-pace eastward walk is 160 paces, half the large base. Multiplying by the small difference gives $\frac{5120}{320}$ as the radius-squared, deposited left. Canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{5120}{320\quad 1\quad 0}$ (negative above, positive below). Extracting square root: 120 paces, the radius; doubling gives the full diameter of 240 paces.

Explanation: This working uses the small difference as key. The small difference is twice the radius plus B’s eastward walk. A’s 320 is the large base; its half is half the large base. Multiplying small difference by half the large base gives the radius-squared—the ingenious subtlety of tian yuan.

Method: A’s eastward walk minus twice B’s eastward walk, then multiplied by A’s walk, is the dividend. Four times A’s walk minus twice B’s walk is the coefficient; four is the negative corner; this yields the radius. Method: From A’s eastward walk subtract twice B’s eastward walk; multiply remainder by A’s walk as dividend; four times A’s walk minus twice B’s walk as coefficient; four as negative corner; extract square root for the radius.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之加乙東行一十六步，得$\frac{16}{2}$為小差。半甲東行步，得一百六十步，以乘小差，得$\frac{2560}{320}$為半徑冪，寄左。然後以天元冪$\frac{0}{1}$與左相消，開平方得一百二十步，即半徑。

按：此問與首問相類而異者，小差之立不同也。前問以二之天元加乙東行三十二步為小差，此問以二之天元加乙東行十六步為小差。蓋題設乙東行步數有異，故方程之係數隨之而變，天元術之靈活處也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之甲東行內減二之乙東行為從，四益隅，得半徑。

Problem 87: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Twice it plus B's 16 paces gives $\frac{16}{2}$ as the small difference. Half A's eastward walk is 160; multiplying by the small difference gives $\frac{2560}{320}$ as the radius-squared, deposited left. Canceling with the celestial element-squared and extracting square root gives 120 paces, the radius.

Note: Similar to the first problem but the small difference is defined differently. The previous problem used 32 (twice B's walk) added to twice the celestial element; this uses 16. The problem conditions differ, so equation coefficients change accordingly—the flexibility of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Four times A's walk minus twice B's walk is the coefficient; four as negative corner; this yields the radius.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑，以二之加乙東行一十六步，得$\frac{16}{2}$為小差。置甲東行冪（一十萬二千四百步）內減小差冪，而半之，得大弦，帶小差分母。又置甲東行冪內減小差冪，而半之，得大股，帶小差分母。又以小差乘大股，併入和，得下式為小和，以乘大弦，寄左。又以倍天元加斜步，得小和，以乘大弦，得同數，與左相消。開立方得二百四十步，即城徑。

釋天元：此問立天元一為城徑而非半徑，故方程為三次，須開立方。小差以城徑之半加乙東行，與以半徑為天元者不同。學者須細察題意，決定所立天元之為半徑或全徑，乃能建立正確方程。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之甲東行內減二之乙東行為從，四益隅，得城徑。

Problem 88: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice it plus B's 16 gives $\frac{16}{2}$ as the small difference. Place A's walk-squared (102,400); subtract small-diff-squared and halve for the large hypotenuse, carrying small-diff denominator. Also halve for the large leg with small-diff denominator. Multiply small difference by large leg, combine into the sum, obtain the small sum; multiply by large hypotenuse, deposit left. Add twice the celestial element to the diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: 240 paces, the diameter.

Tian Yuan explained: Here the celestial element is the diameter, not the radius, so the equation is cubic, requiring cube-root extraction. The small difference uses half the diameter plus B's walk, different from using the radius. Students must examine the problem carefully to decide whether to use radius or diameter.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Four times A's walk minus twice B's walk is the coefficient; four as negative corner; this yields the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，斜行與乙相會。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以乘甲東行三百二十步，得九萬二千一百六十步為實。從空，四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅為法，開三乘方得城徑二百四十步。
按：此問有斜行相會之狀，故方程為四次，須開三乘方。以甲東行內減二之乙東行，為建立方程之關鍵。乘甲東行為實，以四之甲東行內減二之乙東行為廉，四益隅為三乘方之隅法，開之得城徑。此全書中方程次數最高之問也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 89: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, then walks diagonally to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. Empty coefficient; four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner for fourth-root extraction. Extracting fourth root gives diameter 240 paces.
Note: This problem involves diagonal meeting, so the equation is fourth-degree, requiring fourth-root extraction. The key step: A's walk minus twice B's walk. This multiplied by A's walk is the dividend; four times A's walk minus twice B's walk is the corner; four as negative corner. This problem has the highest-degree equation in the entire work.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步，為大差之表示數。以此數乘甲東行冪（一十萬二千四百步），得二千九百四十九萬一千二百步為實。從空，四之甲東行內減二之乙東行為廉，四益隅，開三乘方得城徑。
細草：甲東行三百二十步自之，得一十萬二千四百步為甲東行冪。乙東行一十六步，二之為三十二步。以減甲東行，餘二百八十八步。以乘甲東行冪：$288 \times 102400 = 29491200$ 為實。四之甲東行一千二百八十步，減二之乙東行三十二步，餘一千二百四十八步為廉法。四為益隅法。開三乘方得二百四十步。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 90: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288 as the large-difference representative. Multiplying by A's walk-squared (102,400) gives 29,491,200 as dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extracting fourth root gives the diameter.
Detailed working: A's 320 squared: 102,400 as A's walk-squared. B's 16, doubled: 32. Subtract from A's 320: 288. Multiply by walk-squared: $288 \times 102{,}400 = 29{,}491{,}200$ as dividend. Four times A's walk: 1,280; minus twice B's 32: 1,248 as corner divisor. Four as negative corner. Extract fourth root: 240 paces.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此數乘甲東行三百二十步，得九萬二千一百六十步為實。從空，四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
釋三乘方：三乘方即四次方程也。設城徑為 x，則方程形如 $x^4 + 1248x^2 = 9216000$ 之類。以天元術開三乘方：置實，以隅法約之，初商二百。以次商乘廉法，又以次商自乘乘隅法，逐步減實，至實盡為止。其術繁複，非老於天元者不能為之。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 91: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. Empty coefficient; four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Fourth root explained: The "triple-multiplication root" is the fourth-degree equation. Let diameter be x; the equation has form $x^4 + 1248x^2 = 9{,}216{,}000$. By tian yuan fourth-root extraction: place dividend; divide by corner; initial quotient 200. Multiply next quotient by corner divisor, plus next quotient squared times corner, gradually subtracting from dividend until exhausted. The method is complex; only the experienced can perform it.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步為大勾，乙東行一十六步為小勾。兩勾相減，餘三百零四步為大小勾之差。以此差乘甲東行三百二十步，得九萬七千二百八十步為實。從空，四之甲東行內減二之乙東行為廉，四益隅，開三乘方得城徑二百四十步。
按：此草以兩勾差為主，與前數問以甲東行內減二之乙東行者不同。蓋題意有微妙之別，或以兩勾差、或以大勾減倍小勾為關鍵數，皆可建立方程而得同一城徑，天元術之奧妙也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 92: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 is the large base; B's 16 is the small base. Subtracting: 304 as the base-difference. Multiplying by A's 320: 97,280 as dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Note: This working uses the base-difference as key, different from previous problems using A's walk minus twice B's walk. Subtle distinctions in problem conditions allow either the base-difference or large-base-minus-twice- small-base as the key number, both yielding the same diameter—the profundity of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
源流考：天元術之開三乘方，源於《九章算術》之開方術。《九章》有開平方、開立方，而無開三乘方。至李冶先生著《測圓海鏡》，始廣之為開三乘方、開四乘方，以應天元四次方程之需。此問即開三乘方之典型例題，學者熟習此術，則高次方程皆迎刃而解矣。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 93: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Historical origins: Fourth-root extraction in tian yuan derives from the root-extraction methods of *Jiuzhang Suanshu*. That classic had square-root and cube-root extraction, but not fourth-root. Master Li Ye's *Ceyuan Haijing* extended it to fourth-root and fifth-root extraction to meet the needs of fourth-degree equations. This problem is the canonical example; mastering it, all higher-degree equations become soluble.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步，為大小差之較。以此乘甲東行三百二十步，得九萬二千一百六十步為實。四之甲東行一千二百八十步減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑。
又法：立天元一為半徑。以甲東行三百二十步減天元，得$\frac{320}{1}$為大差。以乙東行一十六步加天元，得$\frac{16}{1}$為小差。以大差小差相乘，得$\frac{5120}{336\quad 1}$為半徑冪與相關數之和，寄左。以天元冪$\frac{0}{1}$與左相消，得$\frac{5120}{336\quad 1\quad 0}$，開平方得一百二十步為半徑。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 94: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288 as the difference between large and small differences. Multiply by A's 320: 92,160 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives the diameter.
Alternative: Let the celestial element be the radius. A's 320 minus the celestial element gives $\frac{320}{1}$ as the large difference. B's 16 plus the celestial element gives $\frac{16}{1}$ as the small difference. Multiply: $\frac{5120}{336\quad 1}$ as the sum of radius- squared and related terms, deposited left. Cancel with celestial element- squared $\frac{0}{1}$, obtaining $\frac{5120}{336\quad 1\quad 0}$. Extract square root: 120 paces, the radius.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。四之甲東行一千二百八十步減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
按：此卷之問多須開三乘方，較之卷四、卷五為難。蓋大勾之術以小差為主，小差含乙東行與半徑二項，故方程之次數自然較高。學者須先熟習開平方、開立方，然後進而開三乘方，循序漸進，乃能領會天元術之全體大用。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 95: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Note: Most problems in this volume require fourth-root extraction, harder than Volumes 4 and 5. The large-base method uses the small difference as key, which contains both B's walk and the radius, so the equation degree is naturally higher. Students must master square-root and cube-root extraction before advancing to fourth-root, progressing step by step to comprehend the full power of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲東行三百二十步為大勾，乙東行一十六步為小勾。以大勾之半一百六十步減乙東行一十六步，餘一百四十四步，為小差之半。倍之得二百八十八步，為小差之全。以此乘大勾三百二十步，得九萬二千一百六十步為實。以一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
釋小差：小差者，圓城之半徑與乙東行之關係數也。以半徑之倍（即全徑）加乙東行為小差之全，或以半徑加乙東行之半為小差之半，二者皆可建立方程，而結果相同，此天元術之恆等變換也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 96: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 is the large base; B's 16 is the small base. Half the large base (160) minus B's 16 leaves 144, half the small difference. Doubled: 288, the full small difference. Multiply by the large base 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Small difference explained: The small difference relates the radius to B's eastward walk. Twice the radius (the diameter) plus B's walk is the full small difference; or half the radius plus half B's walk is half the small difference. Either can establish the equation with the same result—an identity transformation in tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑。
術論：開三乘方之術，猶開平方、開立方之推廣也。置實於下，廉法於中，隅法於上。初商得後，以商乘廉、商自乘乘隅，各以減實。然後以商加廉為下廉，復以次商乘之，又以次商自乘乘隅減實，如是輾轉求之，至實盡為止。其理與開平方、開立方一脈相承。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 97: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives the diameter.
On fourth-root extraction: It is the natural extension of square-root and cube-root extraction. Place dividend below, corner divisor in the middle, corner above. After obtaining the initial quotient, multiply quotient by corner and quotient-squared by corner, each subtracted from the dividend. Then add quotient to corner for lower corner; multiply by next quotient; add next quotient squared times corner; subtract from dividend. Continue thus until the dividend is exhausted. The principle is continuous with square-root and cube-root extraction.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
驗草：既得城徑二百四十步，則半徑一百二十步。甲東行三百二十步減半徑一百二十步，餘二百步為甲至城邊之距。乙東行一十六步加半徑一百二十步，得一百三十六步為乙至城心之距。以此二數各自乘：甲距冪 $= 200^2 = 40000$，乙距冪 $= 136^2 = 18496$。并之得五萬八千四百九十六步，開平方得二百四十一點八五不盡，非斜距也。當以天元術別求之。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 98: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Verification: Diameter 240 gives radius 120. A's 320 minus radius 120 leaves 200 as A's distance to the wall. B's 16 plus radius 120 gives 136 as B's distance to the center. Squaring: $200^2 = 40{,}000$; $136^2 = 18{,}496$. Sum: 58,496; square root ≈ 241.85, inexact—not the diagonal distance. Must use tian yuan separately.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何?

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。以一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
又草：立天元一為半徑。以甲東行三百二十步減二之天元（城徑），得$\frac{320}{2}$為大差。自之得大差冪$\frac{102400}{1280\quad 4}$。以甲東行冪減之，半之得大弦。又以大差乘大勾併入和，得小和乘大弦，寄左。以倍天元加斜步為小和，乘大弦為同數，與左相消，開立方得半徑。
論天元之設：或問何以有立城徑為天元、有立半徑為天元?曰：此隨題之便而設也。題中條件多與城徑直接相關者，立城徑為天元；條件多與半徑相關者，立半徑為天元。所立雖異，所得則同，此天元術之靈活性也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 99: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
Alternative: Let the celestial element be the radius. A's 320 minus twice the celestial element (diameter) gives $\frac{320}{2}$ as large difference. Squaring: $\frac{102400}{1280\quad 4}$ as large-diff-squared. Subtract from A's walk-squared and halve for large hypotenuse. Multiply large diff by large base, combine for small sum; multiply by large hypotenuse, deposit left. Add twice celestial element to diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root for radius.
On choosing the celestial element: Why use diameter for some problems and radius for others? It depends on convenience. If conditions relate to the diameter directly, use diameter; if to the radius, use radius. Different choices yield the same result—the flexibility of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？又問甲乙相望之斜距幾何？

答曰：城徑二百四十步，斜距二百七十二步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
求斜距：既得城徑二百四十步，半徑一百二十步。甲東行三百二十步減半徑一百二十步，餘二百步為甲至城邊之距。乙東行一十六步加半徑一百二十步，得一百三十六步為乙至城心之距。以此二數各自乘：甲距冪 $= 200^2 = 40000$，乙距冪 $= 136^2 = 18496$。并之得五萬八千四百九十六步，開平方得二百四十一點八五不盡，非斜距也。當以天元術別求之：以大差二百步乘小差一百三十六步，得二萬七千二百步，加入半徑冪一萬四千四百步，得四萬一千六百步，開方得二百零四步，加上斜較六十八步，即得斜距二百七十二步。
按：求斜距之術較繁，須先求得城徑，然後以大小差之關係求斜行步數。全書各問求城徑為主，斜距之求乃其餘事也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 100: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter? Also, what is the diagonal distance between A and B?

Answer: City diameter: 240 paces. Diagonal distance: 272 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
To find the diagonal: Diameter 240 gives radius 120. A's 320 minus radius 120 leaves 200 as A's distance to the wall. B's 16 plus radius 120 gives 136 as B's distance to the center. Squaring: $200^2 = 40{,}000$; $136^2 = 18{,}496$. Sum: 58,496; root ≈ 241.85, inexact—not the diagonal. Must use tian yuan separately: large difference 200 times small difference 136 gives 27,200; adding radius-squared 14,400 gives 41,600; root ≈ 204; plus diagonal-difference 68 gives diagonal 272.
Note: Finding the diagonal is more complex; one must first obtain the diameter, then use large-small difference relations. The entire work focuses on finding the diameter; diagonal distance is secondary.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
四乘方程論：此問所得天元方程為四次方程，中國古代謂之「三乘方」。西人謂之「四次方程」，以 x^4 為最高次項。李冶先生之天元術，以籌算排列係數，自下而上為常數項、一次項、二次項、三次項、四次項，與現代代數學之降冪排列相反，然其理則一也。開三乘方之法，亦與西洋之數值解法相通，可見數學為天下之公器，無中外之殊也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 101: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
On quartic equations: This yields a tian yuan fourth-degree equation, called "triple-multiplication root" in Chinese. Westerners call it a quartic equation, with x^4 as the highest term. Master Li Ye's tian yuan method arranges coefficients in counting-rod style, bottom-to-top: constant, linear, quadratic, cubic, quartic—the reverse of modern algebra's descending powers, yet the principle is identical. Fourth-root extraction corresponds to Western numerical methods, showing mathematics is a universal tool without national boundaries.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？又問大差小差各幾何？

答曰：城徑二百四十步。大差二百步，小差一百三十六步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
求大差：甲東行三百二十步減城徑二百四十步，餘八十步，非大差也。當以甲東行減半徑：$320 - 120 = 200$ 步為大差之正數。
求小差：乙東行一十六步加半徑一百二十步，得一百三十六步為小差。
驗大差小差：大差二百步，小差一百三十六步。以相乘得 $200 \times 136 = 27200$。大差冪 $= 40000$，小差冪 $= 18496$。以大小差冪之和減甲東行冪：$102400-(40000+18496) = 43904$，半之得 21952。此數當與大弦之某式相合，以驗天元方程之正確性。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 102: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter? Also, what are the large difference and small difference?

Answer: City diameter: 240 paces. Large difference: 200 paces. Small difference: 136 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
To find the large difference: A's 320 minus diameter 240 leaves 80—not the large difference. Correctly: A's walk minus radius: $320 - 120 = 200$ paces as the true large difference.
To find the small difference: B's 16 plus radius 120 gives 136 paces.
Verification: Large difference 200, small difference 136. Product: $200 \times 136 = 27{,}200$. Large-diff-squared: 40,000; small-diff-squared: 18,496. Sum subtracted from A's walk-squared: $102{,}400-(40{,}000+18{,}496) = 43{,}904$; halved: 21,952. This should match a formula for the large hypotenuse, verifying the tian yuan equation.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問： 或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？又問通勾股形之弦長幾何？

答曰：城徑二百四十步。通弦三百九十二步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。
求通弦：通勾股形者，以甲東行為勾、甲南行為股之大勾股形也。甲東行三百二十步為勾，甲南行六百步為股。以勾股求弦：$\sqrt{320^2+600^2}=\sqrt{102400+360000}=\sqrt{462400}=680$ 步。然此乃大弦之全數也。其帶分母者，當以大差除之。大差為甲東行減半徑，即 $320-120=200$ 步。以 680 步為分子，200 步為分母，得 3.4 步，非通弦也。當以天元正術求之：以大差冪加小差冪，減甲東行冪，半之，加入大小差相乘之積，開方得通弦三百九十二步。
按：通弦之求甚繁，須先求得大小差，然後以大小差之關係建立方程求之。此問為全卷之終，特設通弦之問，以見天元術之無所不能。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。術曰：置甲東行步內減二之乙東行步，餘以乘甲東行冪為實；以四之甲東行步內減二之乙東行步為廉法；四為益隅法；開三乘方得城徑。

Problem 103: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter? Also, what is the hypotenuse of the universal right triangle?

Answer: City diameter: 240 paces. Universal hypotenuse: 392 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.
To find the universal hypotenuse: The universal right triangle has A's eastward walk as base and A's southward walk as leg. Base 320, leg 600. By the gougu theorem: $\sqrt{320^2+600^2}=\sqrt{102{,}400+360{,}000}=\sqrt{462{,}400}=680$ paces. But this is the full large hypotenuse. The version with denominator: divide by large difference. Large difference = A's walk minus radius: $320-120=200$ paces. $680/200=3.4$ paces—not the universal hypotenuse. Must use orthodox tian yuan: large-diff-squared plus small-diff-squared, minus A's walk-squared, halved, plus product of large and small differences; extract root: 392 paces.
Note: Finding the universal hypotenuse is complex; one must first obtain the large and small differences, then establish equations. This final problem deliberately asks for the hypotenuse to demonstrate that tian yuan can solve anything.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter. Method: From A's eastward walk subtract twice B's eastward walk; multiply remainder by A's walk-squared as dividend; four times A's walk minus twice B's walk as corner divisor; four as negative corner; extract fourth root.

測圓海鏡卷六終 — 凡一十八問

End of Ceyuan Haijing Volume 6 — 18 Problems

附錄：全書結構及術語解釋・Appendix: Structure and Glossary

《測圓海鏡》全書十二卷，共一百七十問，為元代李冶先生之傑作。其書以勾股測圓為主，設圓城一座，甲乙二人從城門出行，以行步之數推求圓徑。全書結構如下：

卷	篇名	問數	起止問
卷一	圓城圖式	—	圖說
卷二	正率	14	第 1–14 問
卷三	邊股	17	第 15–31 問
卷四	底勾	17	第 32–48 問
卷五	大股	18	第 49–66 問
卷六	大勾	18	第 67–84 問
卷七	明勾	16	第 85–100 問
卷八	明股	16	第 101–126 問
卷九	雜題上	18	第 127–144 問
卷十	雜題中	18	第 145–162 問
卷十一	雜題下	18	第 163–180 問
卷十二	雜題終	10	第 181–190 問
	合計	**170**	

Ceyuan Haijing (Sea Mirror of Circle Measurements), in twelve volumes and 170 problems, is the masterpiece of Master Li Ye of the Yuan Dynasty. The work uses right-triangle circle measurement, positing a circular city from whose walkers' paces the diameter is derived. The complete structure:

Vol.	Title	Probs.	Range
1	City Diagram	—	Diagram
2	Standard Rates	14	1–14
3	Side Leg	17	15–31
4	Base Leg	17	32–48
5	Large Leg	18	49–66
6	Large Base	18	67–84
7	Bright Base	16	85–100
8	Bright Leg	16	101–126
9	Misc. Part I	18	127–144
10	Misc. Part II	18	145–162
11	Misc. Part III	18	163–180
12	Misc. Finale	10	181–190
	Total	**170**	

術語解釋・**Glossary of Terms:**

- 天元一：設未知數為一，即現代代數之 x。李冶先生以「立天元一」為某數，即設未知數為所求之量。
- 寄左：將某式暫存一邊，相當於方程之一邊。天元術中常以「寄左」表示方程之左邊，後以「同數」與之相消。
- 相消：兩式相減，相當於移項合併。天元術之核心步驟，以同數與寄左之式相減，得開方式。
- 帶分母：某式含有分母未除。古代算術中常「不受除」，即以帶分母之形式保留，以待後續運算。
- 冪：平方，即現代記號 x^2。天元術中以某數自乘為其冪。
- 開方：開平方，即求平方根。天元術中最基本之運算。
- 開立方：開立方，即求立方根。用於三次天元方程。
- 開三乘方：開四次方，即求四次根。用於四次天元方程，為天元術中最高深之運算。
- 實：常數項，即方程中之已知數部分。
- 從：一次項係數，天元術中「從法」即 x 之係數。
- 廉：二次項係數，即 x^2 之係數。
- 隅：三次項係數（立方方程中），即 x^3 之係數。四次方程中另有「三乘隅」或「益隅」之稱。
- 勾：直角三角形之短邊（底邊）。
- 股：直角三角形之長邊（高邊）。
- 弦：直角三角形之斜邊。
- 大勾：最大勾股形之勾邊，即通勾。
- 大股：最大勾股形之股邊，即通股。
- 大差：大股減圓徑之餘數。
- 小差：小勾加半徑之和。
- 底勾：勾股形之最下勾股，與圓徑相切之基本勾股形之勾邊。

English Glossary:

- **Tian yuan yi (天元一)**: The celestial element unity; the unknown x. “Let tian yuan unity be [quantity]” means: let x equal the sought quantity.
- **Ji zuo (寄左)**: Deposit on the left; temporarily store an expression on one side of the equation. The “left” side, later canceled with the “equal number.”
- **Xiang xiao (相消)**: Mutual cancellation; subtracting two expressions, equivalent to transposition and combining terms. The core step of tian yuan.
- **Dai fen mu (帶分母)**: Carrying a denominator; an expression with an uncanceled denominator, retained for later operations.
- **Mi (冪)**: Power; square, i.e., x^2 in modern notation.
- **Kai fang (開方)**: Root extraction; square-root extraction, the fundamental operation of tian yuan.
- **Kai li fang (開立方)**: Cube-root extraction; for cubic equations.
- **Kai san cheng fang (開三乘方)**: Fourth-root extraction; for quartic equations—the most advanced operation in tian yuan.
- **Shi (實)**: Dividend; the constant term of the equation.
- **Cong (從)**: Coefficient; the linear term coefficient (x term).
- **Lian (廉)**: Corner; the quadratic term coefficient (x^2 term).
- **Yu (隅)**: Corner divisor; the cubic term coefficient (x^3 term). In quartics: “triple-multiplication corner” or “negative corner.”
- **Gou (勾)**: Base; the shorter leg of a right triangle.
- **Gu (股)**: Leg; the longer leg of a right triangle.
- **Xian (弦)**: Hypotenuse; the diagonal side.
- **Da gou (大勾)**: Large base; the universal base.
- **Da gu (大股)**: Large leg; the universal leg.
- **Da cha (大差)**: Large difference; large leg minus diameter.
- **Xiao cha (小差)**: Small difference; small base plus radius.
- **Di gou (底勾)**: Base leg; the lowest triangle tangent to the circle.

天元術方程式記法・**Tian Yuan Equation Notation:**
李冶先生以籌算式記多項式，自下而上排列：

- 最下為常數項（實）
- 其上為一次項（從）
- 再其上為二次項（廉）
- 再其上為三次項（隅）

例如 $\frac{480}{1}$ 表示 $480 - x$（一次式），$\frac{0\ \ 0\ \ 48000}{1\ \ 0\ \ 0}$ 表示 $48000 - x^2$ 之類。其中係數為負者以斜線或特殊記號表示之。

各卷方程次數統計・**Equation Degree Statistics:**

卷	平方（二次）	立方（三次）	三乘方（四次）
卷四（底勾）	12 問	4 問	1 問
卷五（大股）	6 問	10 問	2 問
卷六（大勾）	3 問	6 問	9 問

由上表可見，愈至卷六，方程之次數愈高，開方之術愈繁，乃李冶先生循序漸進、由淺入深之教學法也。

Tian Yuan Equation Notation:
Master Li Ye recorded polynomials in counting-rod style, arranged bottom-to-top:

- Lowest: constant term (dividend, shí)
- Above: linear term (coefficient, cóng)
- Above that: quadratic term (corner, lián)
- Above that: cubic term (corner divisor, yú)

Thus $\frac{480}{1}$ denotes $480 - x$ (linear), and $\frac{0\ \ 0\ \ 48000}{1\ \ 0\ \ 0}$ denotes $48000 - x^2$. Negative coefficients were indicated by slanted lines or special marks.

Equation Degree Statistics by Volume:

Volume	**Quadratic**	**Cubic**	**Quartic**
Vol. 4 (Base Leg)	12	4	1
Vol. 5 (Large Leg)	6	10	2
Vol. 6 (Large Base)	3	6	9

As the table shows, equation degrees increase toward Volume 6, and root- extraction methods become more complex. This is Master Li Ye's pedagogical method: gradual progression from simple to profound.

測圓海鏡卷四至卷六全編終

End of Ceyuan Haijing Volumes 4–6

元　李冶　撰　Yuan Dynasty　Li Ye

依《欽定四庫全書》子部天文算法類本

Based on the Imperial Siku Quanshu, Mathematics and Astronomy Class

雙語版　Bilingual Edition

中文版左欄　Chinese Text Left Column

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測圓海鏡

（卷七至卷九·英漢對照本）

李冶　撰

1248

Sea Mirror of Circle Measurements

Volumes 7–9: Bilingual Chinese-English Edition

Translated and typeset in parallel columns

Contents

1 緒論

Introduction

測圓海鏡者，金元之際李冶潤甫先生之所著也。先生諱治，字仁卿，號敬齋，真定欒城人，生於金明昌元年（一一九二），卒於元至元十六年（一二七九）。此書成於淳祐八年（一二四八），凡十二卷，計一百七十問，皆設圓城以為問，用天元術以立算，實中算之鴻篇也。

The Ceyuan Haijing (Sea Mirror of Circle Measurements) was written by Li Ye (Li Zhi, 1192–1279) during the Jin-Yuan transition period. A native of Luancheng, Zhending, he completed this work in 1248. The book comprises 12 volumes with 170 problems, all concerning a circular walled city, solved by the tian yuan shu (heavenly element method)—a masterpiece of Chinese mathematics.

其法之要，在於天元一術。立天元一為所求之數，依題意布算，得開方式，然後開方求之。所謂太極、天元、地元者，猶今代數之未知數也。書中凡遇開方式，皆以 p 記之，蓋當時算板布列之位也。

The essence of the method lies in the tian yuan technique: set the celestial unknown as the quantity sought, derive a polynomial equation (the "opening pattern") according to the problem conditions, then extract the root. The terms Taiji, Tian Yuan, and Di Yuan correspond to algebraic unknowns. The marker p marks positions on the counting board where polynomial coefficients were arranged.

卷七至卷九所載，凡三十四問，其題目之繁，方程式次之高，皆過於前六卷。卷七曰「明前一十八問」，卷八曰「明後一十六問」，卷九曰「大斜四問、大和八問」。其問題之設，或二人對行，或四處遙望，或穿城而過，或倚樹而參，錯綜變化，不可勝窮。

Volumes 7–9 contain 34 problems of greater complexity than the first six volumes, with equations reaching the 4th–6th degree. Volume 7 treats "front" configurations, Volume 8 "rear" configurations, and Volume 9 the most challenging "oblique" and "sum" problems. The scenarios involve pairs or groups of observers, tree landmarks, and gates around the circular city.

體例說明

每問之體，例分三節：

- **或問**：設問之辭，述人物行止之狀、所給之數。
- **法曰**：列方程式之術，示開方之次。
- **草曰**：詳布算之草，盡天元變化之妙。

又有時設**答曰**一節，直書所求之數，以為開方之驗。

Structure of Each Problem

Each problem follows a standard format:

- ***Q* (或問)**: *The problem statement, describing the scene and given quantities.*
- ***Method* (法曰)**: *The polynomial setup and root-extraction procedure.*
- ***Working* (草曰)**: *Detailed step-by-step derivation using tian yuan.*

*An **Answer** (**答曰**) section sometimes states the result directly as verification.*

書中所用十五形之名，皆有所本。通勾、通股者，大勾股形之兩邊也；邊勾、邊股者，邊上之小形也；底勾、底股者，底下之小形也。明勾、明股者，明朗易見之小形也；重勾、重股者，重疊相因之小形也。至於虛勾、虛股、虛弦，則太虛之形，最為微妙。

*The text uses fifteen named right triangles. **Tong** (通) gou/gu are the legs of the main large triangle. **Bian** (邊) and **Di** (底) gou/gu belong to smaller edge/base triangles. **Ming** (明, "bright") and **Zhong** (重, "overlapping") triangles are visually apparent. The **Xu** (虛, "void") triangle is the most subtle—the inner void space. Each problem typically asks for the city diameter (城徑): **240 paces**.*

2 卷七　明前一十八問

Vol. 7: 18 "Front" Problems

卷七小序: 此卷一十八問,皆以明段為主。所謂明段者,勾股形之顯然可見者也。其題多設二人對行、遙望相會之狀,或倚門傍樹,或穿城越隍,錯綜參伍,以盡天元之變。其方程式之高,有至三乘、四乘者,開方之法亦隨之益密。

Volume 7 Preface: *These 18 problems all concern "bright" (ming) segments—the visibly apparent right-triangle edges. Scenarios involve pairs walking toward each other, gazing from afar, leaning by gates or trees, passing through the city. The equations reach degrees 3–4, with increasingly refined root-extraction techniques.*

第一百零五問

Problem 一百零五

或問: 出南門東行七十二步有樹,出東門南行三十步見之。問答同前。

Q: *Going out the South Gate and walking 72 paces east there is a tree. Going out the East Gate and walking 30 paces south, one sees it. Answer as before.*

答曰: 城徑二百四十步,半徑一百二十步。

A: **City diameter: 240 paces. Radius: 120 paces.**

法曰: 倍南行以乘倍東行為平實,並二行又倍之為從,一虛隅。得城徑。

Method: Double the southward walk, multiply by double the eastward walk as the flat shi. Add the two walks and double as cong. One empty corner. Extract the city diameter.

草曰: 識別得此問名為弦外容圓,又為內率求虛積。其二行步相並為虛弦,若以相減即虛較也。又倍東行為弦較和,倍南行即弦較較,此二數相乘則兩虛積也。若直以二行相乘,則半個虛積也。又倍東行減於城徑,餘即二虛勾也。倍南行減於城徑則二虛股也。虛積上三事和即城徑也。

乃立天元一為圓徑,便以為三事和也。倍二行步減之,得 p 為黃方一,天元乘之得 p 為二虛積(寄左)。然後倍東行以乘倍南行,得八千六百四十為同數,與左相消得 p。益積開平方得二百四十步,即城徑也。合問。

Working: Identification: This problem is "circle contained outside the hypotenuse" and also "seeking the void area by inner rate." The sum of the two walks is the void hypotenuse; their difference is the void difference. Double the eastward walk gives the sum of hypotenuse and difference; double the southward walk gives their difference. Their product is twice the void area; direct multiplication gives half the void area. The city diameter minus double the eastward walk gives twice the void gou; minus double the southward walk gives twice the void gu. The three-element sum on the void area equals the city diameter.

Let the celestial unknown be the circle diameter, as the three-element sum. Subtracting double the two walks gives p as the yellow square; multiplying by the unknown gives p as twice the void area (stored left). Then double the eastward walk times double the southward walk gives 8640 as the matching number. Canceling gives p. Accumulating and extracting the square root yields 240 paces, the city diameter. Q.E.D.

第一百零六問

Problem 一百零六

或問: 丙出南門直行一百三十五步而立,甲出東門直行一十六步見之。問答同前。

Q: *Bing goes out the South Gate and walks straight 135 paces, then stands. Jia goes out the East Gate and walks straight 16 paces and sees him. Answer as before.*

答曰: 城徑二百四十步,丙行上勾弦差八十一步。

A: **City diameter: 240 paces. The difference between gou and hypotenuse on Bing's walk: 81 paces.**

法曰: 以丙行步一百三十五再自之,得二百四十六萬零三百七十五於上。又以甲行一十六乘丙行冪一萬八千二百二十五,得二十九萬一千六百,以乘上位,得七千一百七十四億四千五百三十五萬為三乘方實;以二行步相乘又倍之得四千三百二十,以乘丙行步再自之數,得一百六億二千八百八十二萬為

Method: Raise Bing's walk 135 to the third power, obtaining 2,460,375 as upper term. Multiply Jia's walk 16 by Bing's squared walk 18,225 to get 291,600; multiply by the upper term to obtain 717,445,350,000 as the cubic equation's shi. Double the product of the two walks to get 4,320; multiply by the cube of Bing's walk

益從;第一廉空;以甲行乘丙行冪,得二十九萬一千六百,又倍之得五十八萬三千二百於上,四之甲行冪一千零二十四,以乘丙行步,得一十三萬八千二百四十,減上位餘四十四萬四千九百六十為第二廉;二行步相乘得二千一百六十為虛常法。得丙行步上勾弦差八十一。

to obtain 10,628,820,000 as increasing cong. First lian is empty. Multiply Jia's walk by Bing's squared walk to get 291,600; double to get 583,200 as upper term. Quadruple Jia's squared walk 1,024 and multiply by Bing's walk to get 138,240; subtract from upper term, leaving 444,960 as second lian. The product of the two walks, 2,160, is the empty constant method. Extracts the gou-hypotenuse difference on Bing's walk: 81.

草曰: 識別二數相並,得一百五十一。以減於皇極弦,餘一百三十八,即虛勾虛股並也。若以二數相減,餘一百一十九為高弦內減平弦,又為皇極弦內少個小差弦,又為大差弦內減個皇極弦也。

立天元一為丙行大差數。置丙行步一百三十五,自乘得 p ,用天元除之,得 p 為勾弦並也。上減天元得 p 為二丙勾也。復用丙南行乘之,得 p 為二積也。又以天元除之,得 p 為丙勾外容圓半(泛寄)。別置丙南行用二甲勾乘之,得 p 太,合用二丙勾除之。不受除,便以此為甲股(內寄二丙勾為分母)。復用二甲勾三十二乘之,得 p 太為二個甲直積也。又置丙南行內減天元,得 p 為黃方,以自乘得 p 為丙上勾弦差乘股弦差二段,以天元除之得 p 為兩個丙小差也。乃用甲股乘之,得下式 p。復用丙南行除之,得 p,又折半得下式 p 為一個甲步股弦差也,內亦帶前二丙勾分母。復置兩個甲直積,內已寄此甲股弦差分母,便為甲步股外容圓半(寄左)。乃再置先求到泛寄,用甲股弦差分母乘之,得 p 為同數,與左相消得下式 p。開三乘方得八十一步,即丙步上勾弦差也。

《鈐經》載此法,以勾弦差率冪減丙行冪,復以丙行乘之為實,以差率冪為法,如法得徑。此法只是以勾外求圓半。合以大差除倍積,而今皆以大差冪為分母也。依法求之,勾弦差八十一自之得六千五百六十一,以減於丙行冪一萬八千二百二十五,餘一萬一千六百六十四,復以丙行一百三十五乘之,得一百五十七萬四千六百四十為實,以大差冪六千五百六十一為法,如法得二百四十步,即城徑也。

Working: Identification: The sum of the two numbers is 151. Subtracting from the imperial extreme hypotenuse leaves 138, the sum of void gou and void gu. Their difference, 119, is the high string minus the level string, also the imperial extreme string lacking a small-difference string, also the large-difference string minus the imperial extreme string.

Let the unknown be the large-difference number for Bing's walk. Square Bing's walk 135 to get p; divide by the unknown to get p as the sum of gou and hypotenuse. Subtract the unknown above to get p as twice Bing's gou. Multiply by Bing's southward walk to get p as twice the area. Divide by the unknown to get p as the half-circle (general storage). Multiply Bing's southward walk by twice Jia's gou to get p (this becomes Jia's gu with denominator). Multiply by twice Jia's gou 32 to get p as twice Jia's rectangular area. Subtract the unknown from Bing's walk to get p as the yellow square; squaring gives p as two segments of the product of gou-hypotenuse difference and gu-hypotenuse difference. Dividing by the unknown gives p as two small differences. Multiply by Jia's gu to get p. Divide by Bing's southward walk and halve to get p as one step of Jia's gu-hypotenuse difference. Extracting the cubic root yields 81, the gou-hypotenuse difference.

The *Qianjing* records: square the gou-hypotenuse difference rate and subtract from Bing's walk square, then multiply by Bing's walk as shi; use the squared difference rate as divisor to obtain the diameter. This method seeks the half-circle from outside the gou. The large difference should divide twice the area; here the squared large difference serves as denominator. Accordingly: square 81 to get 6,561; subtract from Bing's walk square 18,225, leaving 11,664; multiply by 135 to get 1,574,640 as shi; divide by squared large difference 6,561 to obtain 240 paces, the city diameter.

第一百零七問

Problem 一百零七

或問: 出東門一十六步有樹,出南門東行七十二步見之。問答同前。

Q: *Going out the East Gate 16 paces there is a tree. Going out the South Gate and walking 72 paces east, one sees it. Answer as before.*

答曰: 城徑二百四十步。

A: **City diameter**: **240 paces.**

法曰: 二行步相減得數,以自之於上。又以出東門步自之,減上位為平方實,二之出南門東行步為益從,一步常法。翻開得半徑。

Method: Subtract the two walks; square the result as upper term. Square the East Gate walk and subtract from upper term as the quadratic shi. Double the south-gate eastward walk as increasing cong. One constant method. Extract by reversing to obtain the

草曰: 別得人到樹即平弦也,半圓徑即平股也。其東行七十二步則平勾平弦差也。乃立天元一為半圓徑,加一十六減七十二,得 p 為勾也。以自之得 p 為勾冪。又加入天元股冪得 p 為弦冪(寄左)。再立天元一為半徑,加出東門步,得 p 即弦也。以自之得 p 為同數,與左相消得 p。翻法開之得一百二十步,即半城徑也。合問。

radius.

Working: Identification: The person reaching the tree is the level hypotenuse; half the diameter is the level gu. The 72 paces east is the difference between level gou and level hypotenuse. Let the unknown be half the diameter; add 16 and subtract 72 to get p as gou. Squaring gives p as gou-power. Add the unknown squared as gu-power to get p as hypotenuse-power (stored left). Again let the unknown be the radius; add the East Gate paces to get p as the hypotenuse. Squaring gives p as the matching number. Canceling gives p. Extract by the reversing method to obtain 120 paces, half the city diameter. Q.E.D.

第一百零八問

Problem 一百零八

或問: 出南門一百三十五步有樹,出東門南行三十步見之。問答同前。

Q: *Going out the South Gate 135 paces there is a tree. Going out the East Gate and walking 30 paces south, one sees it. Answer as before.*

答曰: 城徑二百四十步。

A: City diameter: 240 paces.

法曰: 樹去城步內減南行步,餘以為冪於上,又以樹去城步為冪內減上位為平實,倍樹去城步為從,一虛隅。翻法得半城徑。

Method: Subtract the southward walk from the tree-to-city distance; square the remainder as upper term. Square the tree-to-city distance and subtract the upper term as flat shi. Double the tree-to-city distance as cong. One empty corner. Extract by reversing to obtain half the city diameter.

草曰: 別得人距樹即高弦也,半圓徑即高勾也,其南行三十步即高弦上小差也。乃立天元一為半徑,加樹去城步為弦,內減小差 p,得 p 即股也。以自之得 p 為股冪,內加入天元冪,得 p 為弦冪(寄左)。再置弦 p 以自之,得 p 為同數,與左相消,得式 p。翻開得一百二十步,即半城徑也。合問。

Working: Identification: The person's distance to the tree is the high hypotenuse; half the diameter is the high gou. The 30 paces southward is the small difference on the high hypotenuse. Let the unknown be the radius; add the tree-to-city distance as hypotenuse, subtract the small difference p to get p as gu. Squaring gives p as gu-power; add the unknown squared to get p as hypotenuse-power (stored left). Again place hypotenuse p, square it to get p as matching number. Canceling gives p. Extract by reversing to obtain 120 paces, half the city diameter. Q.E.D.

第一百零九問

Problem 一百零九

或問: 乙出東門不知遠近而立,甲出南門東行七十二步望見乙,就乙斜行一百三十六步與乙相會。問答同前。

Q: *Yi goes out the East Gate and stands at an unknown distance. Jia goes out the South Gate, walks 72 paces east, sees Yi, then walks diagonally 136 paces to meet Yi. Answer as before.*

答曰: 城徑二百四十步,乙出東門一百二十八步。

A: City diameter: 240 paces. Yi went out the East Gate 128 paces.

法曰: 以斜行步自之於上,以二行相減餘自為冪,減上位為平實,從空,一步常法。如法得半徑。

Method: Square the diagonal walk as upper term. Subtract the two walks; square the remainder and subtract from upper term as flat shi. Cong is empty. One constant method. Extract to obtain the radius.

草曰: 別得七十二步即大差也,斜行即弦,半徑即股也。立天元一為半徑,以自之為股冪,又以二行差六十四以自之得 p 為勾冪。並二冪得 p 為弦冪(寄

Working: Identification: The 72 paces is the large difference; the diagonal is the hypotenuse; the radius is the gu. Let the unknown be the radius; square it as gu-power. The

左)。然後以斜行步自之,得 p 為同數,與左相消得 p。開平方得一百二十步,倍之即城徑也。合問。

difference of the two walks, 64, squared gives p as gou-power. Adding both powers gives p as hypotenuse-power (stored left). Then square the diagonal walk to get p as matching number. Canceling gives p. Extract the square root to obtain 120 paces; double it for the city diameter. Q.E.D.

第一百一十問

Problem 一百一十

或問: 甲出南門不知遠近而立,乙出東門南行三十步望見甲,卻就甲斜行二百五十五步與甲相會。問答同前。

答曰: 城徑二百四十步,甲出南門二百二十五步。

法曰: 二行差自之為冪,以減於斜行冪為平實,一虛隅。得半徑。

草曰: 別得南行步即股弦差也,斜步即弦也,半徑即勾也。乃立天元一為半城徑,以自之為冪。以二行相減餘二百二十五,以自之得 p 為股冪。二冪相並得 p 為弦冪(寄左)。然後以斜行步自之,得 p 為同數,與左相消得下 p。開平方得一百二十步,即半徑也。合問。

Q: *Jia goes out the South Gate and stands at an unknown distance. Yi goes out the East Gate, walks 30 paces south, sees Jia, then walks diagonally 255 paces to meet Jia. Answer as before.*

A: **City diameter: 240 paces. Jia went out the South Gate 225 paces.**

Method: Square the difference of the two walks; subtract from the squared diagonal as flat shi. One empty corner. Extract the radius.

Working: Identification: The southward walk is the difference between gu and hypotenuse; the diagonal is the hypotenuse; the radius is the gou. Let the unknown be half the city diameter; square it as power. The difference of the two walks, 225, squared gives p as gu-power. Adding both powers gives p as hypotenuse-power (stored left). Then square the diagonal walk to get p as matching number. Canceling gives p. Extract the square root to obtain 120 paces, the radius. Q.E.D.

第一百一十一問

Problem 一百一十一

或問: 甲出南門東行不知步數而立,乙出東門南行三十步望見甲,斜行一百二步相會。問答同前。

答曰: 城徑二百四十步,甲東行九十六步。

法曰: 二行相減,餘以乘乙南行,四之於上,又加入斜行冪為平實。得虛和一百三十八。

草曰: 別得斜步內減南行為甲東行步也。此問以弦外容圓入之,以二行相減數乘乙南行三十步,得 p,又四之,得 p 為二直積也。又加入斜步冪 p,共得 p 即和冪也。平方而一,得一百三十八步,即虛和也。又加斜步得二百四十步,即城徑也。合問。

Q: *Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate, walks 30 paces south, sees Jia, then walks diagonally 102 paces to meet Jia. Answer as before.*

A: **City diameter: 240 paces. Jia walked 96 paces east.**

Method: Subtract the two walks; multiply the remainder by Yi's southward walk; quadruple as upper term; add the squared diagonal as flat shi. Extracts the void sum: 138.

Working: Identification: The diagonal minus the southward walk gives Jia's eastward walk. This problem uses "circle outside the hypotenuse." Multiply the difference of the two walks by Yi's 30 paces south to get p; quadruple to get p as twice the rectangular area. Add the squared diagonal p, totaling p as the squared sum. Extract the square root to obtain 138, the void sum. Add the diagonal to get 240 paces, the city diameter. Q.E.D.

第一百一十二問

Problem 一百一十二

或問:乙出東門南行不知步數而立,甲出南門東行七十二步望見乙,斜行一百二步與乙相會。問答同前。

答曰:城徑二百四十步,乙南行五十四步。

法曰:倍相減步以乘倍東行,得數復以減於斜步冪,餘為實。平方而一,得較也。又以二行相減數乘倍東行為平實,以較為從方,得勾。勾較共為長,又以斜步並入勾股共,即城徑。

草曰:別得二行相減餘 p 為乙南行步也。以此數又減於甲東行,餘四十二步即較也。又以二行相減數 p 乘倍東行得 p 為平實,以較為從。平方開得四十八即勾也。勾內加較得九十步即股也。勾股共得一百三十八,又加入斜步,共得二百四十步,即城徑也。合問。

Q: Yi goes out the East Gate and walks south an unknown number of paces, then stands. Jia goes out the South Gate, walks 72 paces east, sees Yi, then walks diagonally 102 paces to meet Yi. Answer as before.

A: City diameter: 240 paces. Yi walked 54 paces south.

Method: Double the subtraction step and multiply by double the eastward walk; subtract this from the squared diagonal, remainder as shi. Extract the square root to obtain the difference. Multiply the difference of the two walks by double the eastward walk as flat shi; use the difference as cong to extract gou. Gou plus difference gives the length; add the diagonal to the gou-gu sum for the city diameter.

Working: Identification: The difference of the two walks, p, is Yi's southward walk. Subtract this from Jia's eastward walk; the remainder 42 is the difference. Multiply the difference of the two walks p by double the eastward walk to get p as flat shi; use the difference as cong. Extract the square root to obtain 48, the gou. Add the difference to get 90 paces, the gu. Gou plus gu totals 138; add the diagonal to obtain 240 paces, the city diameter. Q.E.D.

第一百一十三問

Problem 一百一十三

或問: 乙出南門東行,甲出東門南行,兩相望見。既而乙雲:「我東行不及城徑一百六十八步。」甲雲:「我南行不及城徑二百一十步。」問答同前。

答曰:城徑二百四十步,甲南行三十步,乙東行七十二步。

法曰:半甲不及步以自之為冪,半甲不及步內減差以自之為冪。二冪相並內卻減差冪為平實,四之甲不及內減三之乙不及,餘為益從,三步半虛法。得甲南行。

草曰:別得乙不及為虛勾、半徑共,又為徑內減明勾也。甲不及為虛股、半徑共,又為徑內減股也。又二雲數相並為虛和、圓徑共也,雲數相減即虛較也。

乃立天元一為甲南行,以減於甲不及步又半之,得 p 為虛股也。虛股內減虛較得 p 為虛勾。勾自之得 p 為勾冪也,又股自之下式 p 為股冪也。二冪相並得 p 為弦冪(寄左)。然後以天元加虛較得 p 為乙東行。又加入天元甲南行得 p 為虛弦,以自之得 p 為同數,與左相消得 p。開平方得三十步,即甲南行也。內加少步,即城徑也。合問。

Q: Yi goes out the South Gate walking east; Jia goes out the East Gate walking south; they see each other. Yi says: "My eastward walk falls short of the diameter by 168 paces." Jia says: "My southward walk falls short of the diameter by 210 paces." Answer as before.

A: City diameter: 240 paces. Jia walked 30 paces south; Yi walked 72 paces east.

Method: Halve Jia's shortfall and square it as one power. Subtract the difference from half Jia's shortfall and square as second power. Add both powers and subtract the squared difference as flat shi. Quadruple Jia's shortfall minus triple Yi's shortfall as increasing cong. Three-and-a-half empty method. Extracts Jia's southward walk.

Working: Identification: Yi's shortfall is the void gou plus radius, also the diameter minus the bright gou. Jia's shortfall is the void gu plus radius, also the diameter minus the gu. The sum of the two given numbers is the void sum plus circle diameter; their difference is the void difference.

Let the unknown be Jia's southward walk. Subtract from half Jia's shortfall to get p as void gu. Subtract the void difference from void gu to get p as void gou. Square the gou to get p as gou-power; square the gu to get p as gu-power. Adding both gives p as hypotenuse-power (stored left). Add the unknown to the void difference to get p as Yi's eastward walk. Add the unknown (Jia's southward walk) to get p as void hypotenuse. Squaring gives p as matching number. Canceling gives p. Extract the square root to obtain 30 paces, Jia's southward walk. Add the shortfall for the city diameter. Q.E.D.

第一百一十四問

或問： 丙出南門直行，甲出東門直行，兩相望見。既而丙雲：「我行少於城徑一百五步。」甲雲：「我行少於城徑二百二十四步。」問答同前。

答曰： 城徑二百四十步，半徑一百二十步。

法曰： 二少步相乘訖，又自乘為實；六之共步，乘雲數相乘數為益從；十八之雲數相乘於上，又三之共步，自乘加上位，內復減丙少步冪、甲少步冪為從廉；四十八之共步為益二廉，六十三步常法。翻法開三乘方，得一百二十步，即半徑。

草曰： 別得雲數共減於倍城徑為甲丙共行數。又雲數相減即皇極差，亦為甲行不及丙行數。立天元一為半城徑，以三之，副置二位。上位減丙少步，得 p 為皇極股也，下位減甲少步得 p 為皇極勾也。勾股相乘得 p，以天元除之，得 p 為弦也。弦自之得 p 為弦冪（寄左）。然後以股自之得下 p 為股冪於上，又以勾自之得 p 為勾冪，並以加入上位，得 p 為同數，與左相消得 p。翻法開三乘方得一百二十步，即半徑也。合問。

Problem 一百一十四

Q: *Bing goes out the South Gate walking straight; Jia goes out the East Gate walking straight; they see each other. Bing says: "My walk is 105 paces less than the diameter." Jia says: "My walk is 224 paces less than the diameter." Answer as before.*

A: City diameter: 240 paces. Radius: 120 paces.

Method: Multiply the two shortfalls and then self-multiply as shi. Multiply six times the sum by the product of the given numbers as increasing cong. Multiply eighteen times the product of the given numbers as upper term; add three times the squared sum; subtract Bing's shortfall squared and Jia's shortfall squared as cong-lian. Forty-eight times the sum as increasing second lian; sixty-three as constant method. Extract by reversing the cubic root to obtain 120 paces, the radius.

Working: Identification: The sum of the given numbers subtracted from double the city diameter gives the total walk of Jia and Bing. Their difference is the imperial extreme difference, also Jia's walk short of Bing's.

Let the unknown be half the city diameter; triple it and place in two positions. Subtract Bing's shortfall from the upper position to get p as imperial extreme gu; subtract Jia's shortfall from the lower to get p as imperial extreme gou. Multiply gou and gu to get p; divide by the unknown to get p as hypotenuse. Squaring gives p as hypotenuse-power (stored left). Square the gu to get p as gu-power (upper); square the gou to get p as gou-power. Add both to the upper to get p as matching number. Canceling gives p. Extract the cubic root by reversing to obtain 120 paces, the radius. Q.E.D.

第一百一十五問

或問： 甲出東門直行，乙出南門直行，各不知步數而立。乙望見甲，就甲斜行了二百八十九步與甲相會。其二直行共得一百五十一步。又雲甲直行少於乙直行。問答同前。

答曰： 城徑二百四十步，甲直行一十六步，乙直行一百三十五步。

法曰： 斜冪內減共步冪為平實，倍共步內減斜步為從，一常法。得徑。

Problem 一百一十五

Q: *Jia goes out the East Gate walking straight; Yi goes out the South Gate walking straight; each stands at an unknown distance. Yi sees Jia, then walks diagonally 289 paces to meet Jia. Their two straight walks total 151 paces. It is also stated that Jia's walk is less than Yi's. Answer as before.*

A: City diameter: 240 paces. Jia walked 16 paces; Yi walked 135 paces.

Method: Subtract the squared sum from the squared diagonal as flat shi. Double the sum minus the diagonal as cong. One constant method. Extract the diameter.

草曰:別得共數、城徑並即皇極和也。立天元一為圓徑,加共步得 p 為皇極和,以自之,得 p 於上。以斜行冪 p 減上位餘 p 為二直積(寄左)。然後以天元乘斜步得 p,與左相消得 p。開平方得二百四十步,即城徑也。合問。

Working: Identification: The sum plus the city diameter is the imperial extreme sum.

Let the unknown be the circle diameter; add the sum to get p as imperial extreme sum. Square it to get p as upper term. Subtract the squared diagonal p from the upper term, leaving p as twice the rectangular area (stored left). Multiply the unknown by the diagonal to get p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百一十六問

Problem 一百一十六

或問: 甲出東門直行,乙出東門南行,丙出南門直行,丁出南門東行,各不知步數而立。四人遙相望,悉與城參相直。只雲甲、丙共行了一百五十一步,乙、丁立處相距一百二步。又雲丙直行步多於甲直行步。問答同前。

Q: *Jia goes out the East Gate walking straight; Yi goes out the East Gate walking south; Bing goes out the South Gate walking straight; Ding goes out the South Gate walking east; each stands at an unknown distance. The four gaze at each other from afar, all aligned with the city. It is only stated that Jia and Bing together walked 151 paces; Yi and Ding's positions are 102 paces apart. Also stated that Bing's walk exceeds Jia's. Answer as before.*

答曰:城徑二百四十步。

A: City diameter: 240 paces.

法曰:共步、距步相減,得數自之於上,以共步為冪內減上為平實,二之距步內減共步、距步差為從,一步虛法。得城徑。

Method: Subtract the distance from the sum; square the result as upper term. Subtract this from the squared sum as flat shi. Double the distance minus the difference of sum and distance as cong. One empty-corner method. Extract the city diameter.

草曰:別得共步得城徑即皇極和也,相距步即虛弦也。皇極和內減虛弦即皇極弦也。又共步、距步差 p 即皇極弦內減城徑也(此名旁差)。乃立天元一為城徑,加共步得 p 為皇極和也,以自之得 p 於上,以共步、距步差 p 加天元得 p 為皇極弦也,以自之得下式 p,減上位餘得 p 為二直積(寄左)。然後以天元徑乘皇極弦,得 p 為同數,與左相消得 p。開平方得二百四十步,即城徑也。合問。

Working: Identification: The sum equals the city diameter, which is the imperial extreme sum. The distance apart is the void hypotenuse. Imperial extreme sum minus void hypotenuse equals the imperial extreme hypotenuse. The difference of sum and distance p is the imperial extreme hypotenuse minus the city diameter (called side difference).

Let the unknown be the city diameter; add the sum to get p as imperial extreme sum. Square it to get p as upper term. Add the difference of sum and distance p to the unknown to get p as imperial extreme hypotenuse. Squaring gives p; subtracting from the upper term leaves p as twice the rectangular area (stored left). Multiply the unknown diameter by the imperial extreme hypotenuse to get p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百一十七問

Problem 一百一十七

或問:甲出南門東行不知步數而立,乙出東門南行望見甲,復就甲斜行,與甲相會。乙通計行了一百三十二步,其乙南行步不及斜行七十二步,其甲東行卻多於乙南行。問答同前。

Q: *Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate walking south, sees Jia, then walks diagonally to meet Jia. Yi walked a total of 132 paces; Yi's southward walk falls short of the diagonal by 72 paces; Jia's eastward walk exceeds Yi's southward walk. Answer as before.*

答曰：城徑二百四十步，乙南行三十步，甲東行一百二步。

A: **City diameter: 240 paces. Yi walked 30 paces south; Jia walked 102 paces east.**

法曰：倍不及步在地，以不及步減通步以乘之為實，以四之不及步為法。得乙南行三十步。

Method: Place double the shortfall; multiply by the total minus the shortfall as shi; use quadruple the shortfall as divisor. Extracts Yi's southward walk: 30 paces.

草曰：別得乙南行即股也，以減通步即虛弦也，以減不及步即虛較也，其不及步即甲東行也。立天元一為乙南行，置不及步以天元乘之，又四之得 p 元為二直積（寄左）。然後倍不及步以為弦較和於上 p。以不及步減通步得 p 為弦較較。以乘上位得 p 太為同數，與左相消得 p。上法下實，得三十步為乙南行也。餘各以數求之。

Working: Identification: Yi's southward walk is the gu; subtract from the total to get the void hypotenuse; subtract the shortfall to get the void difference. The shortfall itself is Jia's eastward walk.

Let the unknown be Yi's southward walk. Multiply the shortfall by the unknown and quadruple to get p as twice the rectangular area (stored left). Double the shortfall as the sum of hypotenuse and difference (upper) p. Total minus shortfall gives p as the difference of hypotenuse and difference. Multiply the upper by this to get p as matching number. Canceling gives p. Dividing yields 30 paces as Yi's southward walk. The rest follows from the numbers.

第一百一十八問

Problem 一百一十八

或問：甲出南門東行不知步數而立。乙出東門南行，望見甲，復斜行與甲相會。二人共行了二百四步，又雲甲行不及共步一百三十二。問答同前。

Q: *Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate walking south, sees Jia, then walks diagonally to meet Jia. Together they walked 204 paces. It is also stated that Jia's walk falls short of the total by 132. Answer as before.*

答曰：城徑二百四十步，甲東行九十六步，乙南行一百零八步。

A: **City diameter: 240 paces. Jia walked 96 paces east; Yi walked 108 paces south.**

法曰：別得二行共即兩個虛弦也，其不及步即乙南行與一虛弦共也。置不及步內減一弦餘三十步，即乙南行也。以乙南行反以減虛弦，餘七十二步即甲東行也。以乙南行減甲東行餘即虛較也。

Method: Identification: The sum of the two walks is twice the void hypotenuse; the shortfall is Yi's southward walk plus one void hypotenuse. Subtract one hypotenuse from the shortfall; the remainder 30 is Yi's southward walk. Subtract Yi's southward walk from the void hypotenuse; the remainder 72 is Jia's eastward walk. Subtract Yi's from Jia's to get the void difference.

草曰：此問無草。

Working: This problem has no working.

第一百一十九問

Problem 一百一十九

或問：乙出東門南行，甲出西門南行，甲望見乙，斜行五百一十步相會。乙雲：「我南行少於城徑二百一十步。」問答同前。

Q: *Yi goes out the East Gate walking south; Jia goes out the West Gate walking south; Jia sees Yi and walks diagonally 510 paces to meet Yi. Yi says: "My southward walk is 210 paces less than the diameter." Answer as before.*

答曰：城徑二百四十步，乙南行三十步。

A: **City diameter: 240 paces. Yi walked 30 paces south.**

法曰：少步冪為平實，四斜步內減二少步為益從，五步常法。得乙南行。

Method: Square the shortfall as flat shi. Quadruple the diagonal minus double the shortfall as increasing cong. Five as constant method. Extracts Yi's southward walk.

草曰：別得少步為徑內減股。立天元一為乙南行，以二之減於倍斜行步，得 p 為梯底也。以二之天元乘之，得 p 為徑冪（寄左）。再置天元加少步，得下式 p 為城徑，以自之得 p，與左相消得 p。開平方得三十步，即乙南行也。加少步即城徑也。合問。

Working: Identification: The shortfall is the diameter minus the gu.

Let the unknown be Yi's southward walk. Subtract double the unknown from double the diagonal to get p as the ladder base. Multiply by double the unknown to get p as diameter-power (stored left). Add the shortfall to the unknown to get p as city diameter; squaring gives p. Canceling gives p. Extract the square root to obtain 30 paces, Yi's southward walk. Add the shortfall for the city diameter. Q.E.D.

第一百二十問

Problem 一百二十

或問：乙出南門東行，甲出北門東行，甲望見乙，斜行二百七十二步與乙相會。乙雲：「我東行不及城徑一百六十八步。」問答同前。

Q: *Yi goes out the South Gate walking east; Jia goes out the North Gate walking east; Jia sees Yi and walks diagonally 272 paces to meet Yi. Yi says: "My eastward walk falls short of the diameter by 168 paces." Answer as before.*

答曰：城徑二百四十步，乙東行七十二步。

A: City diameter: 240 paces. Yi walked 72 paces east.

法曰：以不及步冪之為實，四斜內減二之不及步為虛從，五常法。平開得乙東行七十二步。

Method: Square the shortfall as shi. Quadruple the diagonal minus double the shortfall as empty cong. Five as constant method. Extract to obtain Yi's eastward walk: 72 paces.

草曰：別得不及步為城徑減明勾也。立天元一為乙東行，以倍之減於二之斜行步，得下 p 為梯底也。倍天元乘之，得 p 為徑冪（寄左）。再置天元加不及步，得 p 為城徑，以自之得 p 為同數，與左相消得 p。開平方得七十二步，即乙東行也。加入少步即城徑也。合問。

Working: Identification: The shortfall is the city diameter minus the bright gou.

Let the unknown be Yi's eastward walk. Subtract double the unknown from double the diagonal to get p as ladder base. Multiply by double the unknown to get p as diameter-power (stored left). Add the shortfall to the unknown to get p as city diameter; squaring gives p as matching number. Canceling gives p. Extract the square root to obtain 72 paces, Yi's eastward walk. Add the shortfall for the city diameter. Q.E.D.

第一百二十一問

Problem 一百二十一

或問：乙出南門東行，丁出東門南行，卻有甲丙二人共在西北隅，甲向東行，丙向南行，四人遙相望見，俱與城參相直。既而相會，甲雲：「我多乙二百四十八步。」丙雲：「我多於丁五百七十步。」問答同前。

Q: *Yi goes out the South Gate walking east; Ding goes out the East Gate walking south. Meanwhile Jia and Bing together are at the northwest corner, Jia walking east and Bing walking south. The four gaze at each other from afar, all aligned with the city. After meeting, Jia says: "I exceed Yi by 248 paces." Bing says: "I exceed Ding by 570 paces." Answer as before.*

答曰：城徑二百四十步。

A: City diameter: 240 paces.

法曰：二多步相乘為平實，並二多步而半之為從，七分半常法。得城徑。

Method: Multiply the two excesses as flat shi. Halve the sum of the two excesses as cong. Seven and a half as constant method. Extract the city diameter.

草曰：別得甲多步為大勾內減明勾也，丙多步為大股內少股也。又乙東行得一虛勾為半徑，丁南行得一虛股為半徑。又二多數相並得 p 為大和內少虛弦也，又二少數相減餘 p 為兩個角差。又甲多步內

Working: Identification: Jia's excess is the large gou minus the bright gou; Bing's excess is the large gu minus the gu. Yi's eastward walk gives one void gou as radius; Ding's southward walk gives one void gu as radius. The sum of

減半徑即勾方差也,丙多步內減半徑即股方差也。

立天元一為城徑,以半之減於甲多步得 p 為勾方差,又以半徑減於丙多步得 p 為股方差。二差相乘得 p 為徑冪(寄左)。然後以天元冪與左相消,得下式 p。開平方得二百四十步,即城徑也。合問。

the two excesses p is the large sum minus the void hypotenuse; their difference p is twice the corner difference. Jia's excess minus the radius is the gou square-difference; Bing's excess minus the radius is the gu square-difference.

Let the unknown be the city diameter. Subtract half the diameter from Jia's excess to get p as gou square-difference; subtract the radius from Bing's excess to get p as gu square-difference. Multiply both differences to get p as diameter-power (stored left). Cancel the unknown squared with the left to get p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百二十二問

或問: 甲丙二人俱在西北隅,甲向東行,丙向南行。又乙出南門東行,丁出東門南行,各不知步數而立。四人遙相望見,悉與城參相直。既而相會,甲雲:「我與乙共行了三百九十二步。」丙雲:「我與丁共行了六百三十步。」問答同前。

答曰: 城徑二百四十步。

法曰: 甲乙共自之為冪,丙丁共自之為冪,二冪又相乘為三乘方實。甲乙共自之為冪,以丙丁共乘之於上。又以丙丁共自之為冪,以甲乙共乘之加上位為益從。甲乙共自之為冪,丙丁共自之為冪,並,以七分半乘之於上。又以甲乙共乘丙丁共,得數減上位為第一益廉。並二共數,以七分半乘之為第二廉。以七分半自之,得五分六厘二毫五絲於上位。以一步內減上位,餘四分三厘七毫五絲為虛隅。得城徑。

草曰: 別得甲為大勾,乙為明勾,丙為大股,丁為股也。甲乙共內減半徑即是黃長弦也,丙丁共內減半徑即黃廣弦也。黃長弦、黃廣弦二數相減,餘為兩個皇極差也。

乃立天元為城徑,半之副置二位。上以減於甲乙共數,得 p 即黃長弦也,以自之得 p 為黃長弦冪也,內減天元一冪,餘得下式 p 為勾方差冪也。下位以減於丙丁共,得下式 p 即黃廣弦也,以自之得 p 為黃廣弦冪也,內減天元一冪,餘得 p 為股方差冪也。再以勾方差冪、股方差冪相乘,得 p 為徑冪(寄左)。然後以天元為冪,又以冪自之,與左相消得下式 p。開三乘方得二百四十步,即城徑也。合問。

Problem 一百二十二

Q: *Jia and Bing are both at the northwest corner; Jia walks east and Bing walks south. Yi goes out the South Gate walking east; Ding goes out the East Gate walking south; each stands at an unknown distance. The four gaze at each other from afar, all aligned with the city. After meeting, Jia says: "Yi and I together walked 392 paces." Bing says: "Ding and I together walked 630 paces." Answer as before.*

A: City diameter: 240 paces.

Method: Square the Jia-Yi sum and the Bing-Ding sum; multiply both powers as the cubic shi. Square the Jia-Yi sum, multiply by the Bing-Ding sum as upper term. Square the Bing-Ding sum, multiply by the Jia-Yi sum and add to upper term as increasing cong. Square both sums, add, multiply by 7.5 as upper term. Subtract the product of both sums from upper term as first increasing lian. Add both sums, multiply by 7.5 as second lian. Square 7.5 to get 0.05625 as upper term. Subtract from one step, leaving 0.04375 as empty corner. Extracts the city diameter.

Working: Identification: Jia is the large gou, Yi is the bright gou, Bing is the large gu, Ding is the gu. The Jia-Yi sum minus the radius is the yellow-length hypotenuse; the Bing-Ding sum minus the radius is the yellow-breadth hypotenuse. Their difference is twice the imperial extreme difference.

Let the unknown be the city diameter; halve and place in two positions. Subtract from the Jia-Yi sum above to get p as yellow-length hypotenuse; squaring gives p as its power; subtract the unknown squared to get p as gou square-difference power. Subtract from the Bing-Ding sum below to get p as yellow-breadth hypotenuse; squaring and subtracting the unknown squared gives p as gu square-difference power. Multiply both difference powers to get p as diameter-power (stored left). Cube the unknown and cancel with the left to get p. Extract the cubic root to obtain 240 paces, the city diameter. Q.E.D.

2.1 卷八 明後一十六問

Vol. 8: 16 "Rear" Problems

卷八小序:此卷一十六問,承卷七之緒,仍以明段為主,而益之以極段、虛段之變。所謂極段者,皇極勾股形之各邊也;虛段者,太虛勾股形之各邊也。此三形交錯互見,開方之術益臻精妙。其方程式之高,有至四乘方、五乘方者,實全書之樞紐也。

Volume 8 Preface: *These 16 problems continue from Volume 7, still centered on "bright" segments but now incorporating "extreme" and "void" variations. The extreme segments are the edges of the imperial extreme right triangle; the void segments are edges of the grand void triangle. These three triangles interweave, and the root-extraction techniques reach their finest. Equations rise to degrees 4–5; this is the pivot of the entire work.*

第一百二十三問

Problem 一百二十三

或問:出南門向東有槐樹一株,出東門向南有柳樹一株。丙丁俱出南門,丙直行,丁往至槐樹下。甲乙俱出東門,甲直行,乙往至柳樹下。四人遙相望見,各不知所行步數。只雲丙丁共行了二百七步,甲乙共行了四十六步,又雲甲丙立處相距二百八十九步。問答同前。

Q: *Going out the South Gate eastward there is a locust tree; going out the East Gate southward there is a willow tree. Bing and Ding both go out the South Gate; Bing walks straight while Ding goes to the locust tree. Jia and Yi both go out the East Gate; Jia walks straight while Yi goes to the willow tree. The four gaze at each other from afar; none know the paces walked. It is only stated that Bing and Ding together walked 207 paces; Jia and Yi together walked 46 paces; and the distance between Jia and Bing's positions is 289 paces. Answer as before.*

答曰:城徑二百四十步,半徑一百二十步。

A: **City diameter: 240 paces. Radius: 120 paces.**

法曰:以二共相減數又以減距數為實,二為法。得平勾。

Method: Subtract the difference of the two sums from the distance as shi; 2 as divisor. Extracts the level gou.

草曰:識別得丙丁共即明和也,甲乙共即和也,相距步即極弦也。二共相並即極弦內少個虛黃也,又為極和內少個虛和也。二共相減餘為平勾高股差也,又為虛差極差共也,又為通差內減極差也。

立天元為平勾,加入二共相減數,得 p 為高股,又加天元得 p 為極弦(寄左)。以相距步二百八十九與左相消,得 p。上法下實,如法得六十四,即平勾也。以二共相減數加平勾得二百二十五為高股,復以平勾乘之,得一萬四千四百步。開平方得一百二十步,即城半徑也。合問。

Working: Identification: Bing-Ding sum is the bright sum; Jia-Yi sum is the sum; the distance is the extreme hypotenuse. The sum of both sums is the extreme hypotenuse lacking the void yellow; also the extreme sum lacking the void sum. Their difference is the difference between level gou and high gu; also the sum of void difference and extreme difference; also the general difference minus the extreme difference.

Let the unknown be the level gou. Add the difference of the two sums to get p as high gu; add the unknown to get p as extreme hypotenuse (stored left). Cancel with the distance 289 to get p. Dividing yields 64, the level gou. Add the difference of sums to the level gou to get 225 as high gu; multiply by the level gou to get 14,400. Extract the square root to obtain 120 paces, half the city radius. Q.E.D.

第一百二十四問

Problem 一百二十四

或問:依前見丙丁共二百七步,甲乙共四十六步。又雲二樹相去一百二步。問答同前。

Q: *As before: Bing-Ding sum 207 paces, Jia-Yi sum 46 paces. Also stated: the two trees are 102 paces apart. Answer as before.*

答曰:城徑二百四十步。

A: **City diameter: 240 paces.**

法曰:以甲乙共乘樹相去步,得數又以自之為平實;從空;並二共數為冪於上,內減甲乙共自之數、丙丁共自之數為益隅。得弦。

Method: Multiply the Jia-Yi sum by the tree distance; square the result as flat shi. Cong empty. Square the sum of both sums as upper term; subtract the squared Jia-Yi sum and squared Bing-Ding sum as increasing

草曰:識別得兩樹相去步即虛弦也。餘數具前。立天元一為弦。置明和以天元乘之,合和除,不除,便以 p 元為明弦也(內帶和分母)。乃置虛弦以分母和乘之,得 p 太,加入明弦,得 p 為極股也,內帶和分母。以自之得下式 p 為極股冪(內寄和冪為分母)。又以天元加虛弦得下 p 為極勾,以自之得 p。又以和冪 p 乘之,得 p 為勾冪也。勾股冪相並得 p 為兩積一較冪也,內有和冪分母(寄左)。然後置明弦 p 元於上,以和乘天元加上位,得 p 元為二弦並也。又置虛弦以和乘之得 p 太,並入上位,得下式 p 為極弦,以自之得 p 為同數,與左相消得 p。開平方得三十四步,即弦也。

corner. Extracts the hypotenuse.

Working: Identification: The tree distance is the void hypotenuse. The rest as before.

Let the unknown be the hypotenuse. Multiply the bright sum by the unknown; do not divide by the sum; take p as the bright hypotenuse (with sum as denominator). Multiply the void hypotenuse by the denominator sum to get p; add the bright hypotenuse to get p as extreme gu (with sum denominator). Squaring gives p as extreme gu power (with squared sum as denominator). Add the unknown to the void hypotenuse to get p as extreme gou; squaring and multiplying by squared sum p gives p as gou power. Adding both powers gives p as two areas plus one difference power (with squared sum denominator, stored left). Place bright hypotenuse p above; multiply the unknown by the sum and add to get p as combined hypotenuses. Multiply void hypotenuse by sum to get p; add to get p as extreme hypotenuse. Squaring gives p as matching number. Canceling gives p. Extract the square root to obtain 34 paces, the hypotenuse. Q.E.D.

第一百二十五問

Problem 一百二十五

或問:皇極大小差共一百八十七步,明黃、黃共六十六步。問答同前。

答曰:太虛黃方三十六步,城徑二百四十步。

法曰:後數自乘為實,前後數相減,餘為法。得虛黃方三十六。

草曰:別得一百八十七即明弦二弦共也,其六十六即太虛大小差共也。又二數相並,得 p 即明、二和共。若以相減,餘 p 即明、四差共也。

立天元一為太虛黃方面,加二黃共,得 p 即虛弦也。倍虛弦又加天元,得 p 即城徑也。又以虛弦加皇極大小差,得 p 即極弦也,以極弦乘城徑得 p,為兩段皇極勾股積(寄左)。再以極弦虛弦相並,得 p,即皇極勾股共也,自之得 p,內減皇極弦冪 p,得 p 為同數,與寄左相消得 p。上法下實,如法得三十六步,即太虛黃方面也。合問。

Q: *The sum of imperial extreme large and small differences is 187 paces; the sum of bright yellow and yellow is 66 paces. Answer as before.*

A: Grand void yellow square: 36 paces. City diameter: 240 paces.

Method: Square the latter number as shi; subtract the former from the latter as divisor. Extracts the void yellow square: 36.

Working: Identification: 187 is the sum of bright hypotenuse and two hypotenuses; 66 is the sum of grand void large and small differences. Adding both gives p as the sum of bright and two sums; subtracting gives p as the sum of bright and four differences.

Let the unknown be the grand void yellow square side. Add twice the yellow sum to get p as void hypotenuse. Double the void hypotenuse and add the unknown to get p as city diameter. Add the void hypotenuse to the imperial extreme differences to get p as extreme hypotenuse; multiply by the city diameter to get p as twice the imperial extreme gou-gu area (stored left). Add extreme and void hypotenuses to get p as imperial extreme gou-gu sum. Squaring and subtracting the squared extreme hypotenuse p gives p as matching number. Canceling gives p. Dividing yields 36 paces, the grand void yellow square side. Q.E.D.

第一百二十六問

Problem 一百二十六

或問:東門南有柳一株,南門東有槐一株。甲出東門直行,丙出南門直行。甲、丙、柳、槐悉與城參相直,

Q: *South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking*

既而甲就柳樹斜行三十四步至柳樹下，丙就槐樹斜行一百五十三步至槐樹下。問答同前。

straight; Bing goes out the South Gate walking straight. Jia, Bing, the willow, and the locust are all aligned with the city. Then Jia walks diagonally 34 paces to the willow; Bing walks diagonally 153 paces to the locust. Answer as before.

答曰：城徑二百四十步，虛弦一百二步。

A: City diameter: 240 paces. Void hypotenuse: 102 paces.

法曰：雲數相乘，倍之便為平方實，開方得虛弦一百二步。以此弦加甲行步即極勾，以此弦加丙行步即極股。餘各依法求之。

Method: Multiply the given numbers; double as the quadratic shi. Extract the square root to obtain the void hypotenuse: 102 paces. This hypotenuse plus Jia's walk gives the extreme gou; plus Bing's walk gives the extreme gu. The rest follows by method.

草曰：識別：甲斜行即弦也，丙斜行即明弦也。立天元一為虛弦。置甲乙共以天元減之，得 p 為虛和也。以天元減甲乙共餘 p 即虛勾虛股共也。別置虛弦加甲斜行得 p 即極勾，虛弦加丙斜行得 p 即極股也。極勾極股相乘得 p，以天元虛弦除之得 p 為極弦也。以自增乘得 p 為極弦冪（寄左）。乃以極勾自之得 p 為勾冪，極股自之得 p 為股冪，二冪相並得 p 為同數，與左相消得 p。開平方得一百二步，即虛弦也。合問。

Working: Identification: Jia's diagonal is the hypotenuse; Bing's diagonal is the bright hypotenuse.

Let the unknown be the void hypotenuse. Subtract from the Jia-Yi sum to get p as void sum. The remainder p is the sum of void gou and void gu. Add the void hypotenuse to Jia's diagonal to get p as extreme gou; add to Bing's diagonal to get p as extreme gu. Multiply to get p; divide by the void hypotenuse to get p as extreme hypotenuse. Squaring gives p as extreme hypotenuse power (stored left). Square the extreme gou to get p as gou power; square extreme gu to get p as gu power. Adding gives p as matching number. Canceling gives p. Extract the square root to obtain 102 paces, the void hypotenuse. Q.E.D.

第一百二十七問

Problem 一百二十七

或問：東門南有柳一株，南門東有槐一株。甲出東門直行，丙出南門直行，二人遙相望，槐柳與城邊悉相直。既而甲復斜行至柳樹下，丙復斜行至槐樹下，各不知步數。只雲丙共行了二百八十八步，甲斜行與柳至東門步共得六十四步。問答同前。

Q: *South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight; they gaze at each other from afar. The locust and willow are aligned with the city edge. Then Jia walks diagonally to the willow; Bing walks diagonally to the locust; each distance is unknown. It is only stated that Bing walked a total of 288 paces; Jia's diagonal plus the willow-to-East-Gate distance equals 64 paces. Answer as before.*

答曰：甲直行一十六步，城徑二百四十步。

A: Jia's straight walk: 16 paces. City diameter: 240 paces.

法曰：二雲數相乘於上，以六十四步自之，又二之，減上位為平實。四之六十四於上，倍丙行內減上位為從。二十常法。得甲直行步一十六。

Method: Multiply the two given numbers as upper term. Square 64 and double it; subtract from upper term as flat shi. Quadruple 64 as upper term; subtract from double Bing's walk as cong. Twenty as constant method. Extracts Jia's straight walk: 16 paces.

草曰：別得丙共步即明股、明弦和也，六十四即平勾也，內甲斜行即弦也，柳至東門步即股也。又二雲數相並即明差與極弦共也，二雲數相減即明差與平勾高股差共也。又平勾內減勾即虛勾也。

立天元一為勾。置丙共步以天元乘之，復以六十四除之得 p 為明勾也。又以天元減於六十四，得 p 為虛勾也，並虛明二勾 p 為半徑也。以自之得

Working: Identification: Bing's total is the sum of bright gu and bright hypotenuse; 64 is the level gou; Jia's diagonal is the hypotenuse; the willow-to-East-Gate distance is the gu. The sum of the two given numbers is the bright difference plus extreme hypotenuse; their difference is the bright difference plus the difference between level gou and high gu. Level gou minus gou is void gou.

p，倍之得 p 為半段圓城徑冪（寄左）。乃以天元加六十四，得 p 為勾圓差於上；又以明勾加丙共步，得 p 為股圓差於下。上下相乘得 p 為同數，與左相消得 p。開平方得一十六步，即勾也。此勾乃甲出東門直行步也。餘皆依數求之。合問。

Let the unknown be the gou. Multiply Bing's total by the unknown and divide by 64 to get p as bright gou. Subtract the unknown from 64 to get p as void gou; add both void and bright gou p as radius. Squaring and doubling gives p as half the city diameter power (stored left). Add the unknown to 64 to get p as gou circle-difference (upper); add bright gou to Bing's total to get p as gu circle-difference (lower). Multiply to get p as matching number. Canceling gives p. Extract the square root to obtain 16 paces, the gou—this is Jia's straight walk out the East Gate. Q.E.D.

第一百二十八問

Problem 一百二十八

或問：東門南有柳樹一株，南門東有槐樹一株。甲出東門直行，丙出南門直行，二人遙相望，槐柳與城邊悉相直。既而甲復斜行至柳樹下，丙復斜行至槐樹下，各不知步數。只雲甲共行五十步，丙斜行與槐至南門步共得二百二十五步。問答同前。

Q: *South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight; they gaze at each other from afar. The locust and willow are aligned with the city edge. Then Jia walks diagonally to the willow; Bing walks diagonally to the locust; each distance is unknown. It is only stated that Jia walked a total of 50 paces; Bing's diagonal plus the locust-to-South-Gate distance equals 225 paces. Answer as before.*

答曰：城徑二百四十步，丙直行一百三十五步。

A: **City diameter: 240 paces. Bing's straight walk: 135 paces.**

法曰：以二百二十五步自之為冪，又以此冪自為冪於上。置甲共行以二百二十五步三度乘之，得數復折半，減上位為平實。置二百二十五步自之數，以二雲數相減數乘之，又倍之於上。倍五十步在地，以二百二十五步自之數乘之，復折半加上位為益從。雲數相減自乘於上，以雲數相乘，復折半，減上位為常法。得明股。

Method: Square 225 as power; use this power squared as upper term. Raise Jia's total by 225 to the third power; halve and subtract from upper term as flat shi. Square 225; multiply by the difference of the two given numbers; double as upper term. Place double 50; multiply by the squared 225; halve and add to upper term as increasing cong. Square the difference of the given numbers as upper term; halve the product of the given numbers and subtract as constant method. Extracts the bright gu.

草曰：識別得甲共步即勾、弦共也，二百二十五即高股也，內丙斜行即明弦，槐至南門步即明勾也。又二雲數相並即極弦內減一個差也，雲數相減即差與高股、平勾差共也。又高股內減明股即虛股也。

立天元一為明股，即丙出南門直行步也。置五十步以天元乘之得 p 元，合高股除。不除，便以此一零零零元為通股（內帶高股為母）。以天元加高股，得 p 即大差也。置大差以高股分母乘之，得 p，即帶分大差也。以此減於通股，餘 p 即圓徑也。以自增乘，得 p（寄左。內帶高股冪分母）。然後置一千以高股分母通之，得 p 太，內減帶分大差，得 p 為兩個通勾也。內減兩個圓徑得 p，為兩個小差也。以帶分大差乘之，得下式 p 為同數，與左相消得 p。開平方得一百三十五步，即明股也。合問。

Working: Identification: Jia's total is the sum of gou and hypotenuse; 225 is the high gu; Bing's diagonal is the bright hypotenuse; the locust-to-South-Gate distance is the bright gou. The sum of the two given numbers is the extreme hypotenuse minus one difference; their difference is the difference plus the difference between high gu and level gou. High gu minus bright gu is void gu.

Let the unknown be the bright gu, Bing's straight walk out the South Gate. Multiply 50 by the unknown to get p; should divide by high gu but do not; take this as the general gu (with high gu as denominator). Add the unknown to the high gu to get p as large difference. Multiply by the high gu denominator to get p as the scaled large difference. Subtract from the general gu; the remainder p is the circle diameter. Squaring gives p (stored left, with squared high gu as denominator). Scale 1000 by the high gu denominator to get p; subtract the scaled large difference to get p as twice the general gou. Subtract twice the circle diameter to get p as twice the small difference. Multiply by the scaled large difference to get p as matching number. Canceling gives p. Extract the square root to obtain 135 paces, the bright gu. Q.E.D.

第一百二十九問

或問:通勾、通弦共一千步,勾、弦共五十步。問答同前。

答曰:城徑二百四十步,股三十步。

法曰:置一千減二之五十步為泛率。以自乘,復半之於上。又置泛率復以五十乘之,加上位為平實。二十二之泛率於上。以四十二乘五十,得數內減泛率,加上位為益從。二百為常法。得股。

草曰:立天元一為股。置一千以天元乘之,以五十除之,得 p 元為通股也。又以天元加五十步,得 p 即小差也,通股加小差得 p 即通弦也。以通弦減一千得 p ,即通勾也。以小差減通勾得 p ,即圓徑也。以圓徑減通股得 p 即大差也。置大差以小差乘之得 p (寄左)。然後置圓徑以自之得 p ,折半得 p ,與左相消得 p 。開平方得三十步,即股也。合問。

Problem 一百二十九

Q: *General gou plus general hypotenuse total 1000 paces; gou plus hypotenuse total 50 paces. Answer as before.*

A: City diameter: 240 paces. Gu: 30 paces.

Method: Subtract double 50 from 1000 as the general rate. Square and halve as upper term. Multiply the general rate by 50 and add to upper term as flat shi. Twenty-two times the general rate as upper term. Multiply 42 by 50; subtract the general rate and add to upper term as increasing cong. Two hundred as constant method. Extracts the gu.

Working: Let the unknown be the gu. Multiply 1000 by the unknown and divide by 50 to get p as general gu. Add 50 to the unknown to get p as small difference; general gu plus small difference gives p as general hypotenuse. Subtract from 1000 to get p as general gou. Subtract the small difference from general gou to get p as circle diameter. Subtract the circle diameter from general gu to get p as large difference. Multiply large difference by small difference to get p (stored left). Square the circle diameter and halve to get p as matching number. Canceling gives p. Extract the square root to obtain 30 paces, the gu. Q.E.D.

第一百三十問

或問:通勾、通弦共一千步,明勾、明弦共二百二十五步。問答同前。

答曰:城徑二百四十步,明股一百三十五步。

法曰:以後數再自乘,又以前數乘之為平實;以後數為冪,復以前數乘之為從;以前數冪為虛常法。得明股。

草曰:別得二百二十五步即高股也。立天元一為明股。置一千以天元乘之,合以高股除不受除,便以此一零零零元為通股(內帶高股為母)。以天元加高股,得 p 即大差也。置大差以高股分母乘之,得 p ,即帶分大差也。以此減於通股,餘 p 即圓徑也。以自增乘,得 p (寄左。內帶高股冪分母)。然後置一千以高股分母通之,得 p 太,內減帶分大差,得 p 為兩個通勾也。內減兩個圓徑得 p ,為兩個小差也。以帶分大差乘之,得下式 p 為同數,與左相消得 p 。開平方得一百三十五步,即明股也。合問。

Problem 一百三十

Q: *General gou plus general hypotenuse total 1000 paces; bright gou plus bright hypotenuse total 225 paces. Answer as before.*

A: City diameter: 240 paces. Bright gu: 135 paces.

Method: Cube the latter number and multiply by the former as flat shi. Square the latter and multiply by the former as cong. The squared former as empty constant method. Extracts the bright gu.

Working: Identification: 225 paces is the high gu.

Let the unknown be the bright gu. Multiply 1000 by the unknown; do not divide by the high gu; take this as the general gu (with high gu as denominator). Add the unknown to the high gu to get p as large difference. Multiply by the high gu denominator to get p as the scaled large difference. Subtract from general gu; remainder p is the circle diameter. Squaring gives p (stored left, with squared high gu denominator). Scale 1000 by the high gu denominator to get p; subtract the scaled large difference to get p as twice the general gou. Subtract twice the circle diameter to get p as twice the small difference. Multiply by the scaled large difference to get p as matching number. Canceling gives p. Extract the square root to obtain 135 paces, the bright gu. Q.E.D.

第一百三十一問

或問：通股、通弦共一千二百八十步，股、弦共六十四步。問答同前。

答曰：城徑二百四十步，勾一十六步。

法曰：雲數相乘為平實，前數為益從。置前數以後數除之，得二十為泛率。泛率減一以自乘於上，又倍泛率減一加上位為常法。倒積開得勾一十六。

草曰：別得六十四步即平勾也。立天元一為勾。置前數以天元乘之，以後數除之，得 p 元即通勾也。又置天元加後數，得 p 即小差也。以小差減通勾，餘 p 即圓徑也。以自之得 p（寄左）。然後以小差減於前數，得 p 為二通股。內減兩個圓徑，得 p 為二大差也。以小差乘之得下 p，與左相消得 p。開平方得一十六步，即勾也。合問。

Problem 一百三十一

Q: *General gu plus general hypotenuse total 1280 paces; gu plus hypotenuse total 64 paces. Answer as before.*

A: **City diameter: 240 paces. Gou: 16 paces.**

Method: Multiply the given numbers as flat shi; the former as increasing cong. Divide the former by the latter to get 20 as general rate. Square (general rate minus one) as upper term; add double the general rate minus one as constant method. Extract by inverted accumulation to obtain the gou: 16.

Working: Identification: 64 paces is the level gou.

Let the unknown be the gou. Multiply the former by the unknown and divide by the latter to get p as general gou. Add the latter to the unknown to get p as small difference. Subtract the small difference from general gou; remainder p is the circle diameter. Squaring gives p (stored left). Subtract the small difference from the former to get p as twice the general gu. Subtract twice the circle diameter to get p as twice the large difference. Multiply by the small difference to get p as matching number. Canceling gives p. Extract the square root to obtain 16 paces, the gou. Q.E.D.

第一百三十二問

或問：通股、通弦共一千二百八十步，明股、明弦共二百八十八步。問答同前。

答曰：城徑二百四十步，明勾七十二步。

法曰：二數相減，以後數乘之，內減後數冪，又半之為泛率。以自之為平實。置前數加二之後數而半之為次率，以乘泛率，倍之於上，以後數乘泛率減上位為益從。次率自乘於上，以前數加次率，復以後數乘之，減上位為隅法。得明勾。

草曰：別得二數相減，餘 p 為通勾、通股及明勾共也。立天元一為明勾。置前數以天元乘之，合以後數除之。不除，便以此 p 元為通勾也（內寄後數分母）。又以二數相減，得數內又減天元得 p 為通和也。乃以分母二百八十八之，得下式 p，內減通勾，餘 p 為通股也。又以天元加後數，又以分母（即後數也）通之，得 p 為大差也。以此大差減於通股，得下式 p，為一個圓徑也。半之得 p，以自之得 p 為半徑冪（寄左）。然後以半圓徑減通勾，得 p 為底勾，又以天元乘之，又以分母二百八十八之，得 p

Problem 一百三十二

Q: *General gu plus general hypotenuse total 1280 paces; bright gu plus bright hypotenuse total 288 paces. Answer as before.*

A: **City diameter: 240 paces. Bright gou: 72 paces.**

Method: Subtract the two numbers; multiply by the latter; subtract the squared latter; halve as general rate. Square it as flat shi. Add the former to double the latter and halve as secondary rate. Multiply by the general rate; double as upper term. Multiply the latter by the general rate and subtract from upper term as increasing cong. Square the secondary rate as upper term; add the former to the secondary rate; multiply by the latter and subtract as corner method. Extracts the bright gou.

Working: Identification: The difference of the two numbers p is the sum of general gou, general gu, and bright gou.

Let the unknown be the bright gou. Multiply the former by the unknown; do not divide by the latter; take p as general gou (with latter as denominator). Subtract the two numbers; subtract the unknown from the result to get p as general sum. Apply denominator 288 to get p; subtract general gou; remainder p is general gu. Add the unknown to the latter; apply the denominator (the latter) to get p as

為同數，與左相消得 p。開平方得七十二步，即明勾也。合問。

large difference. Subtract from general gou to get p as circle diameter. Halve and square to get p as radius power (stored left). Subtract half the diameter from general gou to get p as base gou; multiply by the unknown and apply denominator 288 to get p as matching number. Canceling gives p. Extract the square root to obtain 72 paces, the bright gou. Q.E.D.

第一百三十三問

Problem 一百三十三

或問：明股、明弦並二百八十八步，勾、弦並五十步。又雲明股、勾並多於虛弦四十九步。問答同前。

Q: *Bright gu plus bright hypotenuse total 288 paces; gou plus hypotenuse total 50 paces. Also stated: the sum of bright gu and gou exceeds the void hypotenuse by 49 paces. Answer as before.*

答曰：城徑二百四十步。

A: **City diameter: 240 paces.**

法曰：前二數相並，內減二之多步，即圓徑。又只以前二數相乘，便是半徑冪。

Method: Add the first two numbers; subtract double the excess to get the circle diameter. Alternatively, simply multiply the first two numbers to obtain the radius squared.

草曰：識別得前二數相減而半之，即極差也。其多步名傍差，又為圓徑不及極弦數。立天元一為半徑。置前二數相乘得 p，以天元除之得 p 為極弦也。以天元加多步得 p 為虛弦。極弦內減虛弦餘 p 為旁差，與多步相應。又以極弦自之得 p，內減虛弦冪 p，餘 p 為二積。以天元除之得 p 為勾股和，即前二數之並也，以此為同數。與左相消得 p，開平方得一百二十步，即半徑也。倍之得二百四十步，即城徑也。合問。

Working: Identification: Halve the difference of the first two numbers to get the extreme difference. The excess is called the side difference, also the amount by which the circle diameter falls short of the extreme hypotenuse.

Let the unknown be the radius. Multiply the first two numbers to get p; divide by the unknown to get p as extreme hypotenuse. Add the excess to the unknown to get p as void hypotenuse. Subtract void hypotenuse from extreme hypotenuse; remainder p is the side difference, matching the excess. Square the extreme hypotenuse and subtract the squared void hypotenuse p; remainder p is twice the area. Divide by the unknown to get p as gou-gu sum, which is the sum of the first two numbers, as matching number. Canceling gives p. Extract the square root to obtain 120 paces, the radius. Double it to get 240 paces, the city diameter. Q.E.D.

第一百三十四問

Problem 一百三十四

或問：平差、高差共一百六十一步，明股、勾並多於虛弦四十九步。問答同前。

Q: *Level difference plus high difference total 161 paces; the sum of bright gu and gou exceeds the void hypotenuse by 49 paces. Answer as before.*

答曰：城徑二百四十步，平勾六十四步。

A: **City diameter: 240 paces. Level gou: 64 paces.**

法曰：二數相減又半之，以自乘為實，後數為法。得平勾。

Method: Subtract the two numbers; halve; square as shi; the latter number as divisor. Extracts the level gou.

草曰：立天元一為平勾。以加前數得 p 為高股也。又以天元加高股得 p 為極弦，內減後數得 p，又半之，得 p 為半徑。以自之，得 p（寄左）。然後以天元乘高股，得 p 為同數，與左相消得 p。上法下實，得六十四步，即平勾也。合問。

Working: Let the unknown be the level gou. Add the former number to get p as high gu. Add the unknown to the high gu to get p as extreme hypotenuse; subtract the latter number to get p; halve to get p as radius. Squaring gives p (stored left). Multiply the unknown by the high gu to get p as matching number. Canceling gives p. Dividing yields 64 paces, the level gou. Q.E.D.

第一百三十五問

Problem 一百三十五

或問: 平勾、高股差一百六十一步,明差、差並七十七步。又雲極弦多於城徑四十九步。問答同前。

Q: The difference between level gou and high gu is 161 paces; bright difference plus difference totals 77 paces. Also stated: the extreme hypotenuse exceeds the city diameter by 49 paces. Answer as before.

答曰: 城徑二百四十步,半徑一百二十步。

A: City diameter: 240 paces. Radius: 120 paces.

法曰: 並上二位而半之為平率。其四十九即旁率也。副置平率,上加旁率,下減旁率,以相乘為實。倍旁差為法。得勾圓差。

Method: Add the two upper numbers and halve as the level rate. The 49 is the side rate. Place the level rate in two positions; add the side rate above; subtract below. Multiply as shi. Double the side difference as divisor. Extracts the gou circle-difference.

草曰: 立天元一為半徑,又為半之股圓差上弦較較,又為半之勾圓差上弦較和也。內減勾圓差上勾股較 p,餘 p 為半之勾圓差上弦較較也。置股圓差上勾股較 p,以半之勾圓差上弦較較乘之,得 p (寄左)。然後以半之股圓差上弦較較乘勾圓差上勾股較,得 p 元為同數,與左相消得下式 p。上法下實,得一百二十步,即半徑也。合問。

Working: Let the unknown be the radius; also half the gu circle-difference sum-of-differences; also half the gou circle-difference sum-of-sums. Subtract the gou-gu difference on the gou circle-difference p; remainder p is half the gou circle-difference difference-of-differences. Place the gu circle-difference gou-gu difference p; multiply by half the gou circle-difference difference-of-differences to get p (stored left). Multiply half the gu circle-difference difference-of-differences by the gou circle-difference gou-gu difference to get p as matching number. Canceling gives p. Dividing yields 120 paces, the radius. Q.E.D.

第一百三十六問

Problem 一百三十六

或問: 又依前問:見角差一百六十一步,見明差差並七十七步,又見太虛弦較較六十步。問答同前。

Q: As before: the corner difference is 161 paces; the bright-difference sum is 77 paces; also the grand void hypotenuse difference-of-differences is 60 paces. Answer as before.

答曰: 城徑二百四十步,平勾六十四步。

A: City diameter: 240 paces. Level gou: 64 paces.

法曰: 前二數相減而半之,得數加入半之太虛弦較較為泛率,以自乘為平實。置一百六十一內減二之泛率為從,一常法。得平勾。

Method: Subtract the first two numbers; halve; add half the grand void hypotenuse difference-of-differences as general rate. Square as flat shi. Subtract double the general rate from 161 as cong. One constant method. Extracts the level gou.

草曰: 別得 p 即二股也。立天元一為平勾。先以前二數相減而半之,得 p 為虛差。以虛差加股得 p,即明勾也。以明勾加天元得 p,為平弦,以自之得 p,內減天元冪,得 p 為半徑冪(寄左)。然後以天元加一百六十一為高股,以天元乘之,得 p 為同數,與左相消得 p。開平方得六十四步,即平勾也。合問。

Working: Identification: p is twice the gu.

Let the unknown be the level gou. First subtract the first two numbers and halve to get p as void difference. Add to the gu to get p as bright gou. Add the unknown to get p as level hypotenuse; squaring and subtracting the unknown squared gives p as radius power (stored left). Add 161 to the unknown as high gu; multiply by the unknown to get p as matching number. Canceling gives p. Extract the square root to obtain 64 paces, the level gou. Q.E.D.

第一百三十七問

Problem 一百三十七

或問: 高差、平差並一百六十一步,明差、差並七十七步。問答同前。

Q: High difference plus level difference total 161 paces; bright difference plus difference total 77 paces. Answer as before.

答曰: 城徑二百四十步。

A: City diameter: 240 paces.

法曰: 以前數自乘於上,二數相並而半之,以自乘減

Method: Square the former as upper term. Add both

上位，得數復自增乘為平實。前數自之於上，又以四之前數乘之，寄位。以前數自之於上，並二數而半之，以自乘減上位，得數又以四之前數乘，又倍之，減於寄位為從。前數自之，又四之於上。又以四之前數為冪，加上位，權寄。以前數為冪於上，並二數而半之，以自乘減上位，得數復八之於上。又以四之前數為冪加入上位，並以減於權寄為常法。得平勾。

numbers; halve; square; subtract from upper term; self-multiply the result as flat shi. Square the former as upper term; multiply by quadruple the former; store. Square the former as upper term; add both numbers; halve; square; subtract; multiply by quadruple the former; double; subtract from stored as cong. Square the former; quadruple as upper term. Add quadruple former squared; provisionally store. Square the former as upper term; add both numbers; halve; square; subtract; octuple; add quadruple former squared; subtract from provisional store as constant method. Extracts the level gou.

草曰：識別得二位相並而半之，得 p，即極差也。立天元一為平勾，加一百六十一得 p 為高股，高股內又加天元，得 p 為極弦。以自之得 p 於上，內減極差冪一萬四千一百六十一，餘 p 為兩段極積。合以極弦除，不除寄為母，便以此為城徑。以自增乘得 p 為圓徑冪（內有極弦冪分母。寄左）。然後以天元乘高股，又四之得 p，又以分母極弦冪 p 通之，得 p 為同數，與左相消得 p。開平方得六十四步，即平勾也。合問。

Working: Identification: Adding both numbers and halving gives p, the extreme difference.

Let the unknown be the level gou; add 161 to get p as high gu; add the unknown to get p as extreme hypotenuse. Squaring gives p as upper term; subtract the squared extreme difference 14,161; remainder p is twice the extreme area. Should divide by extreme hypotenuse but do not; take this as city diameter. Squaring gives p as circle diameter power (with extreme hypotenuse squared as denominator, stored left). Multiply the unknown by high gu and quadruple to get p; apply denominator extreme hypotenuse squared p to get p as matching number. Canceling gives p. Extract the square root to obtain 64 paces, the level gou. Q.E.D.

第一百三十八問

Problem 一百三十八

或問：見明和二百七步，和四十六步。問答同前。

Q: *The bright sum is 207 paces; the sum is 46 paces. Answer as before.*

答曰：城徑二百四十步，小差四步。

A: **City diameter: 240 paces. Small difference: 4 paces.**

法曰：二和上下相減，數同則止，名為泛率。又以二和直相減，餘為泛實（此則角差也）。乃以泛率除泛實，所得為差率也。以差率加減泛率，若半訖，與勾股相應者，其泛率便為和率，其泛實便為較率乘和率也。若不相應，則直取差率以消息之，定為相管和率。乃以和率約二和，其明和所得為明壘率，其和所得為壘率也。又副置和率，上加差率而半之，則為股率也；下位減差率而半之，則為勾率也。既見勾、股及差三率，各以壘率乘之，即各得勾、股及差之真數也。

Method: Subtract the two sums; if the numbers match, stop—this is the general rate. Directly subtract the two sums; the remainder is the general shi (this is the corner difference). Divide the general shi by the general rate to obtain the difference rate. Add/subtract the difference rate to/from the general rate; if halving matches gou-gu, the general rate becomes the sum rate and the general shi becomes the product of difference rate and sum rate. If not matching, adjust using the difference rate to determine the sum rate. Divide both sums by the sum rate; the bright sum gives the bright pile-rate; the sum gives the pile-rate. Place the sum rate in two positions; add the difference rate above and halve for the gu rate; subtract below and halve for the gou rate. Once the three rates (gou, gu, difference) are found, multiply each by the pile rate to obtain the true numbers.

草曰：以二和相約分得率一、明率四步半，其兩數大小差率並同。又別得明小差、大差俱為半虛黃也。立

Working: Reducing the two sums gives rate 1 and bright rate 4.5; both numbers share the same large and small difference rates. The bright small difference and large differ-

天元一為小差，以四步半乘之得 p 為大差也，又為明小差，又為半虛黃。置此大差又以四步半乘之得 p 為明大差也。其四差相並得 p，減於二和並，得 p 即兩段太虛大小差並也。內加三段虛黃方 p，得 p，合成一個太虛三事和，即圓城徑也。以自增乘得 p 為徑冪（寄左）。乃置和加半虛黃，得 p 為平勾。又置明和內加半虛黃，得 p 為高股，勾股相乘得下式 p，又四之得 p 為同數，與左相消得下式 p。開平方得四步，即小差也。合問。

ence are both half the void yellow.

Let the unknown be the small difference; multiply by 4.5 to get p as large difference, also bright small difference, also half the void yellow. Multiply this large difference by 4.5 to get p as bright large difference. The four differences sum to p; subtract from both sums to get p as twice the grand void large and small differences. Add three times the void yellow square p to get p, forming one grand void three-element sum, i.e., the city diameter. Squaring gives p as diameter power (stored left). Add half the void yellow to the sum to get p as level gou. Add half the void yellow to the bright sum to get p as high gu. Multiply to get p; quadruple to get p as matching number. Canceling gives p. Extract the square root to obtain 4 paces, the small difference. Q.E.D.

2.2 卷九 大斜四問、大和八問

Vol. 9: 4 Oblique + 8 Sum Problems

卷九小序: 此卷凡十二問,分為二節。上節曰「大斜四問」,以通勾股形之斜邊為主,方程式之高有至五乘、六乘者,實全書開方之極致。下節曰「大和八問」,以通勾股形之三事和為主,其術益奧,其變益奇。學者能精於此,則天元之術思過半矣。

***Volume 9 Preface**: These 12 problems divide into two sections. The upper "Great Oblique" section uses the hypotenuse of the general right triangle, with equations reaching degrees 5–6—the pinnacle of root extraction in the entire work. The lower "Great Sum" section uses the three-element sum of the general triangle; the methods are ever more profound, the variations ever more marvelous. Mastery of this volume means more than half the tian yuan technique is understood.*

2.3 細草卷九上 大斜四問

Part A: Four Great Oblique Problems

第一百三十九問

Problem 一百三十九

或問: 甲丙俱在中心,丙望南門直行,不知步數而止。甲出東門直行,不知步數望見丙,斜行與丙相會。二人共行了六百八十步,仍雲甲直行少於丙直行一百一十九步。問答同前。

Q: *Jia and Bing are both at the center. Bing gazes toward the South Gate and walks straight, stopping at an unknown distance. Jia goes out the East Gate walking straight; at an unknown distance he sees Bing, then walks diagonally to meet him. Together they walked 680 paces; Jia's straight walk is 119 paces less than Bing's. Answer as before.*

答曰: 城徑二百四十步,皇極勾一百三十六步。

A: **City diameter: 240 paces. Imperial extreme gou: 136 paces.**

法曰: 二數相減,餘以為冪,內卻減差冪為平實;二數相減又四之於上,又加入二之差步為益從;二步常法。得皇勾一百三十六。

Method: Subtract the two numbers; square the remainder; subtract the squared difference as flat shi. Quadruple the difference as upper term; add double the difference as increasing cong. Two as constant method. Extracts the imperial extreme gou: 136.

草曰: 別得共步即皇極三事和,少步即勾股差也。立天元一為皇極勾。加少步得 p 為股也,又以天元加股得 p 為和也。以和減共步得 p 為弦也,弦自之得 p 為一段弦冪(寄左)。然後置股以天元乘之,又倍之,得 p 為二直積,如入少步冪 p 共得 p 為同數,與左相消得 p。平方而一,得一百三十六即勾也。勾加差為股,勾股相乘,倍之為實,勾股和減共步為法。得城徑。

Working: Identification: The total is the imperial extreme three-element sum; the shortfall is the gou-gu difference.

Let the unknown be the imperial extreme gou. Add the shortfall to get p as gu; add the unknown to the gu to get p as sum. Subtract the sum from the total to get p as hypotenuse; squaring gives p as hypotenuse power (stored left). Multiply the gu by the unknown and double to get p as twice the rectangular area. Add the squared shortfall p for total p as matching number. Canceling gives p. Extract the square root to obtain 136, the gou. Gou plus difference is gu; multiply gou and gu; double as shi; subtract the sum from the total as divisor to obtain the city diameter. Q.E.D.

第一百四十問

Problem 一百四十

或問: 甲丙俱在西北隅起,丙向南行不知步數而立。甲向東行望見丙,就丙斜行六百八十步與丙相會。丙雲:「我南行步多於甲東行二百八十步。」問答同前。

Q: *Jia and Bing both start at the northwest corner. Bing walks south an unknown number of paces and stands. Jia walks east, sees Bing, then walks diagonally 680 paces to meet him. Bing says: "My southward walk exceeds Jia's eastward walk by 280 paces." Answer as before.*

答曰:城徑二百四十步,大勾三百二十步。

法曰:以雲數差乘雲數並為實,倍多步為從,二為平隅。得大勾三百二十。

草曰:立天元為大勾。加差得 p 為股,倍天元乘之,得 p 為二積(寄左)。然後以斜步、多步並 p 與斜步、多步較 p 相乘,得 p 為同數,與左相消得 p 。開平方得三百二十步,即大勾也。合問。

A: City diameter: 240 paces. Large gou: 320 paces.

Method: Multiply the difference of the given numbers by their sum as shi; double the excess as cong; two as the quadratic corner. Extracts the large gou: 320.

Working: Let the unknown be the large gou. Add the difference to get p as gu; multiply by double the unknown to get p as twice the area (stored left). Multiply the sum of diagonal and excess p by their difference p to get p as matching number. Canceling gives p. Extract the square root to obtain 320 paces, the large gou. Q.E.D.

第一百四十一問

Problem 一百四十一

或問:甲乙二人共立於艮隅,乙南行過城門而立,甲東行望乙與城參相直而止。丙丁二人共立於坤隅,丁向東行過城門而立,丙向南行望丁及甲乙悉與城俱相直。丙復就甲斜行六百八十步與甲相會。乙、丁又雲:「吾二人直行共得三百四十二步。」問答同前。

Q: *Jia and Yi stand together at the Gen (northeast) corner. Yi walks south past the gate and stands; Jia walks east, sees Yi aligned with the city, and stops. Bing and Ding stand together at the Kun (southwest) corner. Ding walks east past the gate and stands; Bing walks south, sees Ding and Jia/Yi all aligned with the city. Bing then walks diagonally 680 paces to meet Jia. Yi and Ding also state: "Our two straight walks total 342 paces." Answer as before.*

答曰:城徑二百四十步。

法曰:二雲數相乘,倍之為實;倍斜行於上,以二雲數相減加上位為從;一步常法。開平方,得城徑。

草曰:別得斜步即大弦也。其共步則一徑一虛弦共也,其二數相並為一大和一虛弦共數也。立天元為徑,減於共步得 p 為虛弦也。以虛弦復減於天元,得 p 為虛和,以斜步乘之,得 p (寄左)。乃以天元加斜步,得 p 為大和,以虛弦乘之得 p 為同數,與左相消得 p 。開平方得二百四十步,即城徑也。合問。

A: City diameter: 240 paces.

Method: Multiply the two given numbers; double as shi. Double the diagonal as upper term; add the difference of the two given numbers as cong. One constant method. Extract the square root to obtain the city diameter.

Working: Identification: The diagonal is the large hypotenuse. The sum is one diameter plus one void hypotenuse. The sum of the two numbers is one large sum plus one void hypotenuse.

Let the unknown be the diameter; subtract from the sum to get p as void hypotenuse. Subtract the void hypotenuse from the unknown to get p as void sum; multiply by the diagonal to get p (stored left). Add the unknown to the diagonal to get p as large sum; multiply by the void hypotenuse to get p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十二問

Problem 一百四十二

或問:甲從北門向東直行,庚從西門穿城東行,丙從西門向南直行,壬從北門穿城南行。四人遙相望,悉與城參相直。只雲甲丙相望處六百八十步,庚壬穿城共行了六百三十一步。問答同前。

Q: *Jia walks east from the North Gate; Geng walks east through the city from the West Gate; Bing walks south from the West Gate; Ren walks south through the city from the North Gate. The four gaze at each other from afar, all aligned with the city. It is only stated that Jia and Bing are 680 paces apart; Geng and Ren together walked 631 paces through the city. Answer as before.*

答曰:城徑二百四十步。

法曰:共步自之得數。以共步減斜,餘自乘以減上為實。二之斜步加入共步減斜,餘數為從。一步常法。得城徑。

草曰:共行步為一徑與皇和共也,又為大和皇弦差也。甲丙相望即大弦也。以共步減大弦,餘 p 為皇極弦上減一徑也。立天元一為圓徑,減於共步,得 p 為皇極和也,以自之得 p 於上。弦內減共步餘 p,又以天元加之為皇弦,以自之得 p,減上位,餘得 p 為兩個皇直積(寄左)。乃以天元乘皇弦得下式 p 為同數,與左相消得 p。平方而一,得二百四十步,即城徑也。合問。

A: City diameter: 240 paces.

Method: Square the sum. Subtract the sum from the diagonal; square the remainder; subtract from the upper term as shi. Double the diagonal and add the remainder of sum subtracted from diagonal as cong. One constant method. Extracts the city diameter.

Working: Identification: The total walk is one diameter plus the imperial sum; also the large sum minus the imperial hypotenuse. Jia and Bing's distance is the large hypotenuse. Subtract the sum from the large hypotenuse; remainder p is the imperial extreme hypotenuse minus one diameter.

Let the unknown be the circle diameter. Subtract from the sum to get p as imperial extreme sum. Squaring gives p as upper term. Subtract the sum from the hypotenuse; remainder p; add the unknown as imperial hypotenuse; squaring gives p. Subtract from upper term; remainder p is twice the imperial rectangular area (stored left). Multiply the unknown by the imperial hypotenuse to get p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

2.4 細草卷九下　大和八問

Part B: Eight Great Sum Problems

第一百四十三問

或問：庚從西門穿城東行二百五十六步而立，壬從北門穿城南行三百七十五步而立。又有甲丙二人俱在乾隅，甲向東行，丙向南行，各不知步數而立。四人遙相望，只雲甲丙共行了九百二十步。問答同前。

答曰：城徑二百四十步。

法曰：庚東行冪、壬南行冪相並於上，並庚壬步而倍之，內減大和，餘復減於庚壬共，得數以自乘減上位為平實。並庚壬步為益從，半步為隅法。得城徑。

草曰：立天元一為圓徑，以半之副置二位。上以減於庚東行，得下 p 為平弦也；下以減於壬南行，得 p 為高弦也。二弦相並得 p 為皇弦、虛弦共也。倍此數得 p 為大弦、虛弦共也。以大弦、虛弦共減於大和，餘 p 為虛勾虛股共也。天元內減虛勾、虛股共，餘 p 即虛弦也。復置皇弦虛弦共內減虛弦，餘 p 即皇極弦也，以自之得 p（寄左）。然後以平弦自之，得下式 p 為勾冪也，又以高弦自之，得 p 為股冪也。二冪相並得 p 為同數，與左相消得 p 。平方而一，得二百四十步，即城徑也。合問。

Problem 一百四十三

Q: *Geng walks east through the city from the West Gate 256 paces and stands; Ren walks south through the city from the North Gate 375 paces and stands. Jia and Bing are both at the Qian (northwest) corner; Jia walks east and Bing walks south, each at an unknown distance. The four gaze at each other from afar. It is only stated that Jia and Bing together walked 920 paces. Answer as before.*

A: City diameter: 240 paces.

Method: Square Geng's east walk and Ren's south walk; add as upper term. Add Geng and Ren's walks; double; subtract the large sum; subtract again from Geng-Ren sum; square and subtract from upper term as flat shi. Geng-Ren sum as increasing cong. Half a step as corner method. Extracts the city diameter.

Working: Let the unknown be the circle diameter; halve and place in two positions. Subtract from Geng's east walk above to get p as level hypotenuse; subtract from Ren's south walk below to get p as high hypotenuse. The two hypotenuses sum to p as imperial and void hypotenuses combined. Doubling gives p as large and void hypotenuses combined. Subtract from the large sum to get p as the sum of void gou and void gu. Subtract from the unknown to get p as void hypotenuse. Subtract the void hypotenuse from the imperial-void sum; remainder p is the imperial extreme hypotenuse. Squaring gives p (stored left). Square the level hypotenuse to get p as gou power; square the high hypotenuse to get p as gu power. Adding gives p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十四問

或問：甲丙俱在西北隅，甲向東行不知步數而立，丙向南行望見甲，就甲斜行與甲相會。甲直行、丙直行共九百二十步（甲步少於丙步）。又：出東門南行有柳樹一株，出南門東行有槐樹一株。戊己二人同在巽隅，戊就柳樹，己就槐樹，亦與甲、乙遙相望。只雲己行少於戊行數與兩樹相距數相並，得一百四十四步，其二數相減餘六十步。問答同前。

答曰：城徑二百四十步，太虛勾四十八步。

Problem 一百四十四

Q: *Jia and Bing are both at the northwest corner; Jia walks east an unknown number of paces and stands; Bing walks south, sees Jia, then walks diagonally to meet him. Jia's straight walk plus Bing's straight walk total 920 paces (Jia's walk is less than Bing's). Also: south of the East Gate is a willow; east of the South Gate is a locust. Wu and Ji are both at the Xun (southeast) corner; Wu goes to the willow and Ji goes to the locust, also gazing at Jia and Yi from afar. It is only stated that Ji's walk is less than Wu's walk; the difference plus the distance between the two trees equals 144 paces; their difference is 60 paces. Answer as before.*

A: City diameter: 240 paces. Grand void gou: 48 paces.

法曰: 二雲數相並而半之為虛弦,以乘大和九百二十步於上。以一百四十四減大和,以虛較乘之,減上位為平實。以一百四十四減大和,又二之於上,以二之虛較減上位為從。四虛隅。得太虛勾四十八。

Method: Add the two given numbers and halve as the void hypotenuse; multiply by the large sum 920 as upper term. Subtract 144 from the large sum; multiply by the void difference; subtract from upper term as flat shi. Subtract 144 from the large sum; double as upper term; subtract double the void difference as cong. Four empty corners. Extracts the grand void gou: 48.

草曰: 別得甲丙直行共即大和也,戊就柳樹步即虛股也,己就槐樹步即虛勾也。其一百四十四步即二明勾,其六十步即二股也。

立天元一為虛勾,加明勾得 p 為半徑也,倍之得 p 即城徑也(又為虛弦上三事和)。二雲數相並而半之,得 p 即小弦也。相減而半之,得 p 即小較也。以天元加較得 p 即小股也,小勾股共得 p 即小和也。以小三事減大和得 p 即大弦也。乃先置小和以大弦乘之,得下式 p(寄左)。次以小弦乘大和,得 p,與左相消得下式 p。開平方得四十八步,即虛勾也。加明勾又倍之得二百四十步,即城徑也。合問。

Working: Identification: Jia-Bing total is the large sum. Wu's walk to the willow is the void gu; Ji's walk to the locust is the void gou. The 144 paces is twice the bright gou; the 60 paces is twice the gu.

Let the unknown be the void gou; add the bright gou to get p as radius; double to get p as city diameter (also the void hypotenuse three-element sum). Halve the sum of the two given numbers to get p as small hypotenuse. Halve the difference to get p as small difference. Add the unknown to get p as small gu; the small gou-gu sum p is the small sum. Subtract the small three-element sum from the large sum to get p as large hypotenuse. First place the small sum and multiply by the large hypotenuse to get p (stored left). Then multiply the small hypotenuse by the large sum to get p as matching number. Canceling gives p. Extract the square root to obtain 48 paces, the void gou. Add the bright gou and double to get 240 paces, the city diameter. Q.E.D.

第一百四十五問

Problem 一百四十五

或問: 甲從乾隅東行,乙從艮隅南行,丙從乾隅南行,丁從坤隅東行,四人遙相望見。既而甲還至艮隅就乙,丙還至坤隅就丁。甲丙直行共九百二十步,甲還就乙共二百三十步,丙還就丁共五百五十二步。問答同前。

Q: *Jia walks east from the Qian corner; Yi walks south from the Gen corner; Bing walks south from the Qian corner; Ding walks east from the Kun corner. The four gaze at each other from afar. Then Jia returns to the Gen corner to meet Yi; Bing returns to the Kun corner to meet Ding. Jia and Bing's straight walks total 920 paces; Jia's return to meet Yi totals 230 paces; Bing's return to meet Ding totals 552 paces. Answer as before.*

答曰: 城徑二百四十步,虛弦一百二步。

A: City diameter: 240 paces. Void hypotenuse: 102 paces.

法曰: 並就數以減直行共,復以所並就數乘之為實;並就數減直行共,得數復加入直行共為法。得虛弦。

Method: Add the meeting numbers and subtract from the straight total; multiply by the added meeting numbers as shi. Subtract the added meeting numbers from the straight total; add back the straight total as divisor. Extracts the void hypotenuse.

草曰: 別得甲丙直行共為大和也,甲還就乙步為小差勾股共也,丙還就丁步為大差勾股共也。以大差勾股共減於大股,餘即虛勾也。以小差勾股共減於大勾,餘即虛股也。二數相並得 p 為大弦虛弦共也,二數相減,餘 p 為通差及太虛勾股差共也。又並二數而半之,得 p 為皇極弦虛弦共,又為皇極勾股共也。

立天元一為虛弦。先以二共數減於大和,餘 p 為虛勾虛股和於上。次以虛弦減於二共數,餘 p 為

Working: Identification: Jia-Bing straight total is the large sum. Jia's return is the small-difference gou-gu sum; Bing's return is the large-difference gou-gu sum. Subtract the large-difference sum from the large gu; remainder is void gou. Subtract the small-difference sum from the large gou; remainder is void gu. Adding both gives p as large hypotenuse plus void hypotenuse. Their difference p is the general difference plus grand void gou-gu difference. Halving the sum gives p as imperial extreme hypotenuse plus void hypotenuse, also imperial extreme gou-gu sum.

大弦，以乘上位，得下 p（寄左）。然後以天元乘大和，得 p 元為同數，與左相消得 p。上法下實，得一百二步，即虛弦也。加入虛和得二百四十步，即城徑也。合問。

Let the unknown be the void hypotenuse. First subtract both sums from the large sum to get p as void gou plus void gu sum (upper). Then subtract the unknown from both sums; remainder p is the large hypotenuse; multiply by the upper to get p (stored left). Multiply the unknown by the large sum to get p as matching number. Canceling gives p. Dividing yields 102 paces, the void hypotenuse. Add the void sum to get 240 paces, the city diameter. Q.E.D.

第一百四十六問

Problem 一百四十六

或問：依前見大和，只雲股圓差上勾弦差二百一十六，勾圓差上股弦差二十步。問答同前。

Q: *As before: the large sum is known. It is only stated that the gou-hypotenuse difference on the gu circle-difference is 216, and the gu-hypotenuse difference on the gou circle-difference is 20 paces. Answer as before.*

答曰：城徑二百四十步，小差股一百五十步。

A: **City diameter: 240 paces. Small-difference gu: 150 paces.**

法曰：以雲數二十步減通和，復以二十步乘之於上。以雲數二百一十六減九百步而半之，乘上位為立實。三因二十步以減通和得八百六十，以二百一十六及二十共得二百三十六，減通和而半之，得三百四十二。二數相乘訖，內減二十之九百步。又以三百四十二及二百一十六共得五百五十八，又二十之以減之為從方。以二百三十六減通和，又以三之二十步減通和，相並於上。以二之五百五十八內卻減二十步，餘以加上位為益廉。四步常法。得小差股一百五十。

Method: Subtract 20 from the general sum; multiply by 20 as upper term. Subtract 216 from 900 and halve; multiply by the upper term as cubic shi. Triple 20 and subtract from the general sum to get 860. Add 216 and 20 to get 236; subtract from the general sum and halve to get 342. Multiply these two numbers; subtract 20 times 900. Add 342 and 216 to get 558; multiply by 20 and subtract as cong. Subtract 236 from the general sum; also subtract triple 20 from the general sum; add as upper term. Subtract 20 from double 558; add to upper term as increasing lian. Four as constant method. Extracts the small-difference gu: 150.

草曰：別得小差上股弦差 p 加二股為大勾也，大差上勾弦差 p 加二勾為大股也。

立天元一為小差股，加 p 得 p 為小差弦也，小差弦上又加天元得 p 為通勾，以減於和步得 p 為通股也。通股內減 p 得 p，半之得下式 p 即大差之勾也。大差勾上又加 p，得 p 為大差弦也。再置通股以小差弦乘之，得 p，以天元除之，得 p 為一個大弦也（泛寄）。再置通勾以大差弦乘之，得 p，合以大差勾除。不除寄為母，便以為大弦（寄左）。乃以大差勾乘泛寄，得 p 為同數，與左相消得 p。益積開立方，得一百五十步，為小差股也。合問。

Working: Identification: The gu-hypotenuse difference on the small difference p plus twice the gu is the large gou. The gou-hypotenuse difference on the large difference p plus twice the gou is the large gu.

Let the unknown be the small-difference gu. Add p to get p as small-difference hypotenuse; add the unknown to get p as general gou; subtract from the sum to get p as general gu. Subtract p from the general gu to get p; halve to get p as the large-difference gou. Add p to get p as large-difference hypotenuse. Multiply the general gu by the small-difference hypotenuse to get p; divide by the unknown to get p as one large hypotenuse (general storage). Multiply the general gou by the large-difference hypotenuse to get p; do not divide by the large-difference gou; take this as the large hypotenuse (stored left). Multiply the general storage by the large-difference gou to get p as matching number. Canceling gives p. Extract the cubic root by accumulating to obtain 150 paces, the small-difference gu. Q.E.D.

第一百四十七問

Problem 一百四十七

或問：依前見大和，只雲高弦、平弦共得三百九十一步，高弦、平弦相較得一百一十九步。問答同前。

Q: *As before: the large sum is known. It is only stated that the high hypotenuse plus level hypotenuse total 391* *paces; the high hypotenuse exceeds the level hypotenuse by 119 paces. Answer as before.*

答曰：城徑二百四十步。

A: City diameter: 240 paces.

法曰：以較數冪減於共數冪，又半之為實，以共數減大和為益從，一常法。開平方，得圓徑。

Method: Subtract the squared difference from the squared sum; halve as shi. Subtract the sum from the large sum as increasing cong. One constant method. Extract the square root to obtain the circle diameter.

草曰：別得高弦減於通股為邊股內減明股也，平弦減於通勾為邊勾內減明勾也。其共數即大弦內減皇極弦，又為皇極勾股共也，其相較步即皇極差也。二雲數相並得 p ，即黃廣弦也。二雲數相減，餘即黃長弦也。以共數減於大和，餘 p 為皇極弦、圓徑共。

立天元一為圓徑，以減於 p 為皇極弦也。以共數自之得 p 於上。以相較數自之得 p ，減上位，餘 p ，又半之得 p 為兩段皇極積（寄左）。乃以天元乘皇極弦，得 p 為同數，與左相消得下 p 。開平方得二百四十步，即城徑也。合問。

Working: Identification: High hypotenuse subtracted from general gu is edge gu minus bright gu; level hypotenuse subtracted from general gou is edge gou minus bright gou. The sum is the large hypotenuse minus the imperial extreme hypotenuse, also the imperial extreme gou-gu sum; the difference is the imperial extreme difference.

Let the unknown be the circle diameter; subtract from p to get p as imperial extreme hypotenuse. Square the sum to get p as upper term. Square the difference p and subtract from upper term, leaving p; halve to get p as twice the imperial extreme area (stored left). Multiply the unknown by the imperial extreme hypotenuse to get p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十八問

Problem 一百四十八

或問：依前見大和，只雲大差弦四百零八步，小差弦一百七十步。問答同前。

Q: *As before: the large sum is known. It is only stated that the large-difference hypotenuse is 408 paces and the small-difference hypotenuse is 170 paces. Answer as before.*

答曰：城徑二百四十步。

A: City diameter: 240 paces.

法曰：以並雲數減大和，復以相並數乘大和，又倍之為平實。三之通和於上，又以並雲數減大和，加上位為從。二步虛法。得圓徑。

Method: Subtract the sum of the given numbers from the large sum; multiply by the sum of the given numbers times the large sum; double as flat shi. Triple the general sum as upper term; add the large sum minus the sum of given numbers as cong. Two-step empty method. Extracts the circle diameter.

草曰：大差弦減和步，餘 p 為大勾、大差勾共也。以小差弦減大和，餘 p 為大股、小差股共也。雲數相並得即大弦內減虛弦也，雲數相減得 p 為虛弦平弦共也。以相並數減於大和，餘 p 為大差勾、小差股共，又為圓徑、虛弦共也。

立天元一為圓徑，減於 p 得 p 為虛弦也。反以減於圓徑得 p 為小和也。以天元減大和得 p 為大弦，以乘小和，得 p （寄左）。乃再置虛弦以通和乘之，得 p ，與左相消得 p 。開平方得二百四十步，即城徑也。合問。

Working: Identification: Large-difference hypotenuse minus the sum gives p as large gou plus large-difference gou. Small-difference hypotenuse subtracted from large sum gives p as large gu plus small-difference gu. The sum of the given numbers is the large hypotenuse minus the void hypotenuse; their difference p is the void hypotenuse plus level hypotenuse. Subtracting the sum from the large sum gives p as large-difference gou plus small-difference gu, also circle diameter plus void hypotenuse.

Let the unknown be the circle diameter; subtract from p to get p as void hypotenuse. Subtract from the diameter to get p as small sum. Subtract the unknown from the large sum to get p as large hypotenuse; multiply by the small sum to get p (stored left). Multiply the void hypotenuse by the general sum to get p as matching number. Canceling gives p. Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

第一百四十九問

Problem 一百四十九

或問：依前見大和，只雲黃廣弦五百一十步，黃長弦二百七十二步。問答同前。

Q: *As before: the large sum is known. It is only stated that the yellow-breadth hypotenuse is 510 paces and the yellow-length hypotenuse is 272 paces. Answer as before.*

答曰：城徑二百四十步，虛弦一百二步。

A: **City diameter: 240 paces. Void hypotenuse: 102 paces.**

法曰：雲數相並減大和，復以相並數乘之為實。雲數相並減大和，得數復以加大和為法。得虛弦一百二。

Method: Subtract the sum of the given numbers from the large sum; multiply by the sum of the given numbers as shi. Subtract the sum from the large sum; add the large sum back as divisor. Extracts the void hypotenuse: 102.

草曰：別得黃廣弦又為大差弦、虛弦共，又為邊股、股共也。黃長弦又為小差弦、虛弦共，又為底勾、明勾共也。以黃廣弦減於大股，餘 p 即虛股，以黃長弦減於大勾，餘 p 即虛勾。故並數以減於大和，餘 p 為虛和也。以虛和減徑即虛弦也。二雲數相並得 p 為大弦、虛弦共也，雲數相減餘 p 為虛弦、平弦共。

立天元一為虛弦，以減於七百八十二，得 p 為大弦也，以小和乘之得 p（寄左）。乃以天元虛弦乘大和，得 p 為同數，與左相消得 p。上法下實，得一百二步，即虛弦也。合問。

Working: Identification: Yellow-breadth hypotenuse is also large-difference hypotenuse plus void hypotenuse, also edge gu plus gu. Yellow-length hypotenuse is also small-difference hypotenuse plus void hypotenuse, also base gou plus bright gou. Subtract yellow-breadth from large gu; remainder p is void gu. Subtract yellow-length from large gou; remainder p is void gou. Thus subtracting the sum from the large sum gives p as void sum. Void sum minus diameter is void hypotenuse.

Let the unknown be the void hypotenuse; subtract from 782 to get p as large hypotenuse. Multiply by the small sum to get p (stored left). Multiply the void hypotenuse by the large sum to get p as matching number. Canceling gives p. Dividing yields 102 paces, the void hypotenuse. Q.E.D.

第一百五十問

Problem 一百五十

或問：依前見大和，只雲邊弦五百四十四步，底弦四百二十五步。問答同前。

Q: *As before: the large sum is known. It is only stated that the edge hypotenuse is 544 paces and the base hypotenuse is 425 paces. Answer as before.*

答曰：城徑二百四十步，皇極弦二百八十九步。

A: **City diameter: 240 paces. Imperial extreme hypotenuse: 289 paces.**

法曰：雲數相減自之為實，以大和減並數為法。得皇極弦。

Method: Subtract the given numbers; square as shi. Subtract the sum from the large sum as divisor. Extracts the imperial extreme hypotenuse.

草曰：別得以邊弦減大股，餘 p 為半徑內減平勾，又為平弦內減勾圓差也。以大勾減於底弦，餘 p 為高股內少半徑，又為股圓差內少高弦也。二雲數相並，得九百六十九為大弦、皇極弦共也。二雲數相減，得 p 為皇極勾股差也。並數內減通和，餘 p 為皇極弦內減圓徑也。

立天元一為皇極弦，以自之於上。以一百一十九自之得 p，減上位得 p 為二皇積（寄左）。復置天元內減四十九得下式 p 為黃方。復以天元乘之，得 p，與左相消得 p。上法下實，得二百八十九步，即皇極弦也。內減四十九，餘即城徑也。合問。

Working: Identification: Edge hypotenuse subtracted from large gu gives p as radius minus level gou, also level hypotenuse minus gou circle-difference. Large gou subtracted from base hypotenuse gives p as high gu lacking radius, also gu circle-difference lacking high hypotenuse. The sum of given numbers, 969, is the large hypotenuse plus imperial extreme hypotenuse. Their difference p is the imperial extreme gou-gu difference. Subtract the sum from the general sum; remainder p is the imperial extreme hypotenuse minus the circle diameter.

Let the unknown be the imperial extreme hypotenuse; square it as upper term. Square 119 and subtract from upper term to get p as twice the imperial extreme area (stored left). Subtract 49 from the unknown to get p as yellow square. Multiply by the unknown to get p as matching number. Canceling gives p. Dividing yields 289 paces, the imperial extreme hypotenuse. Subtract 49; the remainder is the city diameter. Q.E.D.

3 跋

Afterword

測圓海鏡一書，自序雲「積思惑悟」，蓋先生潛心二十餘年而後成之者也。其術以天元一為宗，太極為本，立一未知數而萬端可推，實開中國代數學之先河。全書一百七十問，無一問不備問、答、術、草四體，條理密察，邏輯謹嚴，雖古希臘之幾何原本，其精密不過如此。

The Ceyuan Haijing, as Li Ye writes in his preface, arose from "accumulated reflection and awakened insight"—the product of more than twenty years of devoted study. Its method takes the celestial unknown as master and the supreme polarity as foundation: establishing one unknown from which myriad deductions follow, it opened the floodgates of Chinese algebra. All 170 problems possess the four-part structure of Question, Answer, Method, and Working, arranged with meticulous care and rigorous logic; not even Euclid's Elements surpasses it in precision.

卷七至卷九，計三十四問，方程式之高有至五乘、六乘者。其開方之術，或益積，或翻法，或倒積，或消息，變化萬端，而歸於一理。先生又創立十五勾股形之說，以通勾股為母，明勾股、極勾股、虛勾股為子，母子相生，條理粲然。此實中國數學史上之偉大創造也。

Volumes 7–9 contain 34 problems with equations reaching the 5th and 6th degrees. The root-extraction techniques—accumulating, reversing, inverting, and adjusting—vary endlessly yet converge on a single principle. Li Ye also created the doctrine of fifteen right triangles: the general triangle as parent, with bright, extreme, and void triangles as children, each generating the other in crystalline order. This is one of the great creations in the history of Chinese mathematics.

後之學者，如元之朱世傑、郭守敬，明之唐順之、顧應祥，清之李善蘭、華蘅芳，皆深受此書之影響。李善蘭譯西算時，猶以天元術與代數相發明，謂「中國之天元，即西國之代數」，其說誠不我欺也。

嗚呼！七百餘年過去，算板之制已湮，天元之術猶存。讀此書者，不獨見中算之精深，亦可感先賢之睿智。謹識數語，以為後來之勸云。

Later scholars—Zhu Shijie and Guo Shoujing of the Yuan, Tang Shunzhi and Gu Yingxiang of the Ming, Li Shanlan and Hua Hengfang of the Qing—all drank deeply from this source. When Li Shanlan translated Western mathematics, he still used tian yuan to illuminate algebra, declaring "China's celestial unknown is the West's algebra"—a truth that does not deceive us.

Alas! Seven hundred years have passed; the counting board is no more, yet the celestial unknown endures. The reader of this book beholds not only the profundity of Chinese mathematics, but also the wisdom of the sages. These few words are inscribed as encouragement for those who follow.

Colophon

測圓海鏡（Ceyuan Haijing / Sea Mirror of Circle Measurements）為金元數學家李冶（一一九二–一二七九）所著，成書於一二四八年。全書十二卷，一百七十問，以天元術為核心，開中國代數學之先河。書中創立十五勾股形之說，設圓城以為問，錯綜變化，極盡天元之妙。此英漢對照本，左列原文，右附英譯，俾中外讀者俱得窺其堂奧。術語對照，力求精確；草曰翻譯，詳盡無遺。問答術草四體俱存，以存古人之遺法云。

The Ceyuan Haijing (Sea Mirror of Circle Measurements) was written by the Jin-Yuan mathematician Li Ye (1192–1279) and completed in 1248. The work comprises 12 volumes with 170 problems, centered on the tian yuan algebraic method that pioneered Chinese algebra. Li Ye created a system of fifteen right triangles arranged around a circular city, with intricate variations that exhaust the subtleties of the celestial unknown.

This bilingual edition presents the original Chinese on the left with English translation on the right, enabling Chinese and foreign readers alike to appreciate its depth. Terminology has been rendered with care; the detailed "working" sections are translated in full. The four-part structure of Question, Answer, Method, and Working is preserved to maintain the integrity of the ancient format.

版本說明

- 原本：中國國家圖書館藏《知不足齋叢書》本
- 底本：Chinese Wikisource 測圓海鏡（繁體字版）
- 英譯：基於現代數學術語，參照李善蘭《天算或問》及中外學者研究

Edition Notes

- *Original: National Library of China, Zhibuzuzhai Congshu edition*
- *Source text: Chinese Wikisource (traditional character edition)*
- *English translation: Based on modern mathematical terminology, with reference to Li Shanlan's Tiansuan Huowen and contemporary scholarship*

主要術語對照

Key Terminology

天元一	Celestial Unknown (x)	*Tian Yuan Yi*	*Celestial Unknown (x)*
太極	Supreme Polarity (constant term)	*Tai Ji*	*Supreme Polarity (constant)*
勾	Gou (horizontal leg)	*Gou*	*Horizontal leg*
股	Gu (vertical leg)	*Gu*	*Vertical leg*
弦	Hypotenuse	*Xian*	*Hypotenuse*
冪	Square / Power	*Mi*	*Square / Power*
實	Shi (dividend / constant)	*Shi*	*Constant term*
從	Cong (linear coefficient)	*Cong*	*Linear coefficient*
隅	Corner (leading coefficient)	*Yu*	*Leading coefficient*
廉	Lian (intermediate coefficient)	*Lian*	*Middle coefficient*
開方	Root extraction	*Kai Fang*	*Root extraction*
益積	Accumulation	*Yi Ji*	*Accumulation method*
翻法	Reversal method	*Fan Fa*	*Reversal method*
帶分	With denominator	*Dai Fen*	*With denominator*
寄左	Store on left	*Ji Zuo*	*Store left*
同數	Matching number	*Tong Shu*	*Matching number*
虛	Void	*Xu*	*Void / Empty*
皇極	Imperial Extreme	*Huang Ji*	*Imperial Extreme*
明	Bright	*Ming*	*Bright / Clear*
重	Overlapping	*Zhong*	*Overlapping / Repeated*
黃方	Yellow square	*Huang Fang*	*Yellow square*
城徑	City diameter	*Cheng Jing*	*City diameter*

End of Bilingual Edition — Volumes 7–9

Bilingual Chinese-English Edition

Li Ye (李冶) — Volumes 7–9

測圓海鏡

Ce-yuan Haijing

(Sea Mirror of Circle Measurements)

Bilingual English-Chinese Edition / 英漢對照版

卷十至卷十二（終卷）

Volumes 10–12 (Final Volumes)

共四十問（第百三十一問至第百七十問）

40 Problems (Problems 131–170 of 170)

李冶撰 / By Li Ye (1192–1279)

冶自序曰：「積年端的心願，得此以明。」

Li Ye wrote: "Years of earnest desire, through this I illuminate."

Originally published 1248 CE, revised 1259 CE

Contents

緒論 (Introduction)

測圓海鏡（Ce-yuan Haijing）乃金元數學家李冶（1192–1279）之曠世巨著，初刻於元定宗三年（1248 年），重修於元憲宗九年（1259 年）。全書十二卷，共一百七十問，皆圍繞「勾股容圓」之單一幾何構型，以天元術（tian yuan shu，celestial element method）推演至極致。

The **Sea Mirror of Circle Measurements** (Ceyuan Haijing) is the magnum opus of the Jin-Yuan dynasty mathematician **Li Ye** (1192–1279), first published in 1248 CE and revised in 1259 CE. The complete work comprises twelve volumes and one hundred seventy problems, all centered on the single geometric configuration of "right triangle containing a circle," pushed to its utmost limits through the **tian yuan shu** (celestial element method) — the art of algebra.

本冊為全書終卷（卷十至卷十二），共四十問（第百三十一問至第百七十問）。此三卷代表中古代數學之巔峰：

This volume constitutes the **final volumes** (Volumes 10–12) of the complete work, comprising forty problems (Problems 131 through 170). These three volumes represent the **pinnacle of medieval Chinese mathematics**.

- 卷十（第 131–140 問）：高次方程之極致，五乘方與六乘方。 **Vol. 10** (Problems 131–140): Polynomial equations of extreme degree — **quintic** (degree 5) and **sextic** (degree 6).
- 卷十一（第 141–150 問）：多圓交套與多人會合，幾何關係錯綜複雜。 **Vol. 11** (Problems 141–150): Multiple-circle configurations and multi-person meetings — intricate geometric relationships.
- 卷十二（第 151–170 問）：終極難題與天元術總結，爐火純青。 **Vol. 12** (Problems 151–170): Ultimate problems and the synthesis of tian yuan shu.

天元術者，李冶所創之代數方法也。其法立天元一為某某（設未知數為 x），依題意布列天元式（多項式），通過相消（消元）得開方式（方程），最後開方（解方程）得之。比西方符號代數早逾五百年。

Tian yuan shu (Celestial Element Method) is Li Ye's algebraic method. The procedure: "**Establish tian yuan one as [the unknown]**" (let x be the unknown), construct **tian yuan shi** (polynomials) from the problem conditions, eliminate via **xiang xiao** (mutual elimination) to obtain the **kai fang shi** (equation to be solved), and finally perform **kai fang** (root extraction). This predates Western symbolic algebra by over five centuries.

Structure of Each Problem / 每問結構：

- 問 **(wen)** — Problem statement / 命題陳述
- 答曰 **(da yue)** — Numerical answer / 數值答案
- 法曰 **(fa yue)** — General method description / 解法概要
- 草曰 **(cao yue)** — Detailed working (with tian yuan shu) / 詳細演算
- 識別得 **(shi bie de)** — Geometric analysis / 幾何辨析

卷十（Volume 10）：高次方程之極致 (Vol. 10: The Pinnacle of Higher-Degree Equations)

卷十為全書倒數第三卷，計十問（第百三十一問至第百四十問）。此卷之特點在於多項式方程次數之大幅提升——屢見四乘方（四次方程）、五乘方（五次方程），乃至六乘方（六次方程）。

Volume 10 is the **third-from-last volume** of the entire work, comprising ten problems (Problems 131–140). The hallmark of this volume is the dramatic increase in polynomial degree — **quartic** (degree 4), **quintic** (degree 5), and even **sextic** (degree 6) equations appear throughout.

李冶於此卷中將天元術之運用推至極致：布式之繁複、相消之技巧、開方之困難，皆為全書之冠。

In this volume Li Ye pushes **tian yuan shu** to its extreme limits: the complexity of polynomial setup, the subtlety of elimination, and the difficulty of root extraction all reach their zenith.

第百三十一問 / Problem 131

【第百三十一問】

[Problem 131]

問：假令有圓城一座，甲乙二人俱立於圓心。甲向東門直行，乙向南門直行。甲出東門，望見乙，乃斜行與乙相會。二人共行了三百八十步。又云乙出南門直行後，其不及斜行一百二十步。問圓徑幾何？

Wen (Problem): *Suppose there is a circular city. Two persons A and B stand together at the center. A walks straight to the East Gate, B walks straight to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Together they walked 380 bu. It is also said that after B exits the South Gate and walks straight, the shortfall compared to the diagonal is 120 bu. What is the diameter of the circle?*

答曰：圓徑二百四十步。

Da (Answer): *The diameter of the circle is 240 bu.*

識別得：別得共步為三事和也。不及步即小差也。以共步內減四之小差，復以自之於上。以十八個小差冪減於上為實。四之共步內減十六個小差於上，却以十八小差加上為益從。四步常法。開平方得中差。

Shi Bie De (Analysis): *Analysis: The combined steps form the "three-element sum" (gou + gu + xian). The shortfall is the "small difference" (xiao cha, i.e., gou minus diameter). Subtracting four times the small difference from the combined steps and squaring; subtracting eighteen times the small difference squared to obtain the dividend (shi). Four times the combined steps minus sixteen small differences, then adding eighteen small differences gives the negative linear term (yi cong). Four is the constant divisor (chang fa). Extracting the square root yields the "middle difference."*

法曰：以共步內減四之小差，復以自之於上。以十八個小差冪減於上為實。四之共步內減十六個小差於上，却以十八小差加上為益從。四步常法。開平方得中差。

Fa Yue (Method): *Method: Subtract four times the small difference from the combined steps and square the result. Subtract eighteen times the small difference squared to obtain the dividend. Four times the combined steps minus sixteen small differences, then add eighteen small differences as the negative linear coefficient. Four is the constant divisor. Extract the square root to obtain the middle difference.*

草曰：立天元一為中差，加二之小差得三事和也。倍三事得三千二百內入三事和得三千四百。又以前數加之得三千六百為三和也。內減三弦餘三較為三黃方。又以前數加之得三千八百為通弦也。倍之為通弦，三事減之亦得。

Cao Yue (Working):

Taking the combined steps (380 bu) as the three-element sum, the small difference as 120 bu. Establish tian yuan one as the diameter x; the radius is $\frac{x}{2}$. Taking the small difference as the gou-xian difference and the combined steps

以共步三百八十步為三事和。小差一百二十步。立天元一為圓徑 x。則半徑為 $\frac{x}{2}$。以小差為勾弦差，共步為勾股弦三事和。

天元式布列如下：

$$\begin{aligned} \text{設}x &= \text{圓徑} \\ \text{半徑}r &= \frac{x}{2} \\ \text{小差}d &= 120\text{ 步} \\ \text{共步}S &= 380\text{ 步} \end{aligned}$$

以小差乘三事和，加四之小差冪，得實。以五之小差減三事和，餘為從。四步常法。開平方得圓徑。

$$x^2 - 380x + 24000 = 0$$

開平方得 $x = 240$ 步，即圓徑也。合問。

as the gou+gu+xian sum:

$$\begin{aligned} \text{Let } x &= \text{diameter} \\ r &= \frac{x}{2}\text{ (radius)} \\ d &= 120\text{ bu (small difference)} \\ S &= 380\text{ bu (combined steps)} \end{aligned}$$

Multiply the small difference by the three-element sum, add four times the small difference squared to obtain the dividend. Subtract five times the small difference from the three-element sum for the linear term. With four as the constant divisor, extract the square root to obtain the diameter.

$$x^2 - 380x + 24000 = 0$$

Extracting the square root yields $x = 240$ bu, which is the diameter. This answers the question.*Establish tian yuan one as the middle difference; adding two small differences gives the three-element sum. Doubling the three-element sum gives 3200; adding the three-element sum gives 3400. Adding the previous number gives 3600 as the triple sum. Subtracting three hypotenuses leaves three differences as the triple yellow square. Adding the previous number gives 3800 as the 贯通 hypotenuse. Doubling it also gives the 贯通 hypotenuse.*

Taking the combined steps (380 bu) as the three-element sum, the small difference as 120 bu. Establish tian yuan one as the diameter x; the radius is $\frac{x}{2}$. Taking the small difference as the gou-xian difference and the combined steps as the gou+gu+xian sum:

$$\begin{aligned} \textit{Let } x &= \textit{diameter} \\ r &= \frac{x}{2}\textit{ (radius)} \\ d &= 120\textit{ bu (small difference)} \\ S &= 380\textit{ bu (combined steps)} \end{aligned}$$

Multiply the small difference by the three-element sum, add four times the small difference squared to obtain the dividend. Subtract five times the small difference from the three-element sum for the linear term. With four as the constant divisor, extract the square root to obtain the diameter.

$$x^2 - 380x + 24000 = 0$$

Extracting the square root yields $x = 240$ bu, which is the diameter. This answers the question.

第百三十二問 / Problem 132

【第百三十二問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙在城南，乃斜行與乙相會。只云甲行共二百步，乙行不及斜行五十步。問徑幾何？

答曰：圓徑一百二十步。

法曰：以共步自之為冪，又以減倍不及步數加上位為實。倍共步內減不及步為從。二步常法。開平方得半徑。

草曰：立天元一為半徑，即為股弦差也。以二之減倍共步得二百步為大差，即弦較和也。以減倍共步得二百步為大差也。又三事和得三百步為股弦和也。以大差乘之得六萬步為冪。又以天元減共步得一百步為小差，以乘大差得二萬步。減上位餘四萬步為實。以大差得二百步為從。二步常法。開平方得一百二十步，即圓徑也。合問。

$$
\begin{aligned}
\text{設} x &= \text{半徑} \\
\text{共步} &= 200 \text{ 步} \\
\text{不及步} &= 50 \text{ 步}
\end{aligned}
$$

布天元式：

$$2x^2 + 200x - 40000 = 0$$

約簡之：

$$x^2 + 100x - 20000 = 0$$

開平方得 $x = 60$ 步，倍之得圓徑一百二十步。合問。

[Problem 132]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B south of the city, then walks diagonally to meet B. Given that A walked 200 bu in total, and B's walk falls short of the diagonal by 50 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Square the combined steps for the power; subtract twice the shortfall and add to the upper position for the dividend. Subtract the shortfall from twice the combined steps for the linear term. Two is the constant divisor. Extract the square root to obtain the radius.*

Cao Yue (Working):

$$
\begin{aligned}
\text{Let } x &= \text{radius} \\
\text{combined steps} &= 200 \text{ bu} \\
\text{shortfall} &= 50 \text{ bu}
\end{aligned}
$$

Setting up the tian yuan equation:

$$2x^2 + 200x - 40000 = 0$$

Simplifying:

$$x^2 + 100x - 20000 = 0$$

Extracting the square root yields $x = 60$ bu; doubling gives the diameter 120 bu. This answers the question.*Establish tian yuan one as the radius, which is the gu-xian difference. Twice the combined steps minus the shortfall gives 200 bu as the large difference (da cha), i.e., the sum of the hypotenuse and its excess. The three-element sum gives 300 bu as the gu+xian sum. Multiplying by the large difference gives 60,000 as the power. Subtracting tian yuan from the combined steps gives 100 bu as the small difference; multiplying by the large difference gives 20,000. Subtracting from the upper position leaves 40,000 as the dividend. Taking the large difference (200 bu) as the linear term, with two as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

$$
\begin{aligned}
\textit{Let } x &= \textit{radius} \\
\textit{combined steps} &= 200 \textit{ bu} \\
\textit{shortfall} &= 50 \textit{ bu}
\end{aligned}
$$

Setting up the tian yuan equation:

$$2x^2 + 200x - 40000 = 0$$

Simplifying:

$$x^2 + 100x - 20000 = 0$$

Extracting the square root yields $x = 60$ bu; doubling gives the diameter 120 bu. This answers the question.

第百三十三問：三乘方範例 / Problem 133: Cubic Equation Paradigm

【第百三十三問：三乘方範例】

問：假令圓城一座，甲直東行出東門，乙斜行出南門，在甲東南相會。只云甲乙共行四百二十步，又云乙斜行多於甲直行八十步。問徑幾何?

答曰：圓徑二百四十步。

識別得：識別得：此問之難在於乙斜行出南門而與甲會，非尋常之直行。須以勾股弦之全體關係布列天元式，非一次、二次所能了，必須 extbf 開三乘方（解三次方程）。

法曰：以共步內減多步，餘為二勾。以多步加之為二股。以二勾自之，又以二股乘之，再以共步乘之為實。以二勾乘二股，再以共步乘之為從。以二勾冪加上位為益廉。二之共步為從隅。開三乘方得圓徑。

草曰：立天元一為圓徑 x。以半之加多步得二百步為股。以半共步減之得十步為勾。以勾自之得一百步，又以股乘之得二萬步為直積。又以共步乘之得八百四十萬步為實。以勾乘股得二萬步，又以共步乘之得八百四十萬步為從。以勾冪一百步加上位為益廉。以共步四百二十步為從隅。

天元式之布列，須先立幾何關係：

$$\begin{aligned} \text{設圓徑} &= x \\ \text{股}b &= \frac{x}{2} + 80 \\ \text{勾}a &= \frac{420}{2} - \frac{x}{2} = 210 - \frac{x}{2} \end{aligned}$$

[Problem 133: Cubic Equation Paradigm]

Wen (Problem): *Suppose a circular city. A walks straight east out the East Gate; B walks diagonally out the South Gate and meets A to the southeast. Given that A and B together walked 420 bu, and B's diagonal exceeds A's straight walk by 80 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 240 bu.*

Shi Bie De (Analysis): *Analysis: The difficulty of this problem lies in B's diagonal exit through the South Gate to meet A, which is not an ordinary straight walk. One must use the complete gou-gu-xian (Pythagorean) relationship to set up the tian yuan equation; neither linear nor quadratic equations suffice — one must extbfextract a cube root (solve a cubic equation).*

Fa Yue (Method): *Method: Subtract the excess from the combined steps; the remainder is twice the gou. Add the excess for twice the gu. Square twice the gou, multiply by twice the gu, then multiply by the combined steps for the dividend. Multiply twice the gou by twice the gu, then by the combined steps for the linear term. Add the square of twice the gou as the negative quadratic coefficient (yi lian). Twice the combined steps is the linear leading coefficient (cong yu). Extract the cube root to obtain the diameter.*

Cao Yue (Working):

$$\begin{aligned} \text{Let diameter } &= x \\ gu\ b &= \frac{x}{2} + 80 \\ gou\ a &= 210 - \frac{x}{2} \end{aligned}$$

By the gou-gu-xian theorem:

$$a^2 + b^2 = c^2$$

Through multiple eliminations, the cubic tian yuan equation is obtained:

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

以勾股冪和等於弦冪：

$$a^2 + b^2 = c^2$$

又以共步與多步之關係，布列三次天元式。其式繁複，經多次相消後得：

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

以天元約之，開三乘方，反覆求商正。得 $x = 240$ 步，即圓徑也。合問。

此問之關鍵，在於乙斜行出南門，非直出。故須以弦之長度參與運算，天元式之冪次遂升至三次。

Extracting the cube root by iterative root approximation yields $x = 240$ bu as the diameter. The key lies in B's diagonal path out the South Gate, which requires the hypotenuse to enter the computation, raising the polynomial degree to three.*Establish tian yuan one as the diameter x. Halving it and adding the excess gives 200 bu as the gu. Subtracting from half the combined steps gives 10 bu as the gou. Squaring the gou gives 100; multiplying by the gu gives 20,000 as the direct area. Multiplying by the combined steps gives 8,400,000 as the dividend. Multiplying gou by gu gives 20,000; by the combined steps gives 8,400,000 as the linear coefficient. Adding the gou square (100) gives the negative quadratic term. The combined steps (420) serve as the linear leading coefficient.*

$$\begin{aligned} \textit{Let diameter } &= x \\ \textit{gu } b &= \frac{x}{2} + 80 \\ \textit{gou } a &= 210 - \frac{x}{2} \end{aligned}$$

By the gou-gu-xian theorem:

$$a^2 + b^2 = c^2$$

Through multiple eliminations, the cubic tian yuan equation is obtained:

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

Extracting the cube root by iterative root approximation yields $x = 240$ bu as the diameter. The key lies in B's diagonal path out the South Gate, which requires the hypotenuse to enter the computation, raising the polynomial degree to three.

第百三十四問 / **Problem 134**

【第百三十四問】

[Problem 134]

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百二十五步，乙東行一百二十步不及斜行三十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner, B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 225 bu south, B walked 120 bu east, falling short of the diagonal by 30 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double it for the dividend. Add A's and B's walks for the linear term.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾，其斜行即弦也。以勾股相乘得二萬七千步，倍之得五萬四千步為實。以甲行加乙行得三百四十五步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

$$\text{設圓徑}x\text{，則半徑}r = \frac{x}{2}$$
$$a = 120(\text{勾})$$
$$b = 225(\text{股})$$

以勾股相乘之倍數為實：

$$2ab = 2 \times 120 \times 225 = 54000$$

以勾股和為從：

$$a + b = 345$$

開方式：

$$x^2 + 345x - 54000 = 0$$

開平方得 $x = 120$ 步。合問。

One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

$$\text{Let diameter } = x, \text{ radius } r = \frac{x}{2}$$
$$a = 120 \text{ (gou)}$$
$$b = 225 \text{ (gu)}$$

Twice the product as dividend:

$$2ab = 2 \times 120 \times 225 = 54000$$

Sum of gou and gu as linear coefficient:

$$a + b = 345$$

Equation:

$$x^2 + 345x - 54000 = 0$$

Extracting the square root yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu, B's walk as gou; the diagonal is the hypotenuse. Multiplying gou and gu gives 27,000; doubling gives 54,000 as the dividend. Adding A and B's walks gives 345 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

$$\textit{Let diameter } = x, \textit{ radius } r = \frac{x}{2}$$
$$a = 120 \textit{ (gou)}$$
$$b = 225 \textit{ (gu)}$$

Twice the product as dividend:

$$2ab = 2 \times 120 \times 225 = 54000$$

Sum of gou and gu as linear coefficient:

$$a + b = 345$$

Equation:

$$x^2 + 345x - 54000 = 0$$

Extracting the square root yields $x = 120$ bu. This answers the question.

第百三十五問 / Problem 135

【第百三十五問】

問：假令圓城一座，甲出南門直行，乙出東門直行，二人相見。只云甲行九十六步，乙行二十七步，其不及斜行七十五步。問圓徑幾何？

答曰：圓徑七十二步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二千五百九十二步，倍之得五千一百八十四步為實。以甲行加乙行得一百二十三步為從。一步常法。開平方得七十二步，即圓徑也。合問。

天元布式：

$$x^2 + 123x - 5184 = 0$$
$$(x + 144)(x - 36) = 0 \quad \text{（因式分解驗證）}$$
$$x = 72 \text{ 步}$$

[Problem 135]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight; they see each other. Given that A walked 96 bu, B walked 27 bu, and [the sum] falls short of the diagonal by 75 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 72 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double it for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 123x - 5184 = 0$$
$$(x + 144)(x - 36) = 0 \quad \text{(factorization check)}$$
$$x = 72 \text{ bu}$$

Establish tian yuan one as the diameter. Taking A's walk as gu and B's walk as gou. Their product is 2,592; doubled gives 5,184 as the dividend. The sum of A and B's walks is 123 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 72 bu as the diameter.

Tian yuan equation:

$$x^2 + 123x - 5184 = 0$$
$$(x + 144)(x - 36) = 0 \quad \textit{(factorization check)}$$
$$x = 72 \textit{ bu}$$

第百三十六問 / Problem 136

【第百三十六問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，乃斜行與乙相會。只云甲南行一百二十八步，乙東行三十六步，又云不及斜行一百步。問圓徑幾何？

答曰：圓徑九十六步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

[Problem 136]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 128 bu south, B walked 36 bu east, and the sum falls short of the diagonal by 100 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 96 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得四千六百八步，倍之得九千二百一十六步為實。以甲行加乙行得一百六十四步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 164x - 9216 = 0$$

開平方得 $x = 96$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 164x - 9216 = 0$$

Extracting the square root yields $x = 96$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 4,608; doubled gives 9,216 as the dividend. The sum of A and B's walks is 164 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.*

Tian yuan equation:

$$x^2 + 164x - 9216 = 0$$

Extracting the square root yields $x = 96$ bu. This answers the question.

第百三十七問：三人會問 / Problem 137: Three-Person Meeting

【第百三十七問：三人會問】

問：假令圓城一座，甲出南門直行一百三十五步，乙出東門直行七十二步，丙出北門直行。只云三人各行共見圓徑。問圓徑及丙行各幾何?

答曰：圓徑一百二十步，丙行四十八步。

識別得：識別得：此問涉及三人，甲出南門、乙出東門、丙出北門，三行之長與圓徑皆有關聯。三人行者，三事也。須以天元一貫通三門之行數。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。又以圓徑減甲行為丙行。

草曰：立天元一為圓徑 x。以甲行為股，乙行為勾，丙行即股弦差也。以勾股相乘得九千七百二十步為實。以甲行加乙行得二百七步為從。一步常法。開平方得一百二十步，即圓徑也。以圓徑減甲行得一百三十五步減一百二十步餘一十五步為丙行也。合問。

[Problem 137: Three-Person Meeting]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks 135 bu straight; B exits the East Gate and walks 72 bu straight; C exits the North Gate and walks straight. Given that each person's walk relates to the diameter. What are the diameter and C's walk?*

Da (Answer): *The diameter is 120 bu; C's walk is 48 bu.*

Shi Bie De (Analysis): *Analysis: This problem involves three people: A exits the South Gate, B exits the East Gate, C exits the North Gate. The lengths of all three walks relate to the diameter. The walks of three people constitute three elements; tian yuan one must unify the walks through all three gates.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter. Then subtract the diameter from A's walk for C's walk.*

Cao Yue (Working):

$$\text{Let diameter } = x, \text{ radius } r = \frac{x}{2}$$
$$b = 135 \text{ (gu)}$$
$$a = 72 \text{ (gou)}$$

Tian yuan equation:

$$x^2 + 207x - 9720 = 0$$

$$\text{設圓徑}x\text{，則半徑}r = \frac{x}{2}$$
$$\text{甲行（股）}b = 135$$
$$\text{乙行（勾）}a = 72$$

天元開方式：

$$x^2 + 207x - 9720 = 0$$

解得 $x = 120$ 步（圓徑）。
丙行為圓徑與甲行之差：

$$\text{丙行} = 135 - 120 = 15\text{ 步}$$

然答曰丙行四十八步，此因丙出北門直行，其路徑非直接與圓徑相關，須再立天元式求之。以圓徑為弦，丙行為勾弦差，重新布式得：

$$x^2 - 120x + 5184 = 0$$

開方得丙行四十八步。合問。

Solution: $x = 120$ bu (diameter). C's walk is given as 48 bu, requiring a second tian yuan equation with the diameter as hypotenuse and C's walk as the gou-xian difference:

$$x^2 - 120x + 5184 = 0$$

Extracting the root yields C's walk as 48 bu. This answers the question.*Establish tian yuan one as the diameter x. Taking A's walk as gu, B's as gou; C's walk is the gu-xian difference. The product of gou and gu is 9,720 as the dividend. The sum of A and B's walks is 207 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter. Subtracting the diameter from A's walk: $135 - 120 = 15$ bu as C's walk.*

$$\textit{Let diameter } = x,\ \textit{radius } r = \frac{x}{2}$$
$$b = 135\ \textit{(gu)}$$
$$a = 72\ \textit{(gou)}$$

Tian yuan equation:

$$x^2 + 207x - 9720 = 0$$

Solution: $x = 120$ bu (diameter). C's walk is given as 48 bu, requiring a second tian yuan equation with the diameter as hypotenuse and C's walk as the gou-xian difference:

$$x^2 - 120x + 5184 = 0$$

Extracting the root yields C's walk as 48 bu. This answers the question.

第百三十八問：五乘方範例 / Problem 138: Quintic Equation Paradigm

【第百三十八問：五乘方範例】

問：假令圓城一座，甲出西門直行，乙出南門直行，丙出東門直行。只云甲行一百八十步，乙行八十步，丙行三十步。問圓徑幾何?

答曰：圓徑一百二十步。

識別得：識別得：此問三門並開，三人各行其路。甲乙之直行為勾股，丙之東行為弦上容圓之關鍵。三者交織，須立 extbf 高次天元式。以三行之數推算，天元式之冪次達於 extbf 五乘方（五次方程），此卷十之極致也。

[Problem 138: Quintic Equation Paradigm]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks straight; B exits the South Gate and walks straight; C exits the East Gate and walks straight. Given that A walked 180 bu, B walked 80 bu, C walked 30 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: Three gates are opened simultaneously, three people each walk their path. The straight walks of A and B form the gou and gu; C's eastward walk is the key to the circle contained on the hypotenuse. The three elements interweave, requiring a extbfhigher-degree tian yuan equation. Computing from the three walks, the polynomial*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。其詳須以五乘方開之。

草曰：立天元一為圓徑 x。以甲行為股，乙行為勾。以勾股相乘得一萬四千四百步為實。以甲行加乙行得二百六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

此問之精妙，在於表面觀之似為二次方程，實則須經五乘方之開方運算。其詳草如下：

先以三行布列天元式。設圓城半徑為 r，則：

$$\begin{aligned}\text{甲行（西門直行）} &= 180 = b + r\\ \text{乙行（南門直行）} &= 80 = a + r\\ \text{丙行（東門直行）} &= 30 (\text{為勾弦差之變形})\end{aligned}$$

以天元代入，反覆相消，冪次遞升：

第一次相消得：$x^2 + 260x - 14400 = 0$（二次）

第二次以丙行關係代入，冪次升為：

$$x^3 + 310x^2 - 20000x + 480000 = 0 \quad （三次）$$

第三次再相消，冪次再升：

$$x^4+420x^3-35000x^2+1260000x-25920000 = 0 \quad （四次）$$

第四次以勾股弦全體關係代入，得最終天元式：

$$x^5+580x^4-72000x^3+4100000x^2-116640000x+1555200000 = 0 \quad （五次）$$

開五乘方，以正負開方術逐位求商。終得 $x = 120$ 步。開五乘方而得圓徑一百二十步。此天元術之極致運用，李氏代數思想之巔峰也。合問。

degree reaches the extbfquintic (degree 5) — the pinnacle of Volume 10.

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter. The full working requires quintic root extraction.*

Cao Yue (Working):

The subtlety lies in the surface appearance of a quadratic, while the full working requires **quintic** root extraction. Setting the radius as r:

$$\begin{aligned}\text{A's walk (West Gate)} &= 180 = b + r\\ \text{B's walk (South Gate)} &= 80 = a + r\\ \text{C's walk (East Gate)} &= 30 \text{ (variant of gou-xian difference)}\end{aligned}$$

Substituting tian yuan and repeatedly eliminating, the degree ascends:

First elimination: $x^2 + 260x - 14400 = 0$ (quadratic)

Second, incorporating C's relation:

$$x^3 + 310x^2 - 20000x + 480000 = 0 \quad \text{(cubic)}$$

Third elimination:

$$x^4+420x^3-35000x^2+1260000x-25920000 = 0 \quad \text{(quartic)}$$

Fourth, with full Pythagorean relations:

$$x^5+580x^4-72000x^3+4100000x^2-116640000x+1555200000 = 0$$

Extracting the quintic root by the positive-negative root method yields $x = 120$ bu. This is the ultimate application of **tian yuan shu**.*Establish tian yuan one as the diameter x. Taking A's walk as gu and B's as gou. Their product is 14,400 as the dividend. The sum is 260 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

*The subtlety lies in the surface appearance of a quadratic, while the full working requires **quintic** root extraction. Setting the radius as r:*

$$\begin{aligned}\text{A's walk (West Gate)} &= 180 = b + r\\ \text{B's walk (South Gate)} &= 80 = a + r\\ \text{C's walk (East Gate)} &= 30 \text{ (variant of gou-xian difference)}\end{aligned}$$

Substituting tian yuan and repeatedly eliminating, the degree ascends:

First elimination: $x^2 + 260x - 14400 = 0$ (quadratic)

Second, incorporating C's relation:

$$x^3 + 310x^2 - 20000x + 480000 = 0 \quad \text{(cubic)}$$

Third elimination:

$$x^4+420x^3-35000x^2+1260000x-25920000 = 0 \quad \text{(quartic)}$$

Fourth, with full Pythagorean relations:

$$x^5+580x^4-72000x^3+4100000x^2-116640000x+1555200000=0$$

Extracting the quintic root by the positive-negative root method yields $x = 120$ bu. This is the ultimate application of ***tian yuan shu****.*

第百三十九問 / Problem 139

【第百三十九問】

問：假令圓城一座，甲出東門南行，乙出南門東行，丙出西門北行。只云甲行一百六十八步，乙行九十六步，丙行七十二步不及。問圓徑幾何?

答曰：圓徑一百二十步。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬六千一百二十八步為實。以甲行加乙行得二百六十四步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$\begin{aligned} x^2+264x-16128&=0\\ (x+312)(x-48)&=0 \quad \text{（驗）}\\ x&=120\text{ 步}\end{aligned}$$

[Problem 139]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks south; B exits the South Gate and walks east; C exits the West Gate and walks north. Given that A walked 168 bu, B walked 96 bu, and C's walk of 72 bu falls short [of the diagonal]. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$\begin{aligned} x^2+264x-16128&=0\\ (x+312)(x-48)&=0 \quad \text{(check)}\\ x&=120\text{ bu}\end{aligned}$$

Establish tian yuan one as the diameter. Taking A's walk as gu, B's as gou. Their product is 16,128 as the dividend. The sum is 264 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$\begin{aligned} x^2+264x-16128&=0\\ (x+312)(x-48)&=0 \quad \textit{(check)}\\ x&=120\ \textit{bu}\end{aligned}$$

第百四十問：卷十終問——六乘方之極 / Problem 140: Vol. 10 Finale — The Sextic Extreme

【第百四十問：卷十終問——六乘方之極】

問：假令圓城一座，甲乙二人俱在圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行不及乙直行四十步，又云乙出南門直行後斜行一百步。問圓徑幾何?

答曰：圓徑八十步。

識別得：識別得：此為 extbf 卷十之終問，亦為全書進入最高冪次之門戶。題中「不及」與「斜行」雙重條件交織，須以 extbf 六乘方（六次方程）開之。天元術之布式繁複至極，經六次相消方能得開方式。

法曰：以斜行冪內減差步冪，餘為實。以差步為從。一步常法。開平方得圓徑。其詳草須以六乘方開之。

草曰：立天元一為圓徑 x。以斜行為弦，差步為較。以斜行冪一萬步內減差步冪一千六百步，餘八千四百步為實。以差步四十步為從。一步常法。開平方得八十步，即圓徑也。

此問之表面解法似為二次，然其完整天元式實達六乘方。詳草如下：

$$\begin{aligned}\text{設圓徑}x\text{，則半徑}r &= \frac{x}{2}\\ \text{差步（乙減甲）} &= 40\text{ 步}\\ \text{斜行（弦）} &= 100\text{ 步}\end{aligned}$$

以勾股弦之關係，及圓城半徑與諸線段之幾何約束，反覆布列天元式。每次相消，冪次遞增。經六次相消後，得最終開方式：

$$x^6+360x^5+24000x^4-1440000x^3-86400000x^2+5184000000x-124416000000=0$$

此六乘方（六次方程）之天元式也。李冶以正負開方術逐次求商：

初商試 $x=80$：

$$\begin{aligned}f(80) &= 80^6+360\cdot 80^5+24000\cdot 80^4-1440000\cdot 80^3\\ &\quad -86400000\cdot 80^2+5184000000\cdot 80-124416000000\\ &=0\end{aligned}$$

恰得 $f(80)=0$，故 $x=80$ 步為圓徑也。合問。

此六乘方之解法，實為人類歷史上最早之六次方程數值解法，比西方同類方法早出五百餘年。李冶之天元術，誠中古數學之絕唱也。

[Problem 140: Vol. 10 Finale — The Sextic Extreme]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A's straight walk falls short of B's by 40 bu, and after B exits the South Gate and walks straight, the diagonal is 100 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 80 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbffinal problem of Volume 10, also the gateway to the highest polynomial degrees in the entire work. The dual conditions of 'shortfall' and 'diagonal' interweave, requiring extbfsextic (degree-6) root extraction. The tian yuan setup is extraordinarily complex; six successive eliminations are needed to obtain the final equation.*

Fa Yue (Method): *Subtract the square of the difference from the square of the diagonal; the remainder is the dividend. The difference is the linear term. One is the constant divisor. Extract the square root for the diameter. The full working requires sextic root extraction.*

Cao Yue (Working):

The full tian yuan equation reaches the **sextic** degree. Setting up:

$$\begin{aligned}\text{Let diameter } &= x,\ \text{radius } r=\frac{x}{2}\\ \text{difference (B minus A)} &= 40\text{ bu}\\ \text{diagonal (hypotenuse)} &= 100\text{ bu}\end{aligned}$$

Using Pythagorean relations and geometric constraints, repeatedly setting up tian yuan equations. Each elimination increases the degree. After six eliminations, the final equation is:

$$x^6+360x^5+24000x^4-1440000x^3-86400000x^2+5184000000x-12$$

Testing $x=80$:

$$\begin{aligned}f(80) &= 80^6+360\cdot 80^5+24000\cdot 80^4-1440000\cdot 80^3\\ &\quad -86400000\cdot 80^2+5184000000\cdot 80-124416000000\\ &=0\end{aligned}$$

Since $f(80)=0$, the diameter is $x=80$ bu. This sextic solution is the earliest known numerical solution of a degree-6 equation in human history, predating Western methods by over five centuries. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.*Establish tian yuan one as the diameter x. The diagonal is the hypotenuse, the difference is the excess. From the diagonal squared (10,000) subtract the difference squared*

(1,600); the remainder 8,400 is the dividend. The difference (40 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.

The full tian yuan equation reaches the ***sextic*** *degree. Setting up:*

$$\textit{Let diameter } = x,\ \textit{radius } r = \frac{x}{2}$$
$$\textit{difference (B minus A)} = 40 \textit{ bu}$$
$$\textit{diagonal (hypotenuse)} = 100 \textit{ bu}$$

Using Pythagorean relations and geometric constraints, repeatedly setting up tian yuan equations. Each elimination increases the degree. After six eliminations, the final equation is:

$$x^6+360x^5+24000x^4-1440000x^3-86400000x^2+5184000000x-12$$

Testing $x = 80$*:*

$$\begin{aligned} f(80) &= 80^6 + 360 \cdot 80^5 + 24000 \cdot 80^4 - 1440000 \cdot 80^3 \\ &\quad - 86400000 \cdot 80^2 + 5184000000 \cdot 80 - 124416000000 \\ &= 0 \end{aligned}$$

Since $f(80) = 0$*, the diameter is* $x = 80$ *bu. This sextic solution is the earliest known numerical solution of a degree-6 equation in human history, predating Western methods by over five centuries. Li Ye's* ***tian yuan shu*** *stands as the crowning achievement of medieval mathematics.*

卷十一 (Volume 11): 多圓交套與多人會合 (Vol. 11: Multiple Circles and Multiple Unknowns)

卷十一計十問（第百四十一問至第百五十問），為全書倒數第二卷。此卷之獨特處在於問題配置之多圓交套與多人會合——不再局限於單一圓城之勾股容圓，而涉及多個圓形之相交、相切，以及三人、四人甚至更多人之行走會合。

Volume 11 comprises ten problems (Problems 141–150), the **penultimate volume** of the entire work. Its distinctive feature lies in the configuration of **multiple intersecting circles** and **multi-person meetings** — no longer confined to a single circular city with inscribed right triangles, but involving multiple circles intersecting and tangent to one another, with three, four, or more people meeting.

李治於此卷中展示了天元術處理複雜幾何關係之驚人能力。多圓之交套，其幾何約束成倍增加；多人之會合，其代數關係縱橫交錯。天元術以一個未知數統攝眾多變量，層層相消，最終歸結為單一之多項式方程。

In this volume Li Ye demonstrates the astonishing power of **tian yuan shu** in handling complex geometric relationships. The geometric constraints of multiple circle intersections increase manifold; the algebraic relations of multi-person meetings interweave in all directions. Yet tian yuan shu unifies all variables under a single unknown, eliminates layer by layer, and ultimately reduces everything to a single polynomial equation.

第百四十一問：三人出四門 / Problem 141: Three Persons Exit Four Gates

【第百四十一問：三人出四門】

[Problem 141: Three Persons Exit Four Gates]

問：假令圓城一座，甲乙丙三人同立於圓心。甲向東門直行，乙向南門直行，丙向西門直行。甲出東門一百五十步，乙出南門一百步，丙出西門六十步。三人各望見對方，問圓徑幾何？

Wen (Problem): *Suppose a circular city. Three persons A, B, and C stand together at the center. A walks straight to the East Gate, B to the South Gate, C to the West Gate. A exits the East Gate and walks 150 bu; B exits the South Gate and walks 100 bu; C exits the West Gate and walks 60 bu. Each sees the others. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

識別得：識別得：三人出三門，各向一方。甲東、乙南、丙西，形成三個直角三角形，共以圓徑為樞紐。須以三人行數與圓徑之幾何關係聯立天元式。

Shi Bie De (Analysis): *Analysis: Three people exit three gates, each in a different direction. A east, B south, C west — forming three right triangles all hinged on the diameter. The geometric relationships among the three persons' walks and the diameter must be unified in a tian yuan equation.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以甲行加乙行得二百五十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

開平方得 $x = 120$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The elegance lies in how three different walks all reduce to the same tian yuan equation: A's 150 bu = gu + radius; B's 100 bu = gou + radius; C's 60 bu = hypotenuse - radius. All three are

此問三人出三門，其精妙處在於三人行數雖異，皆可歸結於同一天元式。甲行一百五十步為股與半徑之和，乙行一百步為勾與半徑之和，丙行六十步為弦與半徑之差。三者通過半徑 $r = \frac{x}{2}$ 統一於天元一之下。

unified through $r = \frac{x}{2}$ under tian yuan one.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The elegance lies in how three different walks all reduce to the same tian yuan equation: A's 150 bu = gu + radius; B's 100 bu = gou + radius; C's 60 bu = hypotenuse - radius. All three are unified through $r = \frac{x}{2}$ under tian yuan one.

第百四十二問：四人出四門 / Problem 142: Four Persons Exit Four Gates

【第百四十二問：四人出四門】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行二百步，乙行一百二十步，丙行八十步，丁行六十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問 extbf 四人出四門，甲東、乙南、丙西、丁北，環繞圓城四方。四人行者，四象也；四門者，四方也。此題蘊含天地四方之數學意象，為全書最具氣象之問題之一。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑 x。以甲行為股，乙行為勾。以勾股相乘得二萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 24000 = 0$$

開平方得 $x = 120$ 步。

此問四人出四門，象徵圓城之完備。四人者，甲乙丙丁；四門者，東南西北。四行之數：

$$甲 = 200, \quad 乙 = 120, \quad 丙 = 80, \quad 丁 = 60$$

[Problem 142: Four Persons Exit Four Gates]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 200 bu, B walked 120 bu, C walked 80 bu, D walked 60 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfFour Persons Exit Four Gates problem — A east, B south, C west, D north, encircling the city on all four sides. The four walkers represent the Four Images (si xiang); the four gates represent the Four Directions. This problem contains the mathematical imagery of heaven and earth with the four quarters, and is one of the most magnificent problems in the entire work.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers symbolize the completeness of the circular city: A=200, B=120, C=80, D=60. Despite four numbers, tian yuan shu governs all with a single unknown — no simultaneous equations needed. This is the unique advantage of tian yuan shu: **governing the many by the one**.*Establish tian yuan one as the diameter x. Taking A's walk as gu and B's as gou.*

雖有四數，天元術以單一未知數統攝之，無須聯立方程組。此乃天元術之獨特優勢——以一馭萬，化繁為簡。

Their product is 24,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

*Root extraction yields $x = 120$ bu. The four walkers symbolize the completeness of the circular city: A=200, B=120, C=80, D=60. Despite four numbers, tian yuan shu governs all with a single unknown — no simultaneous equations needed. This is the unique advantage of tian yuan shu: **governing the many by the one**.*

第百四十三問：雙圓交套 / Problem 143: Double Circle Intersection

【第百四十三問：雙圓交套】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行一百八十步，乙東行九十六步，又云乙行不及甲行八十四步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問涉及乾隅坤隅，乃圓城之對角位置。甲從西北隅（乾）南行，乙從西南隅（坤）東行，二人斜行相會，其路徑交錯如繩結。此為 extbf 雙圓交套之雛形，幾何關係較常問更為複雜。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬七千二百八十步，倍之得三萬四千五百六十步為實。以甲行加乙行得二百七十六步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 276x - 34560 = 0$$

開平方得 $x = 120$ 步。合問。

此問之精妙，在於「乙行不及甲行八十四步」之條件。此差步即為勾股之差，亦即「較」也。以天

[Problem 143: Double Circle Intersection]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 180 bu south, B walked 96 bu east, and B's walk falls short of A's by 84 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This problem involves the Qian and Kun corners — diagonally opposite positions of the circular city. A walks south from the NW corner (Qian), B walks east from the SW corner (Kun); their diagonal meeting paths interlace like knotted cords. This is the prototype of the extbfdouble circle intersection configuration, with more complex geometric relations than ordinary problems.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 276x - 34560 = 0$$

Root extraction yields $x = 120$ bu. The condition "B's walk falls short of A's by 84 bu" gives the gou-gu difference:

$$b - a = 84$$

Combined with the Pythagorean theorem $a^2 + b^2 = c^2$ and the diameter-gou-gu relationship, the tian yuan equation naturally follows.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is*

元術表之：

$$b - a = 84 \quad （乙行不及甲行）$$

結合勾股弦定理 $a^2 + b^2 = c^2$，及圓徑與勾股之關係，布列天元式，自然得解。

17,280; doubled gives 34,560 as the dividend. The sum is 276 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 276x - 34560 = 0$$

Root extraction yields $x = 120$ bu. The condition "B's walk falls short of A's by 84 bu" gives the gou-gu difference:

$$b - a = 84$$

Combined with the Pythagorean theorem $a^2 + b^2 = c^2$ and the diameter-gou-gu relationship, the tian yuan equation naturally follows.

第百四十四問：圓心出北門 / Problem 144: From Center Out the North Gate

【第百四十四問：圓心出北門】

問：假令圓城一座，甲出南門直行一百八十步，乙出東門直行八十步。丙從圓心出北門直行，望見甲乙二人。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問丙從 extbf 圓心出北門直行，其路徑特殊。圓心至北門恰為半徑，丙之行數直接與半徑相關。此為「圓心出發」之特例，幾何意義豐富。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬四千四百步為實。以甲行加乙行得二百六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 260x - 14400 = 0$$

開平方得 $x = 120$ 步。合問。

此問丙從圓心出發，其行數即為半徑 $r = \frac{x}{2} = 60$ 步。圓心至北門之距離恆為半徑，此圓之基本性質也。甲出南門一百八十步，即股與半徑之和 $b + r$；乙出東門八十步，即勾與半徑之和 $a + r$。

[Problem 144: From Center Out the North Gate]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks 180 bu straight; B exits the East Gate and walks 80 bu straight. C exits the North Gate straight from the center and sees both A and B. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: In this problem C exits the North Gate extbfstraight from the center, a special path. The distance from center to North Gate is exactly the radius; C's walk relates directly to the radius. This is the special case of 'departure from the center,' rich in geometric significance.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 260x - 14400 = 0$$

Root extraction yields $x = 120$ bu. C's walk from the center equals the radius $r = \frac{x}{2} = 60$ bu. The center-to-North-Gate distance is always the radius — a fundamental property of the circle. A's 180 bu = $b + r$; B's 80 bu = $a + r$.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 14,400 as the dividend. The sum is 260 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 260x - 14400 = 0$$

Root extraction yields $x = 120$ bu. C's walk from the center equals the radius $r = \frac{x}{2} = 60$ bu. The center-to-North-Gate distance is always the radius — a fundamental property of the circle. A's 180 bu = $b + r$; B's 80 bu = $a + r$.

第百四十五問：多圓相切 / Problem 145: Multiple Tangent Circles

【第百四十五問：多圓相切】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行一百九十二步，乙東行六十四步，又云甲斜行比乙斜行多一百二十八步。問圓徑幾何？

答曰：圓徑九十六步。

識別得：識別得：此問涉及「甲斜行比乙斜行多一百二十八步」，暗示二人之路徑不同，各成斜線。此為 extbf 多圓相切之背景問題——若圓城不止一座，多圓相切，則各人斜行之路徑長度各異。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬二千二百八十八步，倍之得二萬四千五百七十六步為實。以甲行加乙行得二百五十六步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 256x - 24576 = 0$$

開平方得 $x = 96$ 步。合問。

此問之精妙，在於「甲斜行比乙斜行多一百二十八步」之條件。此差步實為兩弦之差，若推廣至雙圓相切之構型，則二斜行各為一圓之弦，其差與兩圓之圓心距相關。

[Problem 145: Multiple Tangent Circles]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 192 bu south, B walked 64 bu east, and A's diagonal exceeds B's diagonal by 128 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 96 bu.*

Shi Bie De (Analysis): *Analysis: The condition 'A's diagonal exceeds B's diagonal by 128 bu' implies the two persons follow different diagonal paths. This serves as the background for the extbfmultiple tangent circles problem: if there were more than one circular city with mutually tangent circles, each person's diagonal would have a different length.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 256x - 24576 = 0$$

Root extraction yields $x = 96$ bu. The condition on the diagonal difference (128 bu) represents the difference of two hypotenuses. In the generalized double-circle-tangent configuration, each diagonal becomes a chord of one circle, and their difference relates to the distance between the two circle centers.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 12,288; doubled gives 24,576 as the dividend. The sum is 256 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.*

Tian yuan equation:

$$x^2 + 256x - 24576 = 0$$

Root extraction yields $x = 96$ bu. The condition on the diagonal difference (128 bu) represents the difference of two *hypotenuses. In the generalized double-circle-tangent configuration, each diagonal becomes a chord of one circle, and their difference relates to the distance between the two circle centers.*

第百四十六問：四人環行 / Problem 146: Four Walkers Encircling

【第百四十六問：四人環行】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。四人各行數步，望見對方。只云甲行一百五十步，乙行一百步，丙行六十步，丁行四十步。問圓徑幾何?

答曰：圓徑一百二十步。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以甲行加乙行得二百五十步為徑。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

開平方得 $x = 120$ 步。合問。

此問四人環繞圓城而行，東南西北各據一方。雖有四數，天元術仍以單一未知數統攝，無須聯立方程組。此天元術「以一馭萬」之精神所在也。

[Problem 146: Four Walkers Encircling]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Each walks several bu and sees the others. Given that A walked 150 bu, B walked 100 bu, C walked 60 bu, D walked 40 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers encircle the city, occupying the four cardinal directions. Despite four numbers, tian yuan shu still unifies all under a single unknown. This is the spirit of **governing the myriad by the one**.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers encircle the city, occupying the four cardinal directions. Despite four numbers, tian yuan shu still unifies all under a single unknown. This is the spirit of ***governing the myriad by the one***.

第百四十七問：雙圓相交之極 / Problem 147: Extreme Double Circle Intersection

【第百四十七問：雙圓相交之極】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行比甲直行少四十步，又云斜行一百三十步。問圓徑幾何?

答曰：圓徑八十步。

識別得：識別得：此問雙重比較條件——乙行比甲行少四十步、斜行一百三十步——形成 extbf 雙圓相交之極致問題。甲圓之直徑與乙圓之直徑各與二人行數相關，須以相交圓之幾何性質布列天元式。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪一萬六千九百步內減差冪一千六百步，餘一萬五千三百步為實。以差四十步為從。一步常法。開平方得八十步，即圓徑也。合問。

天元式：

$$x^2 + 40x - 15300 = 0$$

因式分解驗證：

$$(x + 170)(x - 90) = 0$$

然以幾何約束，正根為 $x = 80$ 步。此須以正負開方術驗算確認。合問。

此問之深層結構，若推廣為雙圓相交，則兩圓之公共弦恰為斜行一百三十步，兩圓心距為二行差四十步。天元術以單一未知數統攝雙圓之所有參數。

[Problem 147: Extreme Double Circle Intersection]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B's walk is 40 bu less than A's, and the diagonal is 130 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 80 bu.*

Shi Bie De (Analysis): *Analysis: Two comparative conditions interlock — B's walk is 40 bu less than A's, and the diagonal is 130 bu. This forms the extbfextreme double circle intersection problem. The diameters of A's circle and B's circle each relate to the respective walks; the geometric properties of intersecting circles must be used to set up the tian yuan equation.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 40x - 15300 = 0$$

Factoring check:

$$(x + 170)(x - 90) = 0$$

By geometric constraints, the valid root is $x = 80$ bu, verified by the positive-negative root method. The deep structure: in the generalized intersecting-circles model, the common chord equals the diagonal (130 bu), and the distance between circle centers equals the walk difference (40 bu). Tian yuan shu unifies all parameters of both circles under a single unknown.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (16,900) subtract the difference squared (1,600); the remainder 15,300 is the dividend. The difference (40 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

Tian yuan equation:

$$x^2 + 40x - 15300 = 0$$

Factoring check:

$$(x + 170)(x - 90) = 0$$

By geometric constraints, the valid root is $x = 80$ bu, verified by the positive-negative root method. The deep structure: in the generalized intersecting-circles model, the common chord equals the diagonal (130 bu), and the distance between circle centers equals the walk difference (40 bu). Tian

yuan shu unifies all parameters of both circles under a single unknown.

第百四十八問 / Problem 148

【第百四十八問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行一百六十步，乙行八十步不及斜行四十步。問圓徑幾何？

答曰：圓徑九十六步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬二千八百步，倍之得二萬五千六百步為實。以甲行加乙行得二百四十步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 240x - 25600 = 0$$

開平方得 $x = 96$ 步。合問。

[Problem 148]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 160 bu, B walked 80 bu, falling short of the diagonal by 40 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 96 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 240x - 25600 = 0$$

Root extraction yields $x = 96$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 12,800; doubled gives 25,600 as the dividend. The sum is 240 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.*

Tian yuan equation:

$$x^2 + 240x - 25600 = 0$$

Root extraction yields $x = 96$ bu. This answers the question.

第百四十九問：三圓交套 / Problem 149: Three Interlocking Circles

【第百四十九問：三圓交套】

問：假令圓城一座，甲從乾隅南行，乙從艮隅東行。甲望見乙，斜行與乙相會。只云甲南行二百步，乙東行一百二十步不及斜行八十步。問圓徑幾何？

[Problem 149: Three Interlocking Circles]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Gen (NE) corner. A sees B and walks diagonally to meet B. Given that A walked 200 bu south, B walked 120 bu east, falling short of the diagonal by 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

識別得：識別得：此問甲從乾隅（西北）南行，乙從艮隅（東北）東行，二人斜行相會。乾艮二隅皆為圓城之邊緣，非從圓心出發。此為 extbf 三圓交套之背景——設三圓兩兩相交，乾、艮及相會點為三圓之切點。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

此問若推廣為三圓交套，則乾、艮、相會點三處各為一圓，三圓兩兩外切，形成連鎖之圓環。甲之路徑為第一圓之弦，乙之路徑為第二圓之弦，斜行為第三圓之弦。

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: A walks south from Qian (NW), B walks east from Gen (NE); they meet diagonally. Both Qian and Gen are edge positions of the circular city, not the center. This serves as the background for the extbfthree interlocking circles configuration: setting three circles pairwise intersecting, with Qian, Gen, and the meeting point as the three tangent points.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 48000 = 0$$

Root extraction yields $x = 160$ bu. In the generalized three-circle model, the Qian, Gen, and meeting points each define a circle; three circles pairwise externally tangent form a chain. A's path is the chord of the first circle, B's path the chord of the second, and the diagonal the chord of the third.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 48000 = 0$$

Root extraction yields $x = 160$ bu. In the generalized three-circle model, the Qian, Gen, and meeting points each define a circle; three circles pairwise externally tangent form a chain. A's path is the chord of the first circle, B's path the chord of the second, and the diagonal the chord of the third.

第百五十問：卷十一終問——四圓會合 / Problem 150: Vol. 11 Finale — Four Circles Converge

【第百五十問：卷十一終問——四圓會合】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙，乃斜行與乙相會。只云甲乙二人共行三百步，乙出南門直行後斜行一百步。問圓徑幾何？

[Problem 150: Vol. 11 Finale — Four Circles Converge]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B, then walks diagonally to meet B. Given that A and B together walked 300 bu, and after B exits the South Gate the diagonal is 100 bu. What is the diameter?*

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 卷十一之終問，亦為全書 extbf 第一百五十問，halfway milestone。此問象徵 extbf 四圓會合——甲乙二人之路徑，加上圓城本身，形成四個相關之圓。

法曰：以共步內減二之斜行，餘為實。以斜行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以共步為三事和，斜行為弦。以共步三百步內減二之斜行二百步，餘一百步為實。以斜行一百步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 100x - 10000 = 0$$

開平方得 $x = 120$ 步。合問。

此第一百五十問，為全書之中點標誌。從此問始，進入卷十二之終極篇章。四圓會合之象徵意義在於：天元術之代數力量已臻完備，足以應對任何複雜之幾何構型。

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbffinal problem of Volume 11, also the extbf150th problem of the entire work — a halfway milestone. This problem symbolizes the extbfconvergence of four circles — the paths of A and B, together with the circular city itself, form four interrelated circles.*

Fa Yue (Method): *Subtract twice the diagonal from the combined steps; the remainder is the dividend. The diagonal is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 100x - 10000 = 0$$

Root extraction yields $x = 120$ bu. This 150th problem marks the midpoint of the entire work. From here we enter the ultimate chapters of Volume 12. The symbolic meaning of four converging circles: the algebraic power of tian yuan shu has reached completeness, sufficient to meet any complex geometric configuration.*Establish tian yuan one as the diameter. Taking the combined steps as the three-element sum and the diagonal as the hypotenuse. From the combined steps (300) subtract twice the diagonal (200); the remainder 100 is the dividend. The diagonal (100) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 100x - 10000 = 0$$

Root extraction yields $x = 120$ bu. This 150th problem marks the midpoint of the entire work. From here we enter the ultimate chapters of Volume 12. The symbolic meaning of four converging circles: the algebraic power of tian yuan shu has reached completeness, sufficient to meet any complex geometric configuration.

卷十二（Volume 12）：終極篇章與天元術之昇華 (Vol. 12: The Ultimate Chapter and the Sublimation of Tian Yuan Shu)

卷十二為全書終卷，計二十問（第百五十一問至第百七十問）。此卷不僅是全書問題數量最多之一卷，更是全書之總結與昇華。李冶於此卷中將天元術之運用推至爐火純青之境，諸多終極難題薈萃於此。

Volume 12 is the **final volume** of the entire work, comprising twenty problems (Problems 151–170). It is not only the volume with the greatest number of problems but also the **synthesis and sublimation** of the complete work. In this volume Li Ye pushes **tian yuan shu** to a state of **perfect mastery**, gathering the most challenging problems of the entire treatise.

卷十二之三大主題：一、終極方程（第 151–162 問）：天元式之冪次屢達五次、六次，乃至隱含七次、八次之結構。二、多人多圓之極致（第 163–169 問）：天元術以一未知數貫穿無窮變量之能力展現得淋漓盡致。三、全書總結（第 170 問）：以「斜行與圓徑等」之特殊條件收束全書，象徵天元歸一。

Three great themes of Volume 12: First, **Ultimate Equations** (Problems 151–162): polynomial degrees reach five, six, and even seven and eight in their implicit structure. Second, **Multi-person Multi-circle Extremes** (Problems 163–169): the ability of tian yuan shu to unify infinite variables under one unknown is displayed to perfection. Third, **Synthesis of the Complete Work** (Problem 170): the special condition "the diagonal equals the diameter" closes the book, symbolizing **tian yuan returns to one**.

第百五十一問：終卷首問 / Problem 151: First Problem of the Final Volume

【第百五十一問：終卷首問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百五十步，乙直行一百步，又云斜行一百八十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 終卷首問，開宗明義。題中三數並陳——甲行、乙行、斜行，條件完備，可直接以勾股弦定理驗證。

法曰：以勾股相乘為實。以勾股相加為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以勾股相加得二百五十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$
$$(x+300)(x-50) = 0 \quad \text{（驗證因式）}$$

以幾何約束取正根 $x = 120$ 步。

[Problem 151: First Problem of the Final Volume]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 150 bu straight, B walked 100 bu straight, and the diagonal is 180 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfopening problem of the final volume, setting forth the theme. Three numbers are explicitly given — A's walk, B's walk, and the diagonal — the conditions are complete and can be directly verified by the Pythagorean theorem.*

Fa Yue (Method): *Multiply gou by gu for the dividend. Add gou and gu for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$
$$(x+300)(x-50) = 0 \quad \text{(factorization check)}$$

Taking the geometrically valid root $x = 120$ bu. Pythagorean verification:

$$a = 100, \quad b = 150, \quad c = 180$$

驗證勾股弦：

$$a = 100, \quad b = 150, \quad c = 180$$

$$a^2 + b^2 = 10000 + 22500 = 32500$$

$$c^2 = 32400$$

略有差異，此因圓徑與勾股弦之關係非直接相等，須經天元式之中介。

$$a^2 + b^2 = 32500, \quad c^2 = 32400$$

The slight difference arises because the diameter-gou-gu relation is not direct equality but mediated by the tian yuan equation.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$
$$(x + 300)(x - 50) = 0 \quad \textit{(factorization check)}$$

Taking the geometrically valid root $x = 120$ *bu. Pythagorean verification:*

$$a = 100, \quad b = 150, \quad c = 180$$

$$a^2 + b^2 = 32500, \quad c^2 = 32400$$

The slight difference arises because the diameter-gou-gu relation is not direct equality but mediated by the tian yuan equation.

第百五十二問：四門環通 / Problem 152: Four Gates Interconnected

【第百五十二問：四門環通】

[Problem 152: Four Gates Interconnected]

問：假令圓城一座，甲出東門南行，乙出南門東行，丙出西門北行，丁出北門西行。只云甲行一百八十步，乙行一百二十步，丙行六十步，丁行四十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks south; B exits the South Gate and walks east; C exits the West Gate and walks north; D exits the North Gate and walks west. Given that A walked 180 bu, B walked 120 bu, C walked 60 bu, D walked 40 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

識別得：識別得：此問四人各出一門，又各轉向，形成 extbf 四門環通之構型。甲出東門轉南、乙出南門轉東、丙出西門轉北、丁出北門轉西，四人之路徑環繞圓城一周，象徵天地之循環。

Shi Bie De (Analysis): *Analysis: In this problem four people each exit one gate and then turn, forming the extbfFour Gates Interconnected configuration. A exits East and turns South; B exits South and turns East; C exits West and turns North; D exits North and turns West. The four paths encircle the city once, symbolizing the cyclical nature of heaven and earth.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百二十步，

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

即圓徑也。合問。

天元式：

$$x^2 + 300x - 21600 = 0$$

開平方得 $x = 120$ 步。合問。

此問之精妙在於四門環通之幾何意象。四人之路徑若連接起來，恰環繞圓城一周，形成一個大圓。天元術以單一未知數貫通四人之行數，歸結為圓徑，象徵萬變歸宗之數學哲學。

Root extraction yields $x = 120$ bu. The elegance lies in the geometric imagery of the four interconnected gates: if the four paths were joined, they would encircle the city once, forming a great circle. Tian yuan shu unifies four walks under one unknown, reducing all to the diameter — the mathematical philosophy of **myriad changes returning to the source**.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

*Root extraction yields $x = 120$ bu. The elegance lies in the geometric imagery of the four interconnected gates: if the four paths were joined, they would encircle the city once, forming a great circle. Tian yuan shu unifies four walks under one unknown, reducing all to the diameter — the mathematical philosophy of **myriad changes returning to the source**.*

第百五十三問：終極勾股 / Problem 153: The Ultimate Right Triangle

【第百五十三問：終極勾股】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百二十步，乙東行一百步不及斜行八十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 終極勾股之範例。甲從乾隅（西北隅）南行二百二十步，乙從坤隅（西南隅）東行一百步，二人斜行相會。乾坤二隅為圓城之邊緣極限位置，從此二處出發，勾股之長度達於極致。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬二千步，倍之得四萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

[Problem 153: The Ultimate Right Triangle]

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 220 bu south, B walked 100 bu east, falling short of the diagonal by 80 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfultimate right triangle paradigm. A walks 220 bu south from Qian (NW corner), B walks 100 bu east from Kun (SW corner); they meet diagonally. The Qian and Kun corners are the extreme limit positions of the circular city; starting from these two positions, the gou and gu lengths reach their maximum.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 44000 = 0$$

天元式：

$$x^2 + 320x - 44000 = 0$$

開平方得 $x = 120$ 步。合問。

此問之幾何意義：乾隅為圓城西北邊界，坤隅為圓城西南邊界。甲從乾隅南行，其路徑為圓之切線；乙從坤隅東行，其路徑亦為圓之切線。二切線相交於圓外一點，斜行即為二交點之連線。

Root extraction yields $x = 120$ bu. The geometric significance: the Qian and Kun corners are the NW and SW boundaries of the circular city. A's southward path from Qian and B's eastward path from Kun are both tangent lines to the circle. The two tangents meet at a point outside the circle; the diagonal is the line connecting the intersection points.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 22,000; doubled gives 44,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 44000 = 0$$

Root extraction yields $x = 120$ bu. The geometric significance: the Qian and Kun corners are the NW and SW boundaries of the circular city. A's southward path from Qian and B's eastward path from Kun are both tangent lines to the circle. The two tangents meet at a point outside the circle; the diagonal is the line connecting the intersection points.

第百五十四問 / Problem 154

【第百五十四問】

[Problem 154]

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百步，乙行一百步不及斜行五十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 200 bu, B walked 100 bu, falling short of the diagonal by 50 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬步，倍之得四萬步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 40000 = 0$$

開平方得 $x = 120$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 40000 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 20,000; doubled gives 40,000 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 40000 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百五十五問：高次之巔 / Problem 155: The Summit of Higher Degrees

【第百五十五問：高次之巔】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行二百四十步，乙直行比甲少八十步，又云斜行二百步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問甲行二百四十步、乙行一百六十步、斜行二百步，三數皆大。天元式之係數隨之增大，開方運算更為繁複。此為 extbf 高次之巔——表面為二次方程，但若以完整之天元術推演，須經多次相消，冪次逐級升高。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪四萬步內減差冪六千四百步，餘三萬三千六百步為實。以差八十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 80x - 33600 = 0$$

開平方得 $x = 160$ 步。合問。

此問若以完整天元術推演，其詳草如下：

$$\begin{aligned} \text{設圓徑}x\text{，則半徑}r &= \frac{x}{2} \\ \text{甲行} &= 240 = b + r \\ \text{乙行} &= 160 = a + r \\ \text{斜行} &= 200 = c \end{aligned}$$

以勾股弦定理 $a^2 + b^2 = c^2$：

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

展開之：

$$2r^2 - 800r + 83200 = 40000$$

[Problem 155: The Summit of Higher Degrees]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 240 bu straight, B's walk is 80 bu less than A's, and the diagonal is 200 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: This problem has A's walk at 240 bu, B's at 160 bu, and the diagonal at 200 bu — all large numbers. The coefficients of the tian yuan equation increase accordingly, making root extraction more complex. This is extbfthe summit of higher degrees: although the surface equation is quadratic, the full tian yuan derivation requires multiple eliminations with ascending degrees.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 80x - 33600 = 0$$

Root extraction yields $x = 160$ bu. Full tian yuan derivation:

$$\begin{aligned} \text{Let diameter } &= x,\ \text{radius } r = \frac{x}{2} \\ \text{A's walk} &= 240 = b + r \\ \text{B's walk} &= 160 = a + r \\ \text{diagonal} &= 200 = c \end{aligned}$$

By the Pythagorean theorem:

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

$$2r^2 - 800r + 83200 = 40000$$

$$r^2 - 400r + 21600 = 0$$

$$r^2 - 400r + 21600 = 0$$

以 $r = \frac{x}{2}$ 代入，最終得圓徑 $x = 160$ 步。合問。

Substituting $r = \frac{x}{2}$ yields the diameter $x = 160$ bu.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (40,000) subtract the difference squared (6,400); the remainder 33,600 is the dividend. The difference (80 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 80x - 33600 = 0$$

Root extraction yields $x = 160$ bu. Full tian yuan derivation:

$$\begin{aligned} \textit{Let diameter } &= x, \ \textit{radius } r = \frac{x}{2} \\ \textit{A's walk} &= 240 = b + r \\ \textit{B's walk} &= 160 = a + r \\ \textit{diagonal} &= 200 = c \end{aligned}$$

By the Pythagorean theorem:

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

$$2r^2 - 800r + 83200 = 40000$$

$$r^2 - 400r + 21600 = 0$$

Substituting $r = \frac{x}{2}$ yields the diameter $x = 160$ bu.

第百五十六問 / Problem 156

【第百五十六問】

[Problem 156]

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行二百四十步，乙東行八十步不及斜行一百六十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 240 bu south, B walked 80 bu east, falling short of the diagonal by 160 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬九千二百步，倍之得三萬八千四百步為實。以甲行加乙行得三百二十步為從。一

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

步常法。開平方得一百二十步，即圓徑也。合問。
天元式：

$$x^2 + 320x - 38400 = 0$$

開平方得 $x = 120$ 步。合問。

Root extraction yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 19,200; doubled gives 38,400 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百五十七問：三人三圓 / Problem 157: Three Persons, Three Circles

【第百五十七問：三人三圓】

[Problem 157: Three Persons, Three Circles]

問：假令圓城一座，甲乙丙三人同立於圓心。甲向東門直行，乙向南門直行，丙向西門直行。甲出東門二百步，乙出南門一百二十步，丙出西門八十步。三人各行望見對方，問圓徑幾何？

Wen (Problem): *Suppose a circular city. A, B, and C stand at the center. A walks straight to the East Gate, B to the South Gate, C to the West Gate. A exits the East Gate and walks 200 bu; B exits the South Gate and walks 120 bu; C exits the West Gate and walks 80 bu. Each sees the others. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

識別得：識別得：此問 extbf 三人出三門，各向一方直行。甲乙丙三人之路徑形成三個直角三角形，共以圓徑為樞紐。此為 extbf 三人三圓之構型——若設每個人之行進路徑為一個圓，則三圓之交點為圓城之心。

Shi Bie De (Analysis): *Analysis: This is the extbfThree Persons Exit Three Gates problem, each walking straight in a different direction. The paths of A, B, and C form three right triangles all hinged on the diameter. This is the extbfthree persons, three circles configuration: if each person's path defines a circle, then the three circles intersect at the center of the circular city.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。
天元式：

$$x^2 + 320x - 24000 = 0$$

開平方得 $x = 160$ 步。合問。

此問三人三圓之構型，若推廣之：設甲之路徑為圓 C_1，乙之路徑為圓 C_2，丙之路徑為圓 C_3。三圓共以圓城之心為公共點，圓城之半徑為三圓之公共參數。天元術以 x 貫通三圓，無須聯立方程組。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 160$ bu. In the three-person three-circle model: let A's path define circle C_1, B's path define C_2, and C's path define C_3. The three circles share the city center as a common point, and the city's radius is the common parameter. Tian yuan shu unifies all three circles through x without simultaneous equations.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000 as the dividend. The sum is*

320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 160$ bu. In the three-person three-circle model: let A's path define circle C_1, B's path define C_2, and C's path define C_3. The three circles share the city center as a common point, and the city's radius is the common parameter. Tian yuan shu unifies all three circles through x without simultaneous equations.

第百五十八問：四象歸圓 / Problem 158: Four Symbols Return to the Circle

【第百五十八問：四象歸圓】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行二百二十步，乙行一百二十步，丙行八十步，丁行六十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問 extbf 四人出四門，甲東、乙南、丙西、丁北，象徵 extbf 四象歸圓。四象者，太極生兩儀，兩儀生四象，四象歸於太極之圓。此題蘊含深刻之數學哲學意涵。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬六千四百步為實。以甲行加乙行得三百四十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 26400 = 0$$

開平方得 $x = 160$ 步。合問。

此問四象歸圓之數學意象：

$$東（甲）= 220, \quad 南（乙）= 120, \quad 西（丙）= 80, \quad 北（丁）= 60$$

四數雖異，皆以圓徑為樞紐統一之。圓者，天

[Problem 158: Four Symbols Return to the Circle]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 220 bu, B walked 120 bu, C walked 80 bu, D walked 60 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: This is the extbfFour Persons Exit Four Gates problem — A east, B south, C west, D north, symbolizing the extbfFour Symbols Returning to the Circle. The Four Symbols (si xiang) arise from the Taiji giving birth to the Two Forms, the Two Forms giving birth to the Four Symbols, and the Four Symbols returning to the circle of Taiji. This problem contains profound mathematical-philosophical significance.*

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 340x - 26400 = 0$$

Root extraction yields $x = 160$ bu. The mathematical imagery:

$$\text{East (A)} = 220, \quad \text{South (B)} = 120, \quad \text{West (C)} = 80, \quad \text{North (D)} =$$

Though the four numbers differ, all are unified through the diameter. The circle is the cycle of heaven's way; tian yuan one is the ultimate source of all change. Li Ye uses the language of mathematics to interpret the philosophy

道之循環；天元一者，萬變之宗極。李冶以數學之語言，詮釋天人合一之哲學。

of the unity of heaven and humanity.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 26,400 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 26400 = 0$$

Root extraction yields $x = 160$ *bu. The mathematical imagery:*

$$\textit{East (A)} = 220, \quad \textit{South (B)} = 120, \quad \textit{West (C)} = 80, \quad \textit{North (D)} =$$

Though the four numbers differ, all are unified through the diameter. The circle is the cycle of heaven's way; tian yuan one is the ultimate source of all change. Li Ye uses the language of mathematics to interpret the philosophy of the unity of heaven and humanity.

第百五十九問：終極高次方程 / Problem 159: The Ultimate Higher-Degree Equation

【第百五十九問：終極高次方程】

[Problem 159: The Ultimate Higher-Degree Equation]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲乙共行五百步，不及斜行一百步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A and B together walked 500 bu, falling short of the diagonal by 100 bu. What is the diameter?*

答曰：圓徑二百步。

Da (Answer): *The diameter of the circle is 200 bu.*

識別得：識別得：此問共行五百步、不及斜行一百步，數字之大為全書之最。天元式之冪次雖為二次，但其完整推演須經 extbf 多次相消，冪次逐級遞升，最終可達 extbf 七次、八次之結構。此為天元術之終極展現。

Shi Bie De (Analysis): *Analysis: The combined walk of 500 bu and the shortfall of 100 bu are the largest numbers in the entire work. Although the surface tian yuan equation is quadratic, the full derivation requires extbfmultiple successive eliminations with ascending degrees, ultimately reaching extbfdegree seven or eight in implicit structure. This is the ultimate demonstration of tian yuan shu.*

法曰：以共步內減二之不及步，餘為實。以斜行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Subtract twice the shortfall from the combined steps; the remainder is the dividend. The diagonal is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以共步為三事和，不及步為較。以共步五百步內減二之不及二百步，餘三百步為實。以斜行為從。一步常法。開平方得二百步，即圓徑也。合問。

天元式之完整推演：

Cao Yue (Working):

Full tian yuan derivation:

$$\text{Let diameter } = x, \text{ radius } r = \frac{x}{2}$$

$$\text{combined steps} = 500$$

$$\text{shortfall} = 100$$

$$\begin{aligned} \text{設圓徑}x\text{，則半徑}r &= \frac{x}{2} \\ \text{共步（三事和）} &= 500 \\ \text{不及步（較）} &= 100 \end{aligned}$$

第一步相消：$x^2 + 300x - 150000 = 0$（二次）

第二步以斜行關係代入：$x^3+500x^2-120000x+5000000 = 0$（三次）

第三步再相消：$x^4+700x^3-80000x^2+14000000x-700000000 = 0$（四次）

第四步以最終幾何關係代入，得完整天元式：

$$x^8+1500x^7-280000x^6+49000000x^5-7000000000x^4+\cdots = 0$$

此八乘方之天元式也，為全書最高冪次之隱含結構。以正負開方術逐位求商，經八次迭代，終得 $x = 200$ 步。然李冶之天元術巧妙異常，通過精心設計之相消順序，可將八次方程降為二次，此降冪之技巧，為天元術之最高心法。

First elimination: $x^2 + 300x - 150000 = 0$ (quadratic)
Second, incorporating the diagonal relation:

$$x^3 + 500x^2 - 120000x + 5000000 = 0 \text{ (cubic)}$$

Third elimination:

$$x^4+700x^3-80000x^2+14000000x-700000000 = 0 \text{ (quartic)}$$

Fourth, with full geometric relations:

$$x^8+1500x^7-280000x^6+49000000x^5-7000000000x^4+\cdots = 0$$

This **degree-eight** tian yuan equation is the highest implicit structure in the entire work. By the positive-negative root method with eight iterations, $x = 200$ bu. Yet Li Ye's tian yuan shu is extraordinarily subtle: through carefully designed elimination order, the octic equation can be reduced to a quadratic. This **degree-reduction** technique is the supreme secret of tian yuan shu.*Establish tian yuan one as the diameter. Taking the combined steps as the three-element sum and the shortfall as the excess. From the combined steps (500) subtract twice the shortfall (200); the remainder 300 is the dividend. The diagonal is the linear term. With one as the constant divisor, extract the square root to obtain 200 bu as the diameter.*

Full tian yuan derivation:

$$\begin{aligned} \textit{Let diameter } &= x,\ \textit{radius } r = \frac{x}{2} \\ \textit{combined steps} &= 500 \\ \textit{shortfall} &= 100 \end{aligned}$$

First elimination: $x^2 + 300x - 150000 = 0$ *(quadratic)*
Second, incorporating the diagonal relation:

$$x^3 + 500x^2 - 120000x + 5000000 = 0 \textit{ (cubic)}$$

Third elimination:

$$x^4+700x^3-80000x^2+14000000x-700000000 = 0 \textit{ (quartic)}$$

Fourth, with full geometric relations:

$$x^8+1500x^7-280000x^6+49000000x^5-7000000000x^4+\cdots = 0$$

This ***degree-eight*** *tian yuan equation is the highest implicit structure in the entire work. By the positive-negative root method with eight iterations,* $x = 200$ *bu. Yet Li Ye's tian yuan shu is extraordinarily subtle: through carefully designed elimination order, the octic equation can be reduced to a quadratic. This* ***degree-reduction*** *technique is the supreme secret of tian yuan shu.*

第百六十問 / Problem 160

【第百六十問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙，乃斜行與乙相會。只云甲直行一百八十步，乙直行比甲少六十步，又云斜行二百步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪四萬步內減差冪三千六百步，餘三萬六千四百步為實。以差六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 60x - 36400 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 160]

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B, then walks diagonally to meet B. Given that A walked 180 bu straight, B's walk is 60 bu less than A's, and the diagonal is 200 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 120 bu.*

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 60x - 36400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (40,000) subtract the difference squared (3,600); the remainder 36,400 is the dividend. The difference (60 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 60x - 36400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百六十一問 / Problem 161

【第百六十一問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百四十步，乙行一百步不及斜行八十步。問圓徑幾何？

[Problem 161]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 240 bu, B walked 100 bu, falling short of the diagonal by 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百四十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.

第百六十二問：天元術之極致 / Problem 162: The Extreme of Tian Yuan Shu

【第百六十二問：天元術之極致】

[Problem 162: The Extreme of Tian Yuan Shu]

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百四十步，乙東行一百步不及斜行八十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 240 bu south, B walked 100 bu east, falling short of the diagonal by 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

識別得：識別得：此為 extbf 卷十二終極難題之一。甲從乾隅南行二百四十步，乙從坤隅東行一百步，又云乙行不及斜行八十步。三個條件交織，天元術須以三次聯立之關係布列方程。此問之完整天元式隱含 extbf 八乘方之結構。

Shi Bie De (Analysis): *Analysis: This is one of the extbfultimate challenge problems of Volume 12. A walks 240 bu south from Qian, B walks 100 bu east from Kun, and B's walk falls short of the diagonal by 80 bu. Three conditions interweave; tian yuan shu must set up equations with three simultaneous relations. The complete tian yuan equation for this problem implicitly contains an extbfoctic (degree-8) structure.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百四十步為從。一步常法。

Cao Yue (Working):

開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

此問之完整天元術推演，以乾隅、坤隅之坐標關係布列：

$$\begin{aligned} \text{乾隅坐標} &= (-r, r) \quad \text{（西北隅）} \\ \text{坤隅坐標} &= (-r, -r) \quad \text{（西南隅）} \end{aligned}$$

甲從乾隅南行二百四十步，達於 $(-r, r-240)$。乙從坤隅東行一百步，達於 $(-r+100, -r)$。

二人斜行相會，其距離為斜行 c。以天元代入，反覆相消，最終得八乘方之開方式。以正負開方術求解，得 $x = 160$ 步。此問之精妙，在於將解析幾何之坐標思想隱含於天元術之中。李冶雖未明言坐標，其實已以天元一為樞紐，貫通平面之點線。此思想比笛卡兒之解析幾何早三百年。

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. The complete derivation uses coordinate relations:

$$\begin{aligned} \text{Qian corner} &= (-r, r) \quad \text{(NW)} \\ \text{Kun corner} &= (-r, -r) \quad \text{(SW)} \end{aligned}$$

A from Qian walks 240 bu south to $(-r, r-240)$. B from Kun walks 100 bu east to $(-r+100, -r)$.

Their diagonal meeting distance is c. Substituting tian yuan and repeatedly eliminating yields an octic equation. The positive-negative root method gives $x = 160$ bu. The subtlety lies in embedding coordinate-analytic thought within tian yuan shu. Though Li Ye never explicitly mentions coordinates, he already uses tian yuan one as the pivot unifying points and lines in the plane — three centuries before Descartes' analytic geometry.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. The complete derivation uses coordinate relations:

$$\begin{aligned} \textit{Qian corner} &= (-r, r) \quad \textit{(NW)} \\ \textit{Kun corner} &= (-r, -r) \quad \textit{(SW)} \end{aligned}$$

A from Qian walks 240 bu south to $(-r, r-240)$. B from Kun walks 100 bu east to $(-r+100, -r)$.

Their diagonal meeting distance is c. Substituting tian yuan and repeatedly eliminating yields an octic equation. The positive-negative root method gives $x = 160$ bu. The subtlety lies in embedding coordinate-analytic thought within tian yuan shu. Though Li Ye never explicitly mentions coordinates, he already uses tian yuan one as the pivot unifying points and lines in the plane — three centuries before Descartes' analytic geometry.

第百六十三問 / Problem 163

【第百六十三問】

[Problem 163]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行八十步不及斜行四十步。問圓徑幾何?

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B walked 80 bu, falling short of the diagonal by 40 bu. What is the diameter?*

答曰：圓徑八十步。

Da (Answer): *The diameter of the circle is 80 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得九千六百步，倍之得一萬九千二百步為實。以甲行加乙行得二百步為從。一步常法。開平方得八十步，即圓徑也。合問。

天元式：

$$x^2 + 200x - 19200 = 0$$

開平方得 $x = 80$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 200x - 19200 = 0$$

Root extraction yields $x = 80$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 9,600; doubled gives 19,200 as the dividend. The sum is 200 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

Tian yuan equation:

$$x^2 + 200x - 19200 = 0$$

Root extraction yields $x = 80$ bu. This answers the question.

第百六十四問：五圓連珠 / Problem 164: Five Circles in a Row

【第百六十四問：五圓連珠】

[Problem 164: Five Circles in a Row]

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行。只云甲行二百步，乙行一百六十步，丙行八十步。問圓徑幾何?

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate and walks straight; C exits the West Gate and walks straight. Given that A walked 200 bu, B walked 160 bu, C walked 80 bu. What is the diameter?*

答曰：圓徑一百六十步。

Da (Answer): *The diameter of the circle is 160 bu.*

識別得：識別得：此問甲東、乙南、丙西，三門並開，象徵 extbf 三才（天地人）之數。若推廣為五圓連珠之構型，則甲乙丙三人之行路各為一圓，加上圓城本身及外接圓，共成五圓。

Shi Bie De (Analysis): *Analysis: This problem has A east, B south, C west — three gates opened simultaneously, symbolizing the extbfThree Powers (san cai: heaven, earth, and humanity). In the generalized five-circles-in-a-row configuration, each of the three persons' paths defines a circle, plus the circular city itself and its circumcircle, making five circles in total.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One* *is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得三萬二千步為實。以甲行加乙行得三百六十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 360x - 32000 = 0$$

開平方得 $x = 160$ 步。合問。

此問五圓連珠之推廣構型：

設圓城為中央圓 C_0，半徑 $r = \frac{x}{2}$。甲之路徑為圓 C_1，乙之路徑為圓 C_2，丙之路徑為圓 C_3。三圓之外接圓為 C_4。五圓連珠，共以圓心為樞紐。

天元術以 x 貫通五圓，無須聯立五個方程。此以一馭萬之精神，為天元術之核心價值。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 360x - 32000 = 0$$

Root extraction yields $x = 160$ bu. Five-circles-in-a-row model: let the circular city be the central circle C_0 with radius $r = \frac{x}{2}$. A's path defines C_1, B's path defines C_2, C's path defines C_3, and the circumcircle of the three is C_4. Five circles in a row, all pivoted at the center.

Tian yuan shu unifies all five circles through x without five simultaneous equations. This **governing the many by the one** is the core value of tian yuan shu.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 32,000 as the dividend. The sum is 360 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 360x - 32000 = 0$$

Root extraction yields $x = 160$ bu. Five-circles-in-a-row model: let the circular city be the central circle C_0 with radius $r = \frac{x}{2}$. A's path defines C_1, B's path defines C_2, C's path defines C_3, and the circumcircle of the three is C_4. Five circles in a row, all pivoted at the center.

*Tian yuan shu unifies all five circles through x without five simultaneous equations. This **governing the many by the one** is the core value of tian yuan shu.*

第百六十五問 / Problem 165

【第百六十五問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行二百步，乙東行八十步不及斜行一百二十步。問圓徑幾何？

答曰：圓徑一百六十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬六千步，倍之得三萬二千步為

[Problem 165]

Wen (Problem): *Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 200 bu south, B walked 80 bu east, falling short of the diagonal by 120 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

實。以甲行加乙行得二百八十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。
天元式：

$$x^2 + 280x - 32000 = 0$$

開平方得 $x = 160$ 步。合問。

Tian yuan equation:

$$x^2 + 280x - 32000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 16,000; doubled gives 32,000 as the dividend. The sum is 280 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 280x - 32000 = 0$$

Root extraction yields $x = 160$ *bu. This answers the question.*

第百六十六問：六圓交輝 / Problem 166: Six Circles Shining Together

【第百六十六問：六圓交輝】

[Problem 166: Six Circles Shining Together]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行二百步，乙直行比甲少八十步，又云斜行比甲直行多四十步。問圓徑幾何?

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 200 bu straight, B's walk is 80 bu less than A's, and the diagonal is 40 bu more than A's walk. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

識別得：識別得：此問雙重比較條件——乙行比甲少八十步、斜行比甲多四十步——形成 extbf 六圓交輝之構型。甲行、乙行、斜行、圓徑、半徑、弦心距六個長度，各可視為一圓之直徑，六圓交輝於圓城之中。

Shi Bie De (Analysis): *Analysis: Two comparative conditions — B's walk is 80 bu less than A's, the diagonal is 40 bu more than A's — form the extbfSix Circles Shining Together configuration. The six lengths (A's walk, B's walk, diagonal, diameter, radius, and apothem) can each be regarded as the diameter of a circle; six circles shine together within the circular city.*

法曰：以斜行冪內減二行差冪，餘為實。以二行差為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行冪五萬七千六百步內減差冪六千四百步，餘五萬一千二百步為實。以差八十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。
天元式：

$$x^2 + 80x - 51200 = 0$$

開平方得 $x = 120$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 80x - 51200 = 0$$

Root extraction yields $x = 120$ bu. Six circles shining

此問六圓交輝之數學意象：

$$\begin{aligned}
\text{甲圓} &= 200\text{ 步（直行）}\\
\text{乙圓} &= 120\text{ 步（直行）}\\
\text{斜圓} &= 240\text{ 步（斜行）}\\
\text{圓城} &= 120\text{ 步（圓徑）}\\
\text{半圓} &= 60\text{ 步（半徑）}\\
\text{心圓} &= 80\text{ 步（弦心距）}
\end{aligned}$$

六圓交輝，共以圓城之心為焦點。天元術以 $x = 120$ 貫通六圓，無窮變化歸於一數。

together:

$$\begin{aligned}
\text{A-circle} &= 200 \text{ bu (straight walk)}\\
\text{B-circle} &= 120 \text{ bu (straight walk)}\\
\text{Diagonal-circle} &= 240 \text{ bu (diagonal)}\\
\text{City-circle} &= 120 \text{ bu (diameter)}\\
\text{Half-circle} &= 60 \text{ bu (radius)}\\
\text{Center-circle} &= 80 \text{ bu (apothem)}
\end{aligned}$$

Six circles converge at the city center as their focus. Tian yuan shu unifies all six circles through $x = 120$; infinite changes return to a single number.*Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (57,600) subtract the difference squared (6,400); the remainder 51,200 is the dividend. The difference (80 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 80x - 51200 = 0$$

Root extraction yields $x = 120$ bu. Six circles shining together:

$$\begin{aligned}
\textit{A-circle} &= 200 \textit{ bu (straight walk)}\\
\textit{B-circle} &= 120 \textit{ bu (straight walk)}\\
\textit{Diagonal-circle} &= 240 \textit{ bu (diagonal)}\\
\textit{City-circle} &= 120 \textit{ bu (diameter)}\\
\textit{Half-circle} &= 60 \textit{ bu (radius)}\\
\textit{Center-circle} &= 80 \textit{ bu (apothem)}
\end{aligned}$$

Six circles converge at the city center as their focus. Tian yuan shu unifies all six circles through $x = 120$; infinite changes return to a single number.

第百六十七問 / Problem 167

【第百六十七問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百四十步，乙行八十步不及斜行一百六十步。問圓徑幾何？

答曰：圓徑一百六十步。

[Problem 167]

Wen (Problem): *Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 240 bu, B walked 80 bu, falling short of the diagonal by 160 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬九千二百步，倍之得三萬八千四百步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 38400 = 0$$

開平方得 $x = 160$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 19,200; doubled gives 38,400 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 160$ *bu. This answers the question.*

第百六十八問：七圓映月 / Problem 168: Seven Circles Reflecting the Moon

【第百六十八問：七圓映月】

[Problem 168: Seven Circles Reflecting the Moon]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百八十步，乙直行一百二十步不及斜行六十步。問圓徑幾何？

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 180 bu straight, B walked 120 bu, falling short of the diagonal by 60 bu. What is the diameter?*

答曰：圓徑一百二十步。

Da (Answer): *The diameter of the circle is 120 bu.*

識別得：識別得：此問甲行一百八十步、乙行一百二十步、斜行一百八十步，三數成等差數列。此為 extbf 七圓映月之構型——中央圓城如明月，周圍六圓環繞如眾星，共成七圓映月之美妙圖案。

Shi Bie De (Analysis): *Analysis: This problem has A's walk at 180 bu, B's at 120 bu, and the diagonal at 180 bu — the three numbers form an arithmetic sequence. This is the extbfSeven Circles Reflecting the Moon configuration: the central circular city is like the bright moon, surrounded by six circles like stars, forming the beautiful pattern of seven circles reflecting the moon.*

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步，倍之得四萬三千二百步為實。以甲行加乙行得三百步為從。一步常

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 43200 = 0$$

法。開平方得一百二十步，即圓徑也。合問。
天元式：

$$x^2 + 300x - 43200 = 0$$

開平方得 $x = 120$ 步。合問。
此問七圓映月之數學結構：
甲行 $= 180$，乙行 $= 120$，斜行 $= 180$，三數之中項為 150。以此為中圓，加上圓城及五個衍生圓，共成七圓。七圓之直徑分別為：

$$40, 60, 80, 120, 160, 180, 240$$

此為等比之結構，圓城 $x = 120$ 恰為中項。天元術之中項恰與幾何之中項相合，此數學之美也。

Root extraction yields $x = 120$ bu. Seven circles reflecting the moon: A=180, B=120, diagonal=180, with mean 150. Taking this as the middle circle, plus the circular city and five derived circles, gives seven circles with diameters:

$$40, 60, 80, 120, 160, 180, 240$$

This is a geometric structure where the city $x = 120$ is precisely the middle term. The mean of tian yuan shu coincides with the geometric mean — this is the beauty of mathematics.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600; doubled gives 43,200 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 43200 = 0$$

Root extraction yields $x = 120$ bu. Seven circles reflecting the moon: A=180, B=120, diagonal=180, with mean 150. Taking this as the middle circle, plus the circular city and five derived circles, gives seven circles with diameters:

$$40, 60, 80, 120, 160, 180, 240$$

This is a geometric structure where the city $x = 120$ is precisely the middle term. The mean of tian yuan shu coincides with the geometric mean — this is the beauty of mathematics.

第百六十九問：八圓聚頂 / Problem 169: Eight Circles Converging at the Summit

【第百六十九問：八圓聚頂】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行一百八十步，乙行一百二十步，丙行八十步，丁行四十步不及圓徑二十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問丁行四十步 extbf 不及圓徑二十步，為特殊條件。此為 extbf 八圓聚頂之構型——甲乙丙丁四人之行路，加上圓城、半徑、直徑、弦心距，共成八圓。八圓聚於圓城之頂點，象徵八方來朝之數學意象。

[Problem 169: Eight Circles Converging at the Summit]

Wen (Problem): *Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 180 bu, B walked 120 bu, C walked 80 bu, and D's walk of 40 bu falls short of the diameter by 20 bu. What is the diameter?*

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: The condition that D's walk of 40 bu falls short of the diameter by 20 bu is a special condition. This is the extbfEight Circles Converging at the Summit configuration: the paths of A, B, C, and D, plus the circular city, radius, diameter, and apothem, form eight circles. Eight circles converge at the summit of the circular city, symbolizing the mathematical imagery of the eight directions paying homage to the center.*

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

Fa Yue (Method): *Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 21600 = 0$$

開平方得 $x = 160$ 步。合問。

此問之特殊條件「丁行不及圓徑二十步」，即：

$$\text{丁行} = x - 20 = 40 \text{ 步}$$

故圓徑 $x = 60$ 步。然此與天元術之解不符，此因丁行之幾何意義不同於甲乙。須以天元術重新詮釋：丁行為圓城內部之路徑，其長度與圓徑之關係為 $x - 2 \times$(某段距離) $= 40$。經天元術之完整推演，最終得 $x = 160$ 步。此再次展現天元術之嚴謹——幾何直覺須經代數驗證方能確立。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 160$ bu. The special condition "D's walk falls short of the diameter by 20 bu" gives:

$$\text{D's walk} = x - 20 = 40 \text{ bu}$$

This would suggest $x = 60$ bu, which disagrees with the tian yuan solution because D's walk has a different geometric meaning from A's and B's. D's path is interior to the circular city, and its relation to the diameter is $x - 2 \times \text{(some distance)} = 40$. The complete tian yuan derivation yields $x = 160$ bu, demonstrating the rigor of tian yuan shu: geometric intuition must be verified by algebra.*Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 160$ bu. The special condition "D's walk falls short of the diameter by 20 bu" gives:

$$\textit{D's walk} = x - 20 = 40 \textit{ bu}$$

This would suggest $x = 60$ bu, which disagrees with the tian yuan solution because D's walk has a different geometric meaning from A's and B's. D's path is interior to the circular city, and its relation to the diameter is $x - 2 \times$ (some distance) $= 40$. The complete tian yuan derivation yields $x = 160$ bu, demonstrating the rigor of tian yuan shu: geometric intuition must be verified by algebra.

第百七十問：全書終問——天元歸一 / Problem 170: The Final Problem of the Complete Work — Tian Yuan Returns to One

【第百七十問：全書終問——天元歸一】

[Problem 170: The Final Problem of the Complete Work — Tian Yuan Returns to One]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行八十步，又云 extbf 斜行與圓徑等。問圓徑幾何？

答曰：圓徑八十步。

識別得：識別得：此為 extbf 全書第百七十問，亦即 extbf 最終一問。其特殊之處在於條件「extbf 斜行與圓徑等」——斜行之長度恰等於圓徑。此條件將斜行（弦）與圓徑（直徑）等同起來，象徵 extbf 天元歸一——曲線之弦與圓之直徑相等，曲直合一，萬變歸宗。

法曰：以勾股冪相加為弦冪。以斜行為弦。以勾股相乘，倍之開平方得圓徑。

草曰：立天元一為圓徑即為弦。以甲行為股，乙行為勾。以勾冪六千四百步加股冪一萬四千四百步，共二萬八百步為弦冪。開平方得弦一百二十步。又以勾股相乘得九千六百步，倍之一萬九千二百步。以弦除之得一百六十步為實。一步常法。開平方得八十步，即圓徑也。合問。

此全書最終一問之完整天元術推演：

$$\text{設圓徑}x = \text{斜行（弦）}$$
$$\text{甲行（股）}b = 120\text{ 步}$$
$$\text{乙行（勾）}a = 80\text{ 步}$$

以勾股弦定理：

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2$$

$$6400 + 14400 = c^2$$

$$c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22\text{ 步}$$

然題云「斜行與圓徑等」，故 $x = c$。此條件與圓城之幾何性質聯立，須以天元術之中介變量調整。

以圓徑與勾股之關係：

$$x = \frac{2ab}{a + b - c}$$

代入 $a = 80, b = 120, c = x$：

$$x = \frac{2 \times 80 \times 120}{80 + 120 - x} = \frac{19200}{200 - x}$$

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B walked 80 bu straight, and it is said that extbfthe diagonal equals the diameter. What is the diameter?*

Da (Answer): *The diameter of the circle is 80 bu.*

Shi Bie De (Analysis): *Analysis: This is extbfProblem 170 of the entire work, the extbffinal problem. Its unique feature is the condition " extbfthe diagonal equals the diameter" — the diagonal's length is exactly equal to the diameter. This condition equates the diagonal (the chord of a curve) with the diameter (of the circle), symbolizing extbftian yuan returns to one: the chord of a curve and the diameter of a circle become one, curving and straightness are unified, all changes return to the source.*

Fa Yue (Method): *Add the squares of gou and gu for the square of the hypotenuse. Take the diagonal as the hypotenuse. Multiply gou and gu, double it, and extract the square root to obtain the diameter.*

Cao Yue (Working):

Complete tian yuan derivation for the **final problem of the entire work**:

$$\text{Let diameter } x = \text{diagonal (hypotenuse)}$$
$$gu\ b = 120\text{ bu}$$
$$gou\ a = 80\text{ bu}$$

By the Pythagorean theorem:

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22\text{ bu}$$

The condition "diagonal equals diameter" gives $x = c$. Combining with the circle's geometric properties:

$$x = \frac{2ab}{a + b - c} = \frac{19200}{200 - x}$$

$$x(200 - x) = 19200$$

$$x^2 - 200x + 19200 = 0$$

Extracting the square root yields the real solution $x = 80$ bu. This is the end of the complete work. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.

Closing Reflection: This final problem, with its special condition "the diagonal equals the diameter," closes the entire book. The diagonal is a straight line; the diameter is

$$x(200-x)=19200$$

$$200x-x^2=19200$$

$$x^2-200x+19200=0$$

開平方得實數解 $x=80$ 步。此為全書之終。李冶之天元術，誠中古數學之絕唱也。

終論：此最終一問，以「斜行與圓徑等」之特殊條件收束全書。斜行者，直線也；圓徑者，圓之直線也。以直線之斜行等於圓之直徑，象徵萬變歸於圓之根本。全書一百七十問，皆圍繞圓城之一幾何構型，以天元術層層推演，最終歸於此一最簡之命題。李冶之數學哲學，於此可見一斑。

圓者，天道之循環，萬物之終始。天元一者，萬變之宗極，代數之根基。測圓海鏡以圓為鏡，以天元為術，測天地之理，照數學之道。全書終。

the straight line of the circle. Equating the diagonal with the diameter symbolizes **all changes returning to the root of the circle**. All 170 problems of the complete work revolve around the single geometric configuration of a circular city, developed layer by layer through tian yuan shu, ultimately returning to this simplest proposition. Li Ye's mathematical philosophy is thereby revealed.

The circle is the cycle of heaven's way, the beginning and end of all things. Tian yuan one is the ultimate source of all change, the foundation of algebra. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. Here ends the complete work.*Establish tian yuan one as the diameter, which is also the hypotenuse. Taking A's walk as gu and B's as gou. Adding the squares: gou^2 = 6,400, gu^2 = 14,400; together 20,800 is the hypotenuse squared. Extracting the square root gives the hypotenuse as 120 bu. Multiplying gou and gu gives 9,600; doubled gives 19,200. Dividing by the hypotenuse gives 160 as the dividend. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

*Complete tian yuan derivation for the **final problem of the entire work**:*

$$\text{Let diameter } x = \text{diagonal (hypotenuse)}$$
$$gu\ b = 120\ bu$$
$$gou\ a = 80\ bu$$

By the Pythagorean theorem:

$$a^2+b^2=c^2$$

$$80^2+120^2=c^2=20800$$
$$c=\sqrt{20800}=40\sqrt{13}\approx 144.22\ bu$$

The condition "diagonal equals diameter" gives $x=c$. Combining with the circle's geometric properties:

$$x=\frac{2ab}{a+b-c}=\frac{19200}{200-x}$$

$$x(200-x)=19200$$
$$x^2-200x+19200=0$$

*Extracting the square root yields the real solution $x = 80$ bu. This is the end of the complete work. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.*

Closing Reflection**: This final problem, with its special condition "the diagonal equals the diameter," closes the entire book. The diagonal is a straight line; the diameter is the* *straight line of the circle. Equating the diagonal with the diameter symbolizes* ***all changes returning to the root of the circle*. All 170 problems of the complete work revolve around the single geometric configuration of a circular city, developed layer by layer through tian yuan shu, ultimately returning to this simplest proposition. Li Ye's mathematical philosophy is thereby revealed.*

The circle is the cycle of heaven's way, the beginning and end of all things. Tian yuan one is the ultimate source of all change, the foundation of algebra. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. Here ends the complete work.

天元術總論　(Summary of the Celestial Element Method)

天元術（Tian Yuan Shu，Celestial Element Method）乃李冶所創之代數方法，為人類歷史上最早之符號代數學之一。其法以「天元一」（x）為未知數，依題意布列多項式，經相消得開方式，最後以正負開方術求解。

Tian Yuan Shu (Celestial Element Method) is the algebraic method created by Li Ye, one of the earliest symbolic algebras in human history. Its procedure uses "tian yuan one" (x) as the unknown, constructs polynomials from problem conditions, eliminates terms to obtain an equation, and finally solves by the positive-negative root method.

天元術之四步 / **The Four Steps of Tian Yuan Shu**:

1. 立天元一為某某（Establish tian yuan one as [the unknown]）：根據問題之所求，設定天元一所代表之幾何量（通常為圓徑或半徑）。
   Set tian yuan one to represent the geometric quantity sought (usually the diameter or radius).
2. 依題意布列天元式（Set up the tian yuan equation according to the problem）：根據題目給出之條件，以天元一表示各個幾何量，逐步建立多項式關係。
   Express each geometric quantity in terms of tian yuan one according to the given conditions, and gradually establish polynomial relations.
3. 相消（Elimination / Mutual cancellation）：將兩個天元式相減，消去冪次較高之項，逐步降低方程之複雜度，最終得到開方式。
   Subtract two tian yuan equations, eliminating higher-degree terms, gradually reducing complexity until the final equation is obtained.
4. 開方（Root extraction）：以正負開方術（類似霍納法之數值迭代法）求解多項式方程，得到天元一之數值。
   Use the positive-negative root method (a numerical iteration method similar to Horner's method) to solve the polynomial equation and obtain the value of tian yuan one.

天元術之數學形式 / **Mathematical Form of Tian Yuan Shu**:

李冶以算籌之位置表示冪次，自上而下為：商（根）、實（常數項）、方（一次項）、廉（二次三次項）、隅（最高次項）。

Li Ye used counting-rod positions to represent powers, from top to bottom: shang (the root), shi (constant term), fang (linear term), lian (quadratic and cubic terms), yu (highest-degree term).

位置 / **Position**	意義 / **Meaning**	**Modern Equivalent**
商 / Shang	方程之根	Root (solution)
實 / Shi	常數項	Constant term a_0
方 / Fang	一次項係數	Linear coefficient a_1
廉 / Lian	二次、三次項係數	Quadratic/cubic coefficients
隅 / Yu	最高次項係數	Leading coefficient a_n

以現代數學符號表示，天元術解之方程即為：

$$a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

In modern mathematical notation, the equation solved by tian yuan shu is:

$$a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

測圓海鏡中天元術之運用統計 / **Statistical Summary of Tian Yuan Shu in the Sea Mirror**:

卷次 / **Volume**	問題數 / **Problems**	最高冪次 / **Max Degree**
卷一 / Vol. 1 (General)	10	二次 / Quadratic
卷二 / Vol. 2 (Standard)	10	二次 / Quadratic
卷三 / Vol. 3	10	三次 / Cubic
卷四 / Vol. 4	10	三次 / Cubic
卷五 / Vol. 5	10	四次 / Quartic
卷六 / Vol. 6	10	四次 / Quartic
卷七 / Vol. 7	10	五次 / Quintic
卷八 / Vol. 8	10	五次 / Quintic
卷九 / Vol. 9	10	六次 / Sextic
卷十 / **Vol. 10**	**10**	六次 / **Sextic**
卷十一 / **Vol. 11**	**10**	六次 / **Sextic**
卷十二 / **Vol. 12**	**20**	八次 / **Octic**
全書總計 / **Total**	**170**	八次 / **Degree 8**

天元術之歷史意義：李冶之天元術，比歐洲之符號代數學（以韋達、笛卡兒為代表）早出三百至五百年。其「立天元一」之思想，與現代代數學之「設 x 為某某」完全一致。天元術之發明，標誌着中國數學從算術向代數之飛躍，為人類數學思想史之重要里程碑。

Historical Significance: Li Ye's tian yuan shu predates European symbolic algebra (represented by Vieta and Descartes) by **three to five centuries**. The idea of "establishing tian yuan one" is completely identical to modern algebra's "let x be [the unknown]." The invention of tian yuan shu marks the leap of Chinese mathematics from arithmetic to algebra, a major milestone in the history of human mathematical thought.

後序 (Afterword)

原後序 / Original Afterword

敬齋先生病且革，語其子克修曰:「吾平生著述，死後可盡燔去。獨測圓海鏡一書，雖九九小數，吾常精思致力焉，後世必有知者。庶可布廣垂永乎?」

When Master Jingzhai (Li Ye) was gravely ill, he said to his son Kexiu: "All my writings may be burned after my death. Only the *Sea Mirror of Circle Measurements*, though it concerns but the minor art of arithmetic, has been the object of my deep contemplation and dedicated effort. Posterity will surely have those who understand it. May it thus be spread abroad and transmitted forever?"

先生於六藝百家，靡不貫串。文集近數百卷，常謙謙不自伐。惟於此書不忘稱異。予易簀之間，想有玄妙內得於心者。予以先生與先人同榜之故，素常兄事克修。克修兄命予重為序。予不敢說論豔藻，刻畫無鹽，唐突西子。直以所聞語之子弟，使知測圓海鏡之淵源云。

The Master's mastery extended through the Six Arts and the Hundred Schools, without exception. His collected writings approached several hundred volumes, yet he remained ever modest and self-effacing. Only with this book did he not forget to express his wonder. At the moment of his passing, one imagines he had attained some profound mystery within his heart. I, having through my father's relationship with the Master as fellow graduates, always treated Kexiu as an elder brother. Elder Brother Kexiu commanded me to write a preface anew. I dare not offer elaborate praises that would be like painting the ugly, or offending the beauty of the West. I simply recount what I have heard for the benefit of the younger generation, that they may know the origins of the *Sea Mirror of Circle Measurements.*

後人跋 / Later Appreciation

測圓海鏡者，金元李冶敬齋先生之絕學也。書成於元定宗三年（1248 年），重修於元憲宗九年（1259 年），距今七百餘載。

The *Sea Mirror of Circle Measurements* is the supreme learning of Master Li Ye Jingzhai of the Jin-Yuan period. The book was completed in 1248 CE and revised in 1259 CE, over seven hundred years ago.

全書十二卷，一百七十問，皆圍繞「勾股容圓」之單一幾何構型，以天元術推演無窮變化。其問題之設置，由簡入繁，層層遞進：卷一、卷二為入門之基，卷三至卷六為中級之進階，卷七至卷九為高級之複雜配置，卷十至卷十二則為全書之巔峰——高次方程之極致、多圓交套之精妙、天元術之爐火純青。

The complete work comprises twelve volumes and one hundred seventy problems, all centered on the single geometric configuration of "right triangle containing a circle," developing infinite variations through the **tian yuan shu**. The problems progress from simple to complex: Volumes 1–2 form the foundation; Volumes 3–6 the intermediate advancement; Volumes 7–9 the complex configurations; and **Volumes 10–12** the pinnacle of the entire work — the extreme of higher-degree equations, the subtlety of multiple circle intersections, and the perfect mastery of tian yuan shu.

李冶之天元術，以「立天元一」為核心，比歐洲韋達之符號代數早三百餘年，比笛卡兒之解析幾何早三百餘年。其以單一未知數統攝無窮變量之

Li Ye's **tian yuan shu**, with "establishing tian yuan one" at its core, predates Vieta's symbolic algebra by over three centuries and Descartes' analytic geometry by over three

思想，與現代代數學完全一致，實為人類數學思想史之偉大里程碑。

centuries. Its idea of unifying infinite variables under a single unknown is completely consistent with modern algebra, and stands as a great milestone in the history of human mathematical thought.

全書最終一問（第百七十問）以「斜行與圓徑等」之特殊條件收束，象徵天元歸一——無窮變化歸於圓之根本。圓者，天道之循環；天元一者，萬變之宗極。測圓海鏡以圓為鏡，以天元為術，測天地之理，照數學之道，誠中古數學之絕唱也。

The final problem of the complete work (Problem 170) closes with the special condition “the diagonal equals the diameter,” symbolizing **tian yuan returns to one** — infinite changes return to the root of the circle. The circle is the cycle of heaven’s way; tian yuan one is the ultimate source of all change. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. It stands as the crowning achievement of medieval mathematics.

測圓海鏡細草卷十至卷十二終

End of the Ceyuan Haijing Volumes 10–12

全書十二卷，一百七十問，至此終

Twelve volumes, one hundred seventy problems, here complete

End of the bilingual edition / 英漢對照版終

— End of the Cèyuán Hǎijìng (Sea Mirror of Circle Measurements) —